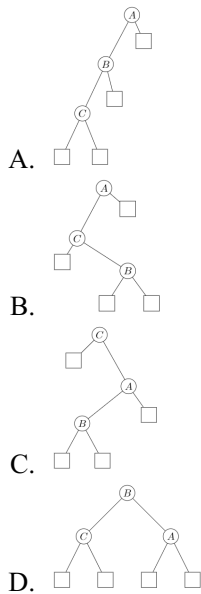


36.0.1 UGC NET CSE | June 2012 | Part 3 | Question: 42



Which of the following binary tree is optimal, if probabilities of successful search and unsuccessful search are same?



ugcnetcse-june2012-paper3

Answer key

36.0.2 UGC NET CSE | June 2012 | Part 3 | Question: 47



The colour of an object is largely determined by its diffuse reflection coefficient. If $K_d = (0.8, 0.4, 0)$, then what shall be the colour of the object, if the light used is blue and magenta?

- A. White and Red B. Red and Blue C. Black and White D. Black and Red

ugcnetcse-june2012-paper3

Answer key

36.0.3 UGC NET CSE | June 2013 | Part 2 | Question: 1



COCOMO stands for

- A. Composite COst MOdel B. Constructive COst MOdel
C. CONstructive COMposite MOdel D. COMprehensive CONstruction MOdel

ugcnetcse-june2013-paper2

Answer key

36.0.4 UGC NET CSE | June 2013 | Part 2 | Question: 2



Match the following :

- | | |
|-----------------|--|
| a. Good Quality | i. Program does not fail for a specified time in a given environment |
| b. Correctness | ii. Meets the functional Requirement |
| c. Predictable | iii. Meets Both functional and non-functional requirements |
| d. Reliable | iv. Process is under statistical control |

Codes :

- A. a-iii; b-ii; c-iv; d-i
C. a-i; b-ii; c-iv; d-iii

- B. a-ii; b-iii; c-iv; d-i
D. a-i; b-ii; c-iii; d-iv

ugcnetcse-june2013-paper2

Answer key 

36.0.5 UGC NET CSE | June 2013 | Part 2 | Question: 4



A good software design must have

- A. High module coupling, High module cohesion
B. High module coupling, Low module cohesion
C. Low module coupling, High module cohesion
D. Low module coupling, Low module cohesion

ugcnetcse-june2013-paper2

Answer key 

36.0.6 UGC NET CSE | June 2013 | Part 2 | Question: 6



When the following code is executed what will be the value of x and y?

```
int x=1, y=0;
```

```
y=x++;
```

- A. 2, 1 B. 2, 2 C. 1, 1 D. 1, 2

ugcnetcse-june2013-paper2

Answer key 

36.0.7 UGC NET CSE | June 2013 | Part 2 | Question: 7



How many values can be held by an array A(-1, m; 1, m)?

- A. m B. m^2 C. $m(m+1)$ D. $m(m+2)$

ugcnetcse-june2013-paper2

Answer key 

36.0.8 UGC NET CSE | June 2013 | Part 2 | Question: 8



What is the result of the expression $(1\&2)+(3/4)$?

- A. 1 B. 2 C. 3 D. 0

ugcnetcse-june2013-paper2

Answer key 

36.0.9 UGC NET CSE | June 2013 | Part 2 | Question: 9



How many times the word 'print' shall be printed by the following program segment?

```
for (i=1, i ≤ 2, i++)
```

```
for (j=1, j ≤ 2, j++)
for (k=1, k ≤ 2, k++)
printf("print\n")
```

- A. 1 B. 3 C. 6 D. 8

ugcnetcse-june2013-paper2

Answer key 

36.0.10 UGC NET CSE | June 2013 | Part 2 | Question: 10



Which of the following is not a type of Database Management System?

- A. Hierarchical B. Network C. Relational D. Sequential

ugcnetcse-june2013-paper2

Answer key 

36.0.11 UGC NET CSE | June 2013 | Part 2 | Question: 11



Manager's salary details are to be hidden from Employee table. This Technique is called as

- A. Conceptual level Datahiding B. Physical level Datahiding
C. External level Datahiding D. Logical level Datahiding

ugcnetcse-june2013-paper2

Answer key 

36.0.12 UGC NET CSE | June 2013 | Part 2 | Question: 12



A Network Schema

- A. restricts to one to many relationship B. permits to many to many relationship
C. Stores Data in a Database D. Stores Data in a Relation

ugcnetcse-june2013-paper2

Answer key 

36.0.13 UGC NET CSE | June 2013 | Part 2 | Question: 13



Which normal form is considered as adequate for usual database design?

- A. 2NF B. 3NF C. 4NF D. 5NF

ugcnetcse-june2013-paper2

Answer key 

36.0.14 UGC NET CSE | June 2013 | Part 2 | Question: 14



If $D_1, D_2, \dots, D_n$ are domains in a relational model, then the relation is a table, which is a subset of

- A. $D_1 + D_2 + \dots + D_n$ B. $D_1 \times D_2 \times \dots \times D_n$
C. $D_1 \cup D_2 \cup \dots \cup D_n$ D. $D_1 - D_2 - \dots - D_n$

ugcnetcse-june2013-paper2

Answer key 

36.0.15 UGC NET CSE | June 2013 | Part 2 | Question: 15



Which of the following addresses is used to deliver a message to the correct application program running on a host?

- A. Port B. IP C. Logical D. Physical

ugcnetcse-june2013-paper2

Answer key 

36.0.16 UGC NET CSE | June 2013 | Part 2 | Question: 16



In _____ substitution, a character in the plaintext is always changed to the same character in ciphertext, regardless of its position in the text.

- A. polyalphabetic
- B. monoalphabetic
- C. transpositional
- D. multialphabetic

ugcnetcse-june2013-paper2

Answer key 

36.0.17 UGC NET CSE | June 2013 | Part 2 | Question: 18



In hierarchical routing with 4800 routers, what region and cluster sizes should be chosen to minimize the size of the routing table for the three-layer hierarchy?

- A. 10 clusters, 24 regions and 20 routers
- B. 12 clusters, 20 regions and 20 routers
- C. 16 clusters, 12 regions and 25 routers
- D. 15 clusters, 16 regions and 20 routers

ugcnetcse-june2013-paper2

Answer key 

36.0.18 UGC NET CSE | June 2013 | Part 2 | Question: 19



In IPv4 header, the _____ field is needed to allow the destination host to determine which datagram a newly arrived fragments belongs to

- A. identification
- B. fragment offset
- C. time to live
- D. header checksum

ugcnetcse-june2013-paper2

Answer key 

36.0.19 UGC NET CSE | June 2013 | Part 2 | Question: 22



Given a grammar $S1 \rightarrow Sc, S \rightarrow SA \mid A, A \rightarrow aSb \mid ab$, there is a rightmost derivation $S1 \Rightarrow Sc \Rightarrow SAc \Rightarrow SaSbc$. Thus, $SaSbc$ is a right sentential form, and its handle is _____.

- A. SaS
- B. bc
- C. Sbc
- D. aSb

ugcnetcse-june2013-paper2

Answer key 

36.0.20 UGC NET CSE | June 2013 | Part 2 | Question: 25



Which is the correct statement(s) for Non Recursive predictive parser?

$S1 : First(\alpha) = \{t \mid \alpha \Rightarrow^* t\beta \text{ for some string } \beta\} \Rightarrow^* t\beta$

$S2 : Follow(X) = \{a \mid S \Rightarrow^* \alpha X a \beta \text{ for some string } \alpha \text{ and } \beta\}$

- A. Both statements S1 and S2 are incorrect
- B. S1 is incorrect S2 is correct
- C. S1 is correct S2 is incorrect
- D. Both statements S1 and S2 are correct

ugcnetcse-june2013-paper2

Answer key 

36.0.21 UGC NET CSE | June 2013 | Part 2 | Question: 26



Given an open address hash table with load factor $\alpha < 1$, the expected number of probes in a successful search is

- A. Atmost $\frac{1}{\alpha} \ln \left(\frac{1-\alpha}{\alpha} \right)$
- B. Atmost $\frac{1}{\alpha} \ln \left(\frac{1}{1-\alpha} \right)$
- C. Atleast $\frac{1}{\alpha} \ln \left(\frac{1}{1-\alpha} \right)$
- D. Atleast $\frac{1}{\alpha} \ln \left(\frac{\alpha}{1-\alpha} \right)$

ugcnetcse-june2013-paper2

Answer key 

36.0.22 UGC NET CSE | June 2013 | Part 2 | Question: 27



For a B-tree of height h and degree t , the total CPU time used to insert a node is

- A. $O(h \log t)$ B. $O(t \log h)$ C. $O(t^2 h)$ D. $O(th)$

ugcnetcse-june2013-paper2

Answer key 

36.0.23 UGC NET CSE | June 2013 | Part 2 | Question: 29



The value of postfix expression: $834+ -382/+*2\$3+$ is

- A. 17 B. 131 C. 64 D. 52

ugcnetcse-june2013-paper2

Answer key 

36.0.24 UGC NET CSE | June 2013 | Part 2 | Question: 31



Repository of information gathered from multiple sources, storing under unified scheme at a single site is called

- A. Data mining B. Meta Data C. Data warehousing D. Database

ugcnetcse-june2013-paper2

Answer key 

36.0.25 UGC NET CSE | June 2013 | Part 2 | Question: 32



The task of correcting and preprocessing data is called as

- A. Data streaming B. Data cleaning
C. Data mining D. Data storming

ugcnetcse-june2013-paper2

Answer key 

36.0.26 UGC NET CSE | June 2013 | Part 2 | Question: 36



A test contains 100 true/false questions. How many different ways can a student answer the question on the test, if the answer may be left blank also.

- A. $^{100}P_2$ B. $^{100}C_2$ C. 2^{100} D. 3^{100}

ugcnetcse-june2013-paper2

Answer key 

36.0.27 UGC NET CSE | June 2013 | Part 2 | Question: 42



A Boolean operator $\ominus$ is defined as follows:

$1 \ominus 1 = 1, 1 \ominus 0 = 0, 0 \ominus 1 = 0$ and $0 \ominus 0 = 1$

What will be the truth value of the expression $(x \ominus y) \ominus z = x \ominus (y \ominus z)$?

- A. Always true B. Always false
C. Sometimes true D. True when x, y, z are all true

ugcnetcse-june2013-paper2

Answer key 

36.0.28 UGC NET CSE | June 2013 | Part 2 | Question: 44

A set of processors $P_1, P_2, \dots, P_k$ can execute in parallel if Bernstein's conditions are satisfied on a pairwise basis; that is $P_1 \parallel P_2 \parallel P_3 \parallel \dots \parallel P_k$ if and only if:

- A. $P_i \parallel P_j$ for all $i \neq j$
- B. $P_i \parallel P_j$ for all $i = j + 1$
- C. $P_i \parallel P_j$ for all $i \leq j$
- D. $P_i \parallel P_j$ for all $i \geq j$

ugcnetcse-june2013-paper2

Answer key

36.0.29 UGC NET CSE | June 2013 | Part 2 | Question: 46

A virtual memory based memory management algorithm partially swaps out a process. This is an example of

- A. short term scheduling
- B. long term scheduling
- C. medium term scheduling
- D. mutual exclusion

ugcnetcse-june2013-paper2

Answer key

36.0.30 UGC NET CSE | June 2013 | Part 2 | Question: 49

Consider the following UNIX command:

`sort<in>temp; head-30<temp; nm temp`

Which of the following functions shall be performed by this command?

- A. Sort, taking the input from "temp", prints 20 lines from temp and delete the file temp
- B. Sort eh file "temp", removes 30 lines from temp and delete the file temp
- C. Sort, taking the input from "in" and writing the output to "temp", then prints 30 lines from temp on terminal. Finally "temp" is removed
- D. Sort, taking the input from "temp", and then prints 30 lines from "temp" on terminal. Finally "temp" is removed.

ugcnetcse-june2013-paper2

Answer key

36.0.31 UGC NET CSE | December 2013 | Part 2 | Question: 23

_____ is often used to prove the correctness of a recursive function.

- A. Diagonalization
- B. Commutivity
- C. Mathematical Induction
- D. Matrix Manipulation

ugcnetcse-dec2013-paper2

Answer key

36.0.32 UGC NET CSE | December 2013 | Part 2 | Question: 40

Answer key

36.0.33 UGC NET CSE | December 2013 | Part 3 | Question: 12

Which is the protocol for performing RPCs between applications in a language and system independent way?

- A. Hyper Text Transmission Protocol (HTTP)
- B. Simple Network Management Protocol (SNMP)
- C. Simple Object Access Protocol (SOAP)
- D. Simple Main Transfer Protocol (SMTP)

ugcnetcse-dec2013-paper3

Answer key 

36.0.34 UGC NET CSE | December 2013 | Part 3 | Question: 20



Match the following control strategies of protocol :

List – I

- a. Forward movement
- b. Unification
- c. Deep back-tracking
- d. Shallow back-tracking

List – II

- i. Variable can be done with a constant, another variable or a function
- ii. The entire conjunctive goal is executed
- iii. Previous sub goal to find alternative solutions
- iv. Chooses sub goal with possible unifier

Codes :

- A. a-iv, b-i, c-ii, d-iii
- C. a-iii, b-i, c-iv, d-ii

- B. a-ii, b-iv, c-i, d-iii
- D. a-ii, b-iii, c-iv, d-i

ugcnetcse-dec2013-paper3

Answer key 

36.0.35 UGC NET CSE | December 2014 | Part 3 | Question: 21



Which of the following statements is not true ?

- A. MPI_Isend and MPI_Irecv are non-blocking message passing routines of MPI.
- B. MPI_Issend and MPI_Ibsend are non-blocking message passing routines of MPI.
- C. MPI_Send and MPI_Recv are non-blocking message passing routines of MPI.
- D. MPI_Ssend and MPI_Bsend are blocking message passing routines of MPI.

ugcnetcse-dec2014-paper3

Answer key 

36.0.36 UGC NET CSE | December 2013 | Part 3 | Question: 27



In alpha-beta pruning, _____ is used to cut off the search at maximizing level only and _____ is used to cut off the search at minimizing level only.

- A. alpha, beta
- B. beta, alpha
- C. alpha, alpha
- D. beta, beta

ugcnetcse-dec2013-paper3

Answer key 

36.0.37 UGC NET CSE | December 2011 | Part 2 | Question: 2



The total number of comparisons in a bubble sort is

- A. $O(\log n)$
- B. $O(n \log n)$
- C. $O(n)$
- D. None of the above

ugcnetcse-dec2011-paper2

Answer key 

36.0.38 UGC NET CSE | June 2011 | Part 2 | Question: 2



The circumference of the two concentric disks are divided into 100 sections each. For the outer disk, 100 of the sections are painted red and 100 of the sections are painted blue. For the inner disk, the sections are painted red and blue in an arbitrary manner. It is possible to align the two disks so that

- A. 100 or more
- B. 125 or more
- C. 150 or more

D. 175 or more

of the sections on the inner disks have their colours matched with the corresponding section on outer disk.

ugcnetcse-june2011-paper2

Answer key 

36.0.39 UGC NET CSE | June 2011 | Part 2 | Question: 5



The number of 1's present in the binary representation of $10 \times 256 + 5 \times 16 + 5$ is

- A. 5 B. 6 C. 7 D. 8

ugcnetcse-june2011-paper2

Answer key 

36.0.40 UGC NET CSE | June 2011 | Part 2 | Question: 6



The hexadecimal number equivalent to $(1762.46)_8$ is

- A. $3F2.89$ B. $3F2.98$ C. $2F3.89$ D. $2F3.98$

ugcnetcse-june2011-paper2

Answer key 

36.0.41 UGC NET CSE | June 2011 | Part 2 | Question: 7



$(A + B)\overline{AB}$ is equivalent to

- A. $A \oplus B$ B. $A \odot B$
C. $(A \oplus B) \odot A$ D. $(A \odot B) \oplus A$

ugcnetcse-june2011-paper2

Answer key 

36.0.42 UGC NET CSE | June 2011 | Part 2 | Question: 8



A latch is constructed using two cross-coupled

- A. AND and OR gates B. AND gates
C. NAND and NOR gates D. NAND gates

ugcnetcse-june2011-paper2

Answer key 

36.0.43 UGC NET CSE | June 2011 | Part 2 | Question: 9



A multiplexer is a logic circuit that

- A. Accepts one input and gives several output B. Accepts many inputs and gives many output
C. Accepts many inputs and gives one output D. Accepts one input and gives one output

ugcnetcse-june2011-paper2

Answer key 

36.0.44 UGC NET CSE | June 2011 | Part 2 | Question: 10



8-bit 1's complement form of -77.25 is

- A. 01001101.0100 B. 01001101.0010
C. 10110010.1011 D. 10110010.1101

ugcnetcse-june2011-paper2

Answer key 

36.0.45 UGC NET CSE | June 2011 | Part 2 | Question: 11

From the point of view of the programmer, what are the major advantages of using a high-level language rather than internal machine code or assembly language?

- A. Program portability
- B. Easy development
- C. Efficiency
- D. None of the above

ugcnetcse-june2011-paper2

Answer key

36.0.46 UGC NET CSE | June 2011 | Part 2 | Question: 12

What features make $C++$ so powerful ?

- A. Easy implementation
- B. Reusing old code
- C. Easy memory management
- D. All of the above

ugcnetcse-june2011-paper2

Answer key

36.0.47 UGC NET CSE | June 2011 | Part 2 | Question: 17

Which of the following is the process by which a user's privileges ascertained ?

- A. Authorization
- B. Authentication
- C. Access Control
- D. None of these

ugcnetcse-june2011-paper2 databases

Answer key

36.0.48 UGC NET CSE | June 2011 | Part 2 | Question: 28

Consider the following statement :

- (i) The bandwidth of the A.M., wave depends on the band width of the modulating signal.
- (ii) The bandwidth of the A.M., wave depends on the modulation index.
- (iii) The bandwidth of the F.M, wave for all practical purpose depends on the amplitude of the carrier.

Of these statements the correct statements are

- A. (i, ii)
- B. (i, iii)
- C. (ii, iii)
- D. All of the above

ugcnetcse-june2011-paper2 computer-networks

Answer key

36.0.49 UGC NET CSE | June 2011 | Part 2 | Question: 29

A smart modem can dial, hangup and answer incoming calls automatically. Can you tell who provides the appropriate instructions to the modem for this purpose ?

- A. Communication software
- B. Error detection protocols
- C. Link access procedure (LAP)
- D. Telecommunications

ugcnetcse-june2011-paper2 computer-networks

Answer key

36.0.50 UGC NET CSE | June 2011 | Part 2 | Question: 33

Which of the following permanent database that has an entry for each terminal symbol ?

- A. Literal table
- B. Identifier table
- C. Terminal table
- D. Source table

ugcnetcse-june2011-paper2 databases

Answer key

36.0.51 UGC NET CSE | August 2016 | Part 2 | Question: 17

Relational database schema normalization is NOT for :

- A. Reducing the number of joins required to satisfy a query.
- B. Eliminating uncontrolled redundancy of data stored in the database.
- C. Eliminating number of anomalies that could otherwise occur with inserts and deletes.
- D. Ensuring that functional dependencies are enforced.

ugcnetcse-aug2016-paper2

Answer key

36.0.52 UGC NET CSE | August 2016 | Part 3 | Question: 7

Which of the following statements is/are True regarding some advantages that an objectoriented DBMS (OODBMS) offers over a relational database ?

- I. An OODBMS avoids the “impedance mismatch” problem.
- II. An OODBMS avoids the “phantom” problem.
- III. An OODBMS provides higher performance concurrency control than most relational databases.
- IV. An OODBMS provides faster access to individual data objects once they have been read from disk.

- A. *II* and *III* only B. *I* and *IV* only C. *I, II*, and *III* only D. *I, III* and *IV* only

ugcnetcse-aug2016-paper3 databases

Answer key

36.0.53 UGC NET CSE | January 2017 | Part 3 | Question: 72

A neuron with inputs has the weight vector $[0.2 \quad -0.1 \quad 0.1]^T$ and a bias $\theta = 0$. If the input vector is $X = [0.2 \quad 0.4 \quad 0.2]^T$. Then the total input to the neuron is:

- A. 0.20 B. 1.0 C. 0.02 D. -1.0

ugcnetcse-jan2017-paper3

Answer key

36.0.54 UGC NET CSE | June 2009 | Part 2 | Question: 11

Assertion(*A*): Bit maps are not often used in memory management .

Reason(*R*): Searching a bitmap for a run of given length is a slow operation.

- A. Both (*A*) and (*R*) are true and (*R*) is correct explanation for (*A*)
- B. Both (*A*) and (*R*) are true but (*R*) is not correct explanation for (*A*)
- C. (*A*) is true (*R*) is false
- D. (*A*) is false (*R*) is true

ugcnetcse-june2009-paper2

36.0.55 UGC NET CSE | June 2009 | Part 2 | Question: 44

The spacing between character pairs is called :

- A. Kerning
- B. x-height
- C. intercap
- D. serif

ugcnetcse-june2009-paper2

Answer key

36.0.56 UGC NET CSE | June 2019 | Part 1 | Question: 48

The present form of Inter University Board that was previously established for promoting cooperation and coordination among Universities is

- A. UGC B. AIU C. NUEPA D. ICSSR

ugcnetcse-june2019-paper1

Answer key

36.0.57 UGC NET CSE | June 2019 | Part 1 | Question: 20

Today's media-society equation is largely

- A. Mystical B. Morally bound
C. Consumer conscious D. Tradition centric

ugcnetcse-june2019-paper1

Answer key

36.0.58 UGC NET CSE | January 2017 | Part 2 | Question: 50

Which of the following statements is/are true w.r.t. Enterprise Resource Planning (ERP)?

- I. ERP automates and integrates majority of business processes
II. ERP provides access to information in a Real Time Environment
III. ERP is inexpensive to implement

- A. I, II and III are false. B. I and II false III true.
C. I and II true III false. D. I true II and III are false.

ugcnetjan2017ii

Answer key

36.0.59 UGC NET CSE | January 2017 | Part 2 | Question: 46

Match the following :

List-I

- a. Affiliate Marketing
b. Viral Marketing
c. Group Purchasing
d. Bartering Online

List-II

- i. Vendors ask partners to place logos on partner's site. If customers click, come to vendors and buy.
ii. Spread your brand on the net by word-of-mouth. Receivers will send your information to their friends.
iii. Aggregating the demands of small buyers to get a large volume. Then negotiate a price.
iv. Exchanging surplus products and services with the process administered completely online by an intermediary. Company receives "points" for its contribution.

Codes :

- A. a-i, b-ii, c-iii, d-iv B. a-i, b-iii, c-ii, d-iv
C. a-iii, b-ii, c-iv, d-i D. a-ii, b-iii, c-i, d-iv

ugcnetjan2017ii

Answer key

36.0.60 UGC NET CSE | January 2017 | Part 3 | Question: 69

At which of the following stage(s), the degeneracy do not occur in transportation problem?

(m, n represents number of sources and destinations respectively)

- i. While the values of dual variables u_i and v_j cannot be computed
- ii. While obtaining an initial solution, we may have less than $m + n - 1$ allocations
- iii. At any stage while moving towards optimal solution, when two or more occupied cells with the same minimum allocation become unoccupied simultaneously.
- iv. At a stage when the no. of +ve allocation is exactly $m + n - 1$.

A. (i), (ii) and (iii)

B. (i),(iii) and (iv)

C. (i) and (iv)

D. (i),(ii),(iii) and (iv)

ugcnetcse-jan2017-paper3

Answer key

36.0.61 UGC NET CSE | January 2017 | Part 3 | Question: 68

With respect to a loop in the transportation table, which one of following is not correct?

- A. Every loop has an odd no.of cells and atleast 5.
- B. Closed loops may or may not be square in shape.
- C. All the cells in the loop that have a plus or minus sign, except the starting cell, must be occupied cells.
- D. Every loop has an even no.of cells and atleast four.

ugcnetcse-jan2017-paper3

Answer key

36.0.62 UGC NET CSE | January 2017 | Part 3 | Question: 67

From the given data below:

$a b b a a b b a a b$

which one of the following is not a word in the dictionary created by LZ-coding (the initial words are a,b)?

A. ab

B. bb

C. ba

D. $baab$

ugcnetcse-jan2017-paper3

Answer key

36.0.63 UGC NET CSE | January 2017 | Part 3 | Question: 65

A t-error correcting q-nary linear code satisfy :

$$M \sum_{i=0}^t \binom{n}{i} (q-1)^i \leq X$$

Where M is the number of code words and X is

A. q^n

B. q^t

C. q^{-n}

D. q^{-t}

ugcnetcse-jan2017-paper3 binary-codes

Answer key

36.0.64 UGC NET CSE | December 2004 | Part 2 | Question: 9

Essential hazards may occur in :

- A. Combinational logic circuits
- B. Synchronous sequential logic circuits
- C. Asynchronous sequential logic circuits working in the fundamental mode
- D. Asynchronous sequential logic circuits working in the pulse mode

Answer key

36.0.65 UGC NET CSE | December 2004 | Part 2 | Question: 8



An example of a connective which is not associative is :

- A. AND B. OR C. EX-OR D. NAND

Answer key

36.0.66 UGC NET CSE | December 2004 | Part 2 | Question: 7



When a tri-state logic device is in the third state, then :

- A. it draws low current B. it does not draw any current
C. it draws very high current D. it presents a low impedance

Answer key

36.0.67 UGC NET CSE | December 2004 | Part 2 | Question: 6



An example of a binary number which is equal to its 2's complement is :

- A. 1100 B. 1001 C. 1000 D. 1111

Answer key

36.0.68 UGC NET CSE | December 2004 | Part 2 | Question: 5



If I_m denotes the set of integers modulo m , then the following are fields with respect to the operations of addition modulo m and multiplication modulo m :

- i. Z_{23}
ii. Z_{29}
iii. Z_{31}
iv. Z_{33}

- A. (i) only B. (i) and (ii) only
C. (i), (ii) and (iii) only D. (i), (ii), (iii) and (iv)

36.0.69 UGC NET CSE | June 2005 | Part 2 | Question: 50



Another name of IEEE 802.11a is :

- A. WECA B. Fast Ethernet C. Wi-Fi 5 D. 802 – 11g

36.0.70 UGC NET CSE | June 2005 | Part 2 | Question: 49



Theoretically, how many 1.25 MHz Carriers are there in a CDMA cell ?

- A. 18 B. 22 C. 9 D. 64

36.0.71 UGC NET CSE | June 2005 | Part 2 | Question: 48



Which interim standard describes inter-switching networking ?

- A. *IS* – 54 B. *IS* – 95 C. *DS* – 45 D. *ANSI* – 41

ugcnetcse-june2005-paper2

Answer key

36.0.72 UGC NET CSE | June 2005 | Part 2 | Question: 47



The frequency reuse plan is divided into cell grouping using how many cells, where the number of cells equals *N* ?

- A. 3 B. 10 C. 7 D. 21

ugcnetcse-june2005-paper2

Answer key

36.0.73 UGC NET CSE | June 2005 | Part 2 | Question: 46



PVM and *MPI* Library functions are implemented on the top of the :

- A. Universal Message Passing (*UMS*) B. Network Interface Driver (*NID*)
C. Media Access Control (*MAC*) D. None of these

ugcnetcse-june2005-paper2

36.0.74 UGC NET CSE | June 2005 | Part 2 | Question: 45



Which type of coupling is least preferred ?

- A. Content coupling B. Data coupling
C. Control coupling D. Common coupling

ugcnetcse-june2005-paper2

Answer key

36.0.75 UGC NET CSE | June 2005 | Part 2 | Question: 44



Which one is measure of software complexity ?

- A. Number of lines of code (*LOC*) B. Number of man years
C. Number of function points (*FP*) D. All of the above

ugcnetcse-june2005-paper2

Answer key

36.0.76 UGC NET CSE | June 2005 | Part 2 | Question: 43



System Development Life-cycle has following stages :

- I. Requirement analysis
II. Coding
III. Design
IV. Testing

Which option describes the correct sequence of stages ?

- A. III, I, IV, II B. II, III, I, IV
C. I, III, IV, II D. None of the above

ugcnetcse-june2005-paper2

Answer key

36.0.77 UGC NET CSE | June 2005 | Part 2 | Question: 41

The capability maturity model (*err*) defines 5 levels :

- | | |
|-------------|------------------|
| (a) Level 1 | (i) Managed |
| (b) Level 2 | (ii) Defined |
| (c) Level 3 | (iii) Repeatable |
| (d) Level 4 | (iv) Initial |
| (e) Level 5 | (v) Optimized |

Correct matching is :

	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>
<i>A</i>	(<i>i</i>)	(<i>ii</i>)	(<i>iii</i>)	(<i>iv</i>)	(<i>v</i>)
<i>B</i>	(<i>iv</i>)	(<i>iii</i>)	(<i>ii</i>)	(<i>i</i>)	(<i>v</i>)
<i>C</i>	(<i>v</i>)	(<i>i</i>)	(<i>iii</i>)	(<i>ii</i>)	(<i>iv</i>)
<i>D</i>	(<i>v</i>)	(<i>ii</i>)	(<i>i</i>)	(<i>iii</i>)	(<i>iv</i>)

ugcnetcse-june2005-paper2

Answer key

36.0.78 UGC NET CSE | June 2005 | Part 2 | Question: 40

Which of the following is not an Unix Command?

- A. Whoami B. wc C. Is D. put

ugcnetcse-june2005-paper2

Answer key

36.0.79 UGC NET CSE | June 2005 | Part 2 | Question: 37

The principle of Locality of reference justifies the use of :

- | | |
|-------------------|---------------------|
| A. Virtual memory | B. Interrupts |
| C. Cache memory | D. Secondary memory |

ugcnetcse-june2005-paper2

Answer key

36.0.80 UGC NET CSE | June 2005 | Part 2 | Question: 34

The dynamic binding occurs during the

- | | |
|-----------------|------------------------|
| A. Compile time | B. Run time |
| C. Linking time | D. Pre-processing time |

ugcnetcse-june2005-paper2

Answer key

36.0.81 UGC NET CSE | June 2005 | Part 2 | Question: 29

Which protocol is used to encapsulate a data packet created of a higher *OSI* model layer ?

- A. *HDLC* B. *SDLC* C. *LAPB* D. *LAPD*

ugcnetcse-june2005-paper2

Answer key

36.0.82 UGC NET CSE | June 2005 | Part 2 | Question: 28

Which type of links are used for a connection between two *DTE* devices ?

- A. *X.21* B. Frame Relay C. *ATM* D. Modem

ugcnetcse-june2005-paper2

Answer key

36.0.83 UGC NET CSE | June 2005 | Part 2 | Question: 27

Which type of Bridge would be used to connect an Ethernet Segment with a token ring Segment ?

- A. Transparent Bridge B. Source-Route Bridge
C. Translation Bridge D. None of these

ugcnetcse-june2005-paper2

Answer key

36.0.84 UGC NET CSE | June 2005 | Part 2 | Question: 26

Which of the following are Data link layer standard ?

1. Ethernet
2. *HSSI*
3. Frame Relay
4. 10-base *T*
5. Token ring

- A. 1,2 B. 1,3,5 C. 1,3,4,5 D. 1,2,3,4,5

ugcnetcse-june2005-paper2

Answer key

36.0.85 UGC NET CSE | June 2005 | Part 2 | Question: 25

The algorithm that will efficiently sort an array that is nearly sorted except for the interchange of some adjacent pairs of numbers like {1, 3, 2, 5, 4, 6} is :

- A. Quick sort B. Bubble sort C. Merge sort D. Selection sort

ugcnetcse-june2005-paper2

Answer key

36.0.86 UGC NET CSE | June 2005 | Part 2 | Question: 24

If the postfix form of a string is $ABC+ - D^*$, the actual string is :

- A. $(A - (B + C)) * D$ B. $((A - B) + C) * D$
C. $((A + B) - C) * D$ D. $(A + (B - C)) * D$

ugcnetcse-june2005-paper2

Answer key

36.0.87 UGC NET CSE | June 2005 | Part 2 | Question: 22

Which of the following statement is false ?

- A. Every tree is a bipertite graph B. A tree contains a cycle
C. A tree with n nodes contains $n - 1$ edges D. A tree is a connected graph

ugcnetcse-june2005-paper2

Answer key

36.0.88 UGC NET CSE | June 2005 | Part 2 | Question: 20

A *WINDOW* into a portion of a data base is :

- A. Schema B. View C. Query D. Data Dictionary

ugcnetcse-june2005-paper2

Answer key

36.0.89 UGC NET CSE | June 2005 | Part 2 | Question: 18

A black hole in a *DFD* is a :

- A. A data store with no in bound flows B. A data store with only in bound flows
C. A data store with more than one in bound flow D. None of these

ugcnetcse-june2005-paper2

Answer key

36.0.90 UGC NET CSE | June 2005 | Part 2 | Question: 17

Which of the following tools is not required during system analysis phase of system development Life cycle ?

- A. CASE Tool B. RAD Tool
C. Reverse engineering tool D. None of these

ugcnetcse-june2005-paper2

Answer key

36.0.91 UGC NET CSE | June 2005 | Part 2 | Question: 16

An Entity-relationship diagram is a tool to represent :

- A. Data model B. Process model C. Event model D. Customer model

ugcnetcse-june2005-paper2

Answer key

36.0.92 UGC NET CSE | June 2005 | Part 2 | Question: 14

If a data-item is declared as a protected access specifier then it can be accessed :

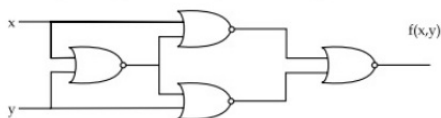
- A. Anywhere in the program B. By the base and derived classes
C. Only by base class D. Only by derived class

ugcnetcse-june2005-paper2

Answer key

36.0.93 UGC NET CSE | June 2005 | Part 2 | Question: 10

Identify the logic function performed by the circuit shown



- A. Exclusive-OR B. AND C. Exclusive-NOR D. NOR

ugcnetcse-june2005-paper2

Answer key

36.0.94 UGC NET CSE | June 2005 | Part 2 | Question: 8

Which of the following logic has the maximum fan out ?

A. *RTL*

B. *ECL*

C. *NMOS*

D. *CMOS*

ugcnetcse-june2005-paper2

Answer key 

36.0.95 UGC NET CSE | June 2005 | Part 2 | Question: 6



$(101011)_2 = (53)_b$, then ' b ' is equal to :

A. 4

B. 8

C. 10

D. 16

ugcnetcse-june2005-paper2

Answer key 

36.0.96 UGC NET CSE | June 2005 | Part 2 | Question: 5



Minimum number of individual shoes to be picked up from a dark room (containing 10 pair of shoes) if we have to get atleast one proper pair :

A. 2

B. 20

C. 11

D. None of these

ugcnetcse-june2005-paper2

Answer key 

36.0.97 UGC NET CSE | June 2005 | Part 2 | Question: 4



Consider the relation on the set of non-negative integers defined by $x \equiv y$ if and only if:

A. $x \bmod 3 = 3 \bmod y$

B. $3 \bmod x \equiv 3 \bmod y$

C. $x \bmod 3 = y \bmod 3$

D. None of the above

ugcnetcse-june2005-paper2

Answer key 

36.0.98 UGC NET CSE | June 2005 | Part 2 | Question: 2



Identify the language which is not context – free.

A. $L = \{wwR \mid w \in \{0,1\}^*\}$

B. $L = \{a^n b^n \mid n \geq 0\}$

C. $L = \{ww \mid w \in \{0,1\}^*\}$

D. $L = \{a^n b^m c^m d^n \mid n, m \geq 0\}$

ugcnetcse-june2005-paper2

Answer key 

36.0.99 UGC NET CSE | June 2005 | Part 2 | Question: 1



Which of the following is not true ?

A. Power of deterministic automata is equivalent to power of non-deterministic automata

B. Power of deterministic pushdown automata is equivalent to power of non-deterministic pushdown automata

C. Power of deterministic turing machine is equivalent to power of non-deterministic turing machine

D. All the above

ugcnetcse-june2005-paper2

Answer key 

36.0.100 UGC NET CSE | December 2005 | Part 2 | Question: 30



Encryption and decryption are the functions of the _____ layer of OSI model :

A. transport

B. session

C. router

D. presentation

ugcnetcse-dec2005-paper2

Answer key

36.0.101 UGC NET CSE | December 2005 | Part 2 | Question: 29



_____ provide a method to recover data that has been delivered but not get used :

- A. Segmentation
- B. Concatenation
- C. Transalation
- D. Synchronization

ugcnetcse-dec2005-paper2

Answer key

36.0.102 UGC NET CSE | December 2005 | Part 2 | Question: 28



Which of following is an example of a client - server model :

- A. DNS
- B. FTP
- C. TELNET
- D. All the above

ugcnetcse-dec2005-paper2

36.0.103 UGC NET CSE | December 2005 | Part 2 | Question: 27



Which of following file retrieval methods use hypermedia ?

- A. HTML
- B. Veronica
- C. WAIS
- D. HTTP

ugcnetcse-dec2005-paper2

Answer key

36.0.104 UGC NET CSE | December 2005 | Part 2 | Question: 26



The data unit in the TCP/IP application Layer is called a _____ .

- A. message
- B. segment
- C. datagram
- D. frame

ugcnetcse-dec2005-paper2

Answer key

36.0.105 UGC NET CSE | December 2005 | Part 2 | Question: 21



In what tree, for every node the height of its left subtree and right subtree differ at least by one:

- A. Binary search tree
- B. AVL-tree
- C. Threaded binary tree
- D. Complete tree

ugcnetcse-dec2005-paper2

36.0.106 UGC NET CSE | December 2005 | Part 2 | Question: 20



A locked file can be :

- A. accessed by only one user
- B. modified by users with the correct password
- C. is used to hide sensitive information
- D. both *B* and *C*

ugcnetcse-dec2005-paper2

36.0.107 UGC NET CSE | December 2005 | Part 2 | Question: 19



An embedded printer provides :

- A. Physical record key
- B. An inserted Index
- C. A secondary access path
- D. All the above

ugcnetcse-dec2005-paper2

36.0.108 UGC NET CSE | December 2005 | Part 2 | Question: 17



One approach to standarolizing storing of data :

- A. MIS
C. Structured Programming

- B. CODASYL
D. None of the above

ugcnetcse-dec2005-paper2

Answer key 

36.0.109 UGC NET CSE | December 2005 | Part 2 | Question: 16



A schema describes :

- A. data elements
C. record relationship
B. records and files
D. all of the above

ugcnetcse-dec2005-paper2

Answer key 

36.0.110 UGC NET CSE | December 2005 | Part 2 | Question: 14



Function over loading done at :

- A. Runtime
C. Linking time
B. Compile time
D. Switching from function to function

ugcnetcse-dec2005-paper2

Answer key 

36.0.111 UGC NET CSE | December 2005 | Part 2 | Question: 13



Data members and member function of a class by default is respectively :

- A. private and public
B. public
C. public and private
D. private

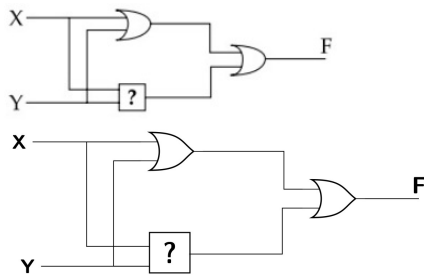
ugcnetcse-dec2005-paper2

Answer key 

36.0.112 UGC NET CSE | December 2005 | Part 2 | Question: 9



Consider the following circuit :



to make it a Tautology the ? should be :

- A. NAND gate
B. AND gate
C. OR gate
D. EX-OR gate

ugcnetcse-dec2005-paper2

Answer key 

36.0.113 UGC NET CSE | December 2005 | Part 2 | Question: 8



Consider the following sequence of instructions :

$a = a \oplus b, b = a \oplus b, a = b \oplus a$ This Sequence

- A. retains the value of the a and b
C. swap a and b
B. complements the value of a and b
D. negates values of a and b

ugcnetcse-dec2005-paper2

Answer key 

36.0.114 UGC NET CSE | December 2005 | Part 2 | Question: 3

Let A and B be two arbitrary events, then :

- A. $P(A \cap B) = P(A)P(B)$
- B. $P(P \cup B) = P(A) + P(B)$
- C. $P(A \cup B) \leq P(A) + P(B)$
- D. $P(A/B) = P(A \cap B) + P(B)$

ugcnetcse-dec2005-paper2

Answer key

36.0.115 UGC NET CSE | December 2005 | Part 2 | Question: 2

If the proposition $7P \Rightarrow Q$ is true, then the truth value of the proportion $7PV(P \Rightarrow Q)$ is :

- A. True
- B. Multi – Valued
- C. False
- D. Can not determined

ugcnetcse-dec2005-paper2

36.0.116 UGC NET CSE | December 2005 | Part 2 | Question: 1

T is a graph with n vertices. T is connected and has exactly $n - 1$ edges, then :

- A. T is a tree
- B. T contains no cycles
- C. Every pairs of vertices in T is connected by exactly one path
- D. All of these

ugcnetcse-dec2005-paper2

Answer key

36.0.117 UGC NET CSE | June 2006 | Part 2 | Question: 30

The program used to determine the round – trip delay between a workstation and a destination adress is :

- A. Tracert
- B. Traceroute
- C. Ping
- D. Pop

ugcnetcse-june2006-paper2

Answer key

36.0.118 UGC NET CSE | June 2006 | Part 2 | Question: 29

What is the function of a translating bridge?

- A. Connect similar remote LANs
- B. Connect similar local LANs
- C. Connect different types of LANs
- D. Translate the network addresses into a layer 2 address

ugcnetcse-june2006-paper2

Answer key

36.0.119 UGC NET CSE | June 2006 | Part 2 | Question: 28

What is the difference between the Ethernet frame preamble field and the IEEE 802.3 preamble and start of frame Delimiter fields?

- A. 1 byte
- B. 1 bit
- C. 4 bits
- D. 16 bits

ugcnetcse-june2006-paper2

Answer key

36.0.120 UGC NET CSE | June 2006 | Part 2 | Question: 27

How much bandwidth is required to send 132 voice- grade channels by FDM on an international satellite system?

- A. 500 MHz B. 10 MHz C. 1320 MHz D. 50 MHz

ugcnetcse-june2006-paper2

36.0.121 UGC NET CSE | June 2006 | Part 2 | Question: 26

What is the transmission signal coding method for a T_1 carrier called?

- A. Binary B. NRZ C. Bipolar D. Manchester

ugcnetcse-june2006-paper2

Answer key

36.0.122 UGC NET CSE | June 2006 | Part 2 | Question: 25

Application of data structure queue is :

- A. Levelwise printing of tree B. Implementation of priority queues
C. Function call implementation D. Depth first search in a graph

ugcnetcse-june2006-paper2

Answer key

36.0.123 UGC NET CSE | June 2006 | Part 2 | Question: 24

If the post fix form of a string is $ABC + -D^*$, The actual string is:

- A. $(A - (B + C)) * D$ B. $((A - B) + C) * D$ C. $((A + B) - C) * D$ D. $(A + (B - C)) * D$

ugcnetcse-june2006-paper2

Answer key

36.0.124 UGC NET CSE | June 2006 | Part 2 | Question: 23

Which of the following statement is false?

- A. Every tree is a bipartite graph B. A tree contains a cycle
C. A tree with n nodes contains $(n - 1)$ edges D. A tree is connected graph

ugcnetcse-june2006-paper2

Answer key

36.0.125 UGC NET CSE | June 2006 | Part 2 | Question: 22

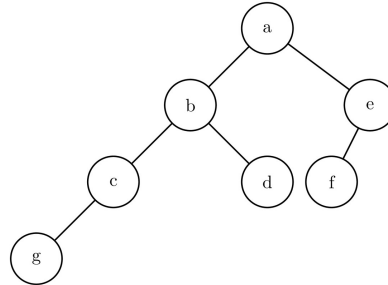
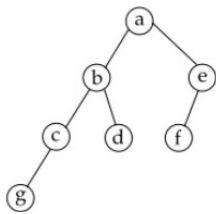
Preorder is also known as:

- A. Depth first order B. Breadth first order
C. Topological order D. Linear order

ugcnetcse-june2006-paper2

36.0.126 UGC NET CSE | June 2006 | Part 2 | Question: 21

In the balanced binary tree given below, how many nodes will become unbalanced when a node is inserted as a child of the node "g".



- A. 1 B. 3 C. 7 D. 8

ugcnetcse-june2006-paper2

36.0.127 UGC NET CSE | June 2006 | Part 2 | Question: 20



A subclass having more than one super class is called_____.

- A. Category B. Classification
C. Combination D. Partial participation

ugcnetcse-june2006-paper2

36.0.128 UGC NET CSE | June 2006 | Part 2 | Question: 19



Which data management language component enabled the DBA to define the schema components?

- A. DML B. Subschema DLL C. Schema DLL D. All of these

ugcnetcse-june2006-paper2

36.0.129 UGC NET CSE | June 2006 | Part 2 | Question: 18



Which of the following statement is wrong?

- A. 2– phase locking protocol suffers from dead locks
B. Time-Stamp protocol suffers from more abort
C. Time-Stamp protocol suffers from cascading rollbacks whereas 2– phase locking protocol do not
D. None of these

ugcnetcse-june2006-paper2

Answer key

36.0.130 UGC NET CSE | June 2006 | Part 2 | Question: 17



Immediate updates as a recovery protocol is preferable, when:

- A. Database reads more than writes B. Writes are more than reads
C. It does not matter as it is good in both the situations D. There are only writes

ugcnetcse-june2006-paper2

36.0.131 UGC NET CSE | June 2006 | Part 2 | Question: 16



A relation $R = \{A, B, C, D, E, F\}$ is given with following set of functional dependencies:

$F = \{A \rightarrow B, AD \rightarrow C, B \rightarrow F, A \rightarrow E\}$

Which of the following is candidate key?

- A. A B. AC C. AD D. None of these

ugcnetcse-june2006-paper2

Answer key 

36.0.132 UGC NET CSE | June 2006 | Part 2 | Question: 15



The function *sprintf()* works like *printf()*, but operates on :

- A. Data in a file B. `stderr` C. `stdin` D. string

ugcnetcse-june2006-paper2

Answer key 

36.0.133 UGC NET CSE | June 2006 | Part 2 | Question: 14



define $\max(x, y)x = (x > y)?x : y$ is a macro definition, which can find the maximum of two numbers x and y if:

- A. x and y are both integers only B. x and y are both declared as float only
C. x and y are both declared as double only D. x and y are both integers, float or double

ugcnetcse-june2006-paper2

36.0.134 UGC NET CSE | June 2006 | Part 2 | Question: 13



If the following loop is implemented

```
{
    int num =0;

    do {-- num; printf ("%d", num); num++;}

    while (num>=0)
}
```

- A. the loop will run infinitely many times
B. the program will not enter the loop
C. there will be compilation error reported
D. a run time error will be reported

ugcnetcse-june2006-paper2

36.0.135 UGC NET CSE | June 2006 | Part 2 | Question: 12



A static variable is one :

- A. Which cannot be initialized
B. Which is initialized once at the commencement of execution and cannot be changed at runtime
C. Which retains its value throughout the life of the program
D. Which is the same as an automatic variable but is placed at the head of a program

ugcnetcse-june2006-paper2

36.0.136 UGC NET CSE | June 2006 | Part 2 | Question: 11



What is the output of the following program segment?

```
main ()
{
    int count, digit=0;
    count=1;
```



```
while (digit<=9)
{
printf ("%d/n", ++count); ++digit;
}
}
```

- A. 10 B. 9 C. 12 D. 11

ugcnetcse-june2006-paper2

36.0.137 UGC NET CSE | June 2006 | Part 2 | Question: 10



Upto how many variables, can the Karnaugh map be used?

- A. 3 B. 4 C. 5 D. 6

ugcnetcse-june2006-paper2

Answer key

36.0.138 UGC NET CSE | June 2006 | Part 2 | Question: 9



In a weighted code with weight 6, 4, 2, -3 the decimal 5 is represented by:

- A. 0101 B. 0111 C. 1011 D. 1000

ugcnetcse-june2006-paper2

Answer key

36.0.139 UGC NET CSE | June 2006 | Part 2 | Question: 8



Which of the following logic has the maximum fan out?

- A. RTL B. ECL C. NMOS D. CMOS

ugcnetcse-june2006-paper2

36.0.140 UGC NET CSE | June 2006 | Part 2 | Question: 7



Which of the following expression removes static hazard from a two level AND-OR gate implementation of $xy + z\bar{x}$

- A. $xy + z\bar{x}$ B. $xy + z\bar{x} + wyz$ C. $xy + z\bar{x} + yz$ D. $xy + z\bar{x} + wz$

ugcnetcse-june2006-paper2

Answer key

36.0.141 UGC NET CSE | June 2006 | Part 2 | Question: 6



The number of 1's present in the binary representation of $(3 \times 512 + 7 \times 64 + 5 \times 8 + 3)_{10}$ is:

- A. 8 B. 9 C. 10 D. 11

ugcnetcse-june2006-paper2

36.0.142 UGC NET CSE | June 2006 | Part 2 | Question: 5



Let $A = \{x | -1 < x < 1\} = B$. The function $f(x) = \frac{x}{2}$ from A to B is:

- A. injective B. surjective
C. both injective and surjective D. neither injective nor surjective

ugcnetcse-june2006-paper2

Answer key

ugcnetcse-june2006-paper2

ugcnetcse-june2006-paper2



ugcnetcse-june2006-paper2



ugcnetcse-dec2005-paper2



uqcnetcse-dec2005-paper2

ugcnetcse-dec2005-paper2



uqcnetcse-dec2005-paper2

Answer key

36.0.150 UGC NET CSE | December 2005 | Part 2 | Question: 46



Which of the following transmission media is used in Blue tooth Technology :

- A. Radio links
- B. Microwave links
- C. VSAT Communication
- D. Fiber - optic

ugcnetcse-dec2005-paper2

36.0.151 UGC NET CSE | December 2005 | Part 2 | Question: 45



Sliding window concept of software project management is :

- A. Preparation of comprehensible plan
- B. Preparation of the various stages of development
- C. Ad-hoc planning
- D. Requirement analysis

ugcnetcse-dec2005-paper2

Answer key

36.0.152 UGC NET CSE | December 2005 | Part 2 | Question: 44



Successive layer of design in software using bottom-up design is called :

- A. Layer of Definition
- B. Layer of Construction
- C. Layer of abstraction
- D. None of the above

ugcnetcse-dec2005-paper2

Answer key

36.0.153 UGC NET CSE | December 2005 | Part 2 | Question: 43



The Reliability of the software is directly dependent upon :

- A. Quality of the design
- B. Programmer's experience
- C. Number of error
- D. Set of user requirements

ugcnetcse-dec2005-paper2

Answer key

36.0.154 UGC NET CSE | December 2005 | Part 2 | Question: 42



The lower degree of cohesion is :

- A. logical cohesion
- B. coincidental cohesion
- C. procedural cohesion
- D. communicational cohesion

ugcnetcse-dec2005-paper2

Answer key

36.0.155 UGC NET CSE | December 2005 | Part 2 | Question: 41



The testing of software against SRS is called :

- A. Acceptance testing
- B. Integration testing
- C. Regression testing
- D. Series testing

ugcnetcse-dec2005-paper2

Answer key

36.0.156 UGC NET CSE | December 2005 | Part 2 | Question: 39



Nonmodifiable procedures are called :

- A. Serially usable procedure
- B. Concurrent procedure
- C. Reentrant procedure
- D. Topdown procedure

ugcnetcse-dec2005-paper2

Answer key

36.0.157 UGC NET CSE | December 2005 | Part 2 | Question: 38



Which of the following checks cannot be carried out on the input data to a system?

- A. Consistency check
- B. Syntax check
- C. Range check
- D. All the above

ugcnetcse-dec2005-paper2

Answer key

36.0.158 UGC NET CSE | December 2005 | Part 2 | Question: 37



If you want to execute more than one program at a time, the systems software that are used must be capable of :

- A. word processing
- B. virtual memory
- C. compiling
- D. multitasking

ugcnetcse-dec2005-paper2

Answer key

36.0.159 UGC NET CSE | December 2005 | Part 2 | Question: 36



Producer consumer problem can be solved using :

- A. semaphores
- B. event counters
- C. monitors
- D. all the above

ugcnetcse-dec2005-paper2

Answer key

36.0.160 UGC NET CSE | December 2005 | Part 2 | Question: 33



A general macroprocessor is an in built function of :

- A. Loader
- B. Linker
- C. Editor
- D. Assembler

ugcnetcse-dec2005-paper2

Answer key

36.0.161 UGC NET CSE | December 2005 | Part 2 | Question: 31



The Register or main memory location which contains the effective address of the operand is known as:

- A. Pointer
- B. Indexed register
- C. Special Locations
- D. Scratch Pad

ugcnetcse-dec2005-paper2

Answer key

36.0.162 UGC NET CSE | December 2004 | Part 2 | Question: 48



The cost of a parallel processing is primarily determined by :

- A. Time complexity
- B. Switching complexity
- C. Circuit complexity
- D. None of the above

ugcnetcse-dec2004-paper2

Answer key

36.0.163 UGC NET CSE | December 2004 | Part 2 | Question: 44



In transform analysis, input portion is called :

- A. Afferent branch
- B. Efferent branch
- C. Central Transform
- D. None of the above

ugcnetcse-dec2004-paper2

Answer key

36.0.164 UGC NET CSE | December 2004 | Part 2 | Question: 39



Non modifiable procedures are called

- A. Serially useable procedures
- B. Concurrent procedures
- C. Reentrant procedures
- D. Topdown procedures

ugcnetcse-dec2004-paper2

Answer key

36.0.165 UGC NET CSE | June 2006 | Part 2 | Question: 50



The use of a smart card represents a form of:

- A. password encryption
- B. user-ID encryption
- C. authorization
- D. authentication

ugcnetcse-june2006-paper2

Answer key

36.0.166 UGC NET CSE | June 2006 | Part 2 | Question: 49



Predictive modeling in data mining uses ____ technique.

- A. Statistical
- B. Association Discovery
- C. Value prediction
- D. None of these

ugcnetcse-june2006-paper2

Answer key

36.0.167 UGC NET CSE | June 2006 | Part 2 | Question: 48



Downflow is the process associated with _____ and back up of data in a warehouse.

- A. packaging
- B. archiving
- C. extraction
- D. loading

ugcnetcse-june2006-paper2

Answer key

36.0.168 UGC NET CSE | June 2006 | Part 2 | Question: 47



Bluetooth technology uses the transmission media.

- A. Radio links
- B. Microwave links
- C. VSAT communication
- D. Optical fiber links

ugcnetcse-june2006-paper2

Answer key

36.0.169 UGC NET CSE | June 2006 | Part 2 | Question: 46



MMS (Multimedia Messaging Service) uses _____ types of messages to perform messaging transactions.

- A. 4
- B. 6
- C. 8
- D. 10

ugcnetcse-june2006-paper2

36.0.170 UGC NET CSE | June 2006 | Part 2 | Question: 45



The COCOMO model is used for_____.

- A. software design
- B. software cost estimation
- C. software cost approximation
- D. software analysis

ugcnetcse-june2006-paper2

Answer key

36.0.171 UGC NET CSE | June 2006 | Part 2 | Question: 44



System study yields the following:

- A. Requirement specifications
- B. Prevailing process description
- C. Data source identification
- D. All the above

ugcnetcse-june2006-paper2

36.0.172 UGC NET CSE | June 2006 | Part 2 | Question: 43



In a good software design, ____ coupling is desirable between modules.

- A. Highest
- B. Lowest
- C. Internal
- D. External

ugcnetcse-june2006-paper2

Answer key

36.0.173 UGC NET CSE | June 2006 | Part 2 | Question: 42



In Software Metrics, McCABE'S cyclomatic number is given by following formula:

- A. $c = e - n + 2p$
- B. $c = e - n - 2p$
- C. $c = e + n + 2p$
- D. $c = e - n * 2p$

ugcnetcse-june2006-paper2

36.0.174 UGC NET CSE | June 2006 | Part 2 | Question: 41



In software project planning, work Breakdown structure must be ____.

- A. A graph
- B. A tree
- C. A Euler's graph
- D. None of the above

ugcnetcse-june2006-paper2

36.0.175 UGC NET CSE | June 2006 | Part 2 | Question: 40



The memory allocation scheme subjected to "external" fragmentation is :

- A. Segmentation
- B. Swapping
- C. Demand paging
- D. Multiple contiguous fixed partitions

ugcnetcse-june2006-paper2

Answer key

36.0.176 UGC NET CSE | June 2006 | Part 2 | Question: 39



_____ synchronize critical resources to prevent deadlock.

- A. *P*- operator
- B. *V*- operator
- C. Semaphores
- D. Hard disk

ugcnetcse-june2006-paper2

Answer key

36.0.177 UGC NET CSE | June 2006 | Part 2 | Question: 38



A page fault ____.

- A. is an error in specific page
- B. is an access to the page not currently in main memory
- C. occurs when a page program accesses a page of memory
- D. is reference to the page which belongs to another program

ugcnetcse-june2006-paper2

Answer key

36.0.178 UGC NET CSE | June 2006 | Part 2 | Question: 37



In processor management, round robin method essentially uses the preemptive version of_____.

- A. FILO B. FIFO C. SJF D. Longest time first

ugcnetcse-june2006-paper2

Answer key

36.0.179 UGC NET CSE | June 2006 | Part 2 | Question: 36



The first operating system of Microprocessor is_____.

- A. ATLAS B. CP/M C. SAGE D. T.H.E

ugcnetcse-june2006-paper2

36.0.180 UGC NET CSE | June 2006 | Part 2 | Question: 35



The dynamic binding occurs during the:

- A. Compile time B. Run time
C. Linking time D. Pre-processing time

ugcnetcse-june2006-paper2

Answer key

36.0.181 UGC NET CSE | June 2006 | Part 2 | Question: 34



The action of passing the source program into the proper syntactic class is known as:

- A. Syntax analysis B. Lexical analysis
C. Interpretation analysis D. Uniform symbol generation

ugcnetcse-june2006-paper2

Answer key

36.0.182 UGC NET CSE | June 2006 | Part 2 | Question: 33



YACC builds up _____parsing table.

- A. LALR B. LR C. SLR D. LLR

ugcnetcse-june2006-paper2

Answer key

36.0.183 UGC NET CSE | June 2006 | Part 2 | Question: 32



Tasks done in parsing are:

- A. Check the validity of a source string B. Determine the syntactic structure of a source string
C. Both A and B D. None of these

ugcnetcse-june2006-paper2

36.0.184 UGC NET CSE | June 2006 | Part 2 | Question: 31



Which statement is wrong

- A. If linked origin $\neq$ translated origin, relocation must be performed by the linker
B. If load $\neq$ linked origin, the loader must perform relocation
C. A linker always perform relocation, whereas some loaders do not
D. None of these

ugcnetcse-june2006-paper2

Answer key

36.0.185 UGC NET CSE | December 2006 | Part 2 | Question: 48



An INT file in Windows 95 is :

- A. a program file.
- B. a message file.
- C. a text file.
- D. link file.

ugcnetcse-dec2006-paper2

Answer key

36.0.186 UGC NET CSE | December 2006 | Part 2 | Question: 47



Which of the following is an EDI standard ?

- A. ANSI X.15
- B. ANSI X.14
- C. ANSI X.13
- D. ANSI X.12

ugcnetcse-dec2006-paper2

36.0.187 UGC NET CSE | December 2006 | Part 2 | Question: 46



The frequency band allocated for the downlink in GSM is :

- A. 960 – 985 MHz
- B. 935 – 960 MHz
- C. 920 – 945 MHz
- D. 930 – 955 MHz

ugcnetcse-dec2006-paper2

36.0.188 UGC NET CSE | December 2006 | Part 2 | Question: 14



When one-dimensional character array of unspecified length is assigned an initial value :

- A. an arbitrary character is automatically added to the end of the string
- B. 'o' is added to the end of the string
- C. length of the string is added to the end of the string
- D. 'end' is added to the end of the string

ugcnetcse-dec2006-paper2

36.0.189 UGC NET CSE | December 2006 | Part 2 | Question: 12



Enumeration variables can be used in :

- A. search statement like an integer variable
- B. break statement
- C. preprocessor commands
- D. function statement

ugcnetcse-dec2006-paper2

36.0.190 UGC NET CSE | December 2006 | Part 2 | Question: 10



Which of the following logic is the fastest ?

- A. RTL
- B. ECL
- C. HTL
- D. HCL

ugcnetcse-dec2006-paper2

36.0.191 UGC NET CSE | December 2006 | Part 2 | Question: 5



If $(a^2 - b^2)$ is a prime number where a and $b \in \mathbb{N}$, then :

- A. $a^2 - b^2 = 3$
- B. $a^2 - b^2 = a - b$

C. $a^2 - b^2 = a + b$

D. $a^2 - b^2 = 5$

ugcnetcse-dec2006-paper2

36.0.192 UGC NET CSE | December 2006 | Part 2 | Question: 4



Which of the following is not true ?

- A. B. $A - B = A \cap \sim B$ C. D.

ugcnetcse-dec2006-paper2

36.0.193 UGC NET CSE | June 2007 | Part 2 | Question: 50



Which digital radio technology employs an $N = 1$ frequency – reuse plan?

- A. GSM B. TDMA C. D AMPS D. CDMA

ugcnetcse-june2007-paper2

36.0.194 UGC NET CSE | June 2007 | Part 2 | Question: 49



The U-NII (Unlicensed National Information Infrastructure) band operates at the _____ frequency:

- A. 2.4 GHz B. 33 MHz C. 5 GHz D. 16 GHz

ugcnetcse-june2007-paper2

36.0.195 UGC NET CSE | June 2007 | Part 2 | Question: 48



Dimensional modeling in Data Mining refers to:

- A. view and interrogate data B. define structures and store data
C. retrieve information only D. none of these

ugcnetcse-june2007-paper2

36.0.196 UGC NET CSE | June 2007 | Part 2 | Question: 47



Wireless interconnection to the PSTN are also known as:

- A. Localities B. CLECs C. POPs D. IXCs

ugcnetcse-june2007-paper2

Answer key

36.0.197 UGC NET CSE | June 2007 | Part 2 | Question: 46



Which technical concept sets cellular apart from all preceding mobile/ratio systems?

- A. FM-Transmission B. Duplex Functionality
C. Frequency Reuse D. TDMA Technology

ugcnetcse-june2007-paper2

36.0.198 UGC NET CSE | June 2007 | Part 2 | Question: 45



Which one of the following is a object-oriented approaches:

- A. The Booch method B. The Rambaugh method
C. The Load and Yomdon method D. All of the above

ugcnetcse-june2007-paper2

36.0.199 UGC NET CSE | June 2007 | Part 2 | Question: 44



Problems with waterfall model are:

1. Real projects rarely follow this model proposes

2. It is often difficult for the customer
3. Working model is available only in the end
4. Developers are delayed unnecessarily

Which of the following is true:

- A. 1 and 4 only B. 2 and 3 only C. 1,2 and 3 only D. 1,2,3 and 4

ugcnetcse-june2007-paper2

Answer key 

36.0.200 UGC NET CSE | June 2007 | Part 2 | Question: 42



Difference between flow-chart and data-flow diagram is:

- A. there is no difference B. usage in high level design and low level design
C. control flow and data flow D. used in application programs and system programs

ugcnetcse-june2007-paper2

36.0.201 UGC NET CSE | June 2007 | Part 2 | Question: 41



Which of the following combination is preferred with respect to cohesion and coupling:

- A. low and low B. low and high C. high and low D. high and high

ugcnetcse-june2007-paper2

36.0.202 UGC NET CSE | June 2007 | Part 2 | Question: 40



Cache memory is:

- A. High-Speed Register B. Low-Speed RAM
C. Non-Volatile RAM D. High-speed RAM

ugcnetcse-june2007-paper2

Answer key 

36.0.203 UGC NET CSE | June 2007 | Part 2 | Question: 39



Which page replacement policy suffers from Belady's anomaly?

- A. LRV B. LFU C. FIFO D. OPTIMAL

ugcnetcse-june2007-paper2

Answer key 

36.0.204 UGC NET CSE | June 2007 | Part 2 | Question: 38



The "nice" command is used in Unix:

- A. to decrease the priority of a process B. to increase the priority of a process
C. to get the highest priority D. nothing to do with the priorities

ugcnetcse-june2007-paper2

Answer key 

36.0.205 UGC NET CSE | June 2007 | Part 2 | Question: 36



Part of a program where the shared memory is accessed and which should be executed indivisibly, is called:

- A. Semaphores B. Directory
C. Critical section D. Mutual exclusion

ugcnetcse-june2007-paper2

Answer key 

36.0.206 UGC NET CSE | June 2007 | Part 2 | Question: 35



Moving process from main memory to disk is called:

- A. Caching B. Termination C. Swapping D. Interruption

ugcnetcse-june2007-paper2

Answer key

36.0.207 UGC NET CSE | June 2007 | Part 2 | Question: 34



Symbol table can be used for:

- A. Checking type compatibility B. Suppressing duplication of error message
C. Storage allocation D. All of these above

ugcnetcse-june2007-paper2

Answer key

36.0.208 UGC NET CSE | June 2007 | Part 2 | Question: 33



In an absolute loading scheme, which loader function is accomplished by programmer?

- A. Allocation B. Linking C. Rellocation D. Both A and B

ugcnetcse-june2007-paper2

36.0.209 UGC NET CSE | June 2007 | Part 2 | Question: 32



A Top-down Parse generates:

- A. Right-most derivation B. Right-most derivation, in reverse
C. Left-most derivation D. Left-most derivation, in reverse

ugcnetcse-june2007-paper2

Answer key

36.0.210 UGC NET CSE | June 2007 | Part 2 | Question: 31



The parsing technique that avoids back tracking is:

- A. Top-down parsing B. Recursive-descent parsing
C. Predicative D. Syntax tree

ugcnetcse-june2007-paper2

Answer key

36.0.211 UGC NET CSE | June 2007 | Part 2 | Question: 30



Which of the following protocols is used to prevent looping?

- A. OSPF B. Spanning tree
C. SRB D. Fragment free switching

ugcnetcse-june2007-paper2

36.0.212 UGC NET CSE | June 2007 | Part 2 | Question: 29



What services does the Internet Layer provide?

1. Quality of service
2. Routing
3. Addressing
4. Connection-oriented delivery
5. Framming bits

A. 1,2,3

B. 2,3,4

C. 1,3,4,5

D. 2,3,4,5

ugcnetcse-june2007-paper2

Answer key 

36.0.213 UGC NET CSE | June 2007 | Part 2 | Question: 28



TDM is _____.

A. A primary /secondary type protocol

B. A peer/peer protocol

C. A Non-priority peer/peer protocol

D. A priority type protocol

ugcnetcse-june2007-paper2

36.0.214 UGC NET CSE | June 2007 | Part 2 | Question: 27



Which of the following techniques are used to control data flow?

1. Windowing
2. Routing
3. RPCs
4. Buffering

A. 1,4

B. 2,3,4

C. 1,3,4

D. 1,2,3,4

ugcnetcse-june2007-paper2

Answer key 

36.0.215 UGC NET CSE | June 2007 | Part 2 | Question: 26



In case of Bus/Tree topology signal balancing issue is overcome by:

A. Strong Transmitter

B. Polling

C. Segmentation

D. Modulation

ugcnetcse-june2007-paper2

36.0.216 UGC NET CSE | June 2007 | Part 2 | Question: 25



Application of data structure is queue is:

- A. Level wise printing of tree.
C. Function call implementation

- B. Implementation of priority queues.
D. Depth first search in a graph.

ugcnetcse-june2007-paper2

Answer key 

36.0.217 UGC NET CSE | June 2007 | Part 2 | Question: 24



Which algorithm has some average, worst case and best case time:

- A. Binary search
C. Quick sort

- B. Maximum of n numbers
D. Fibonacci search

ugcnetcse-june2007-paper2

36.0.218 UGC NET CSE | June 2007 | Part 2 | Question: 23



The equivalent postfix express for $d/(e+f)+b*c$ is :

A. $defbc/+$

B. $def+ /bc+*$

C. $def+ /bc*+$

D. None of these

ugcnetcse-june2007-paper2

36.0.219 UGC NET CSE | June 2007 | Part 2 | Question: 22

Pre order is also known as:

- A. Depth first order
- B. Breadth first order
- C. Topological order
- D. Linear order

ugcnetcse-june2007-paper2

36.0.220 UGC NET CSE | June 2007 | Part 2 | Question: 21

The time required to find the shortest path in a graph with n vertices and e edges is:

- A. $O(e)$
- B. $O(n)$
- C. $O(e^2)$
- D. $O(n^2)$

ugcnetcse-june2007-paper2

Answer key

36.0.221 UGC NET CSE | June 2007 | Part 2 | Question: 20

Which statement is false regarding data Independence:

- A. Hierarchical data model suffers from data Independence.
- B. Network model suffers from data Independence.
- C. Relational model suffers only from logical data Independence.
- D. Relational model suffers only from physical data Independence.

ugcnetcse-june2007-paper2

36.0.222 UGC NET CSE | June 2007 | Part 2 | Question: 19

A Relation $R = \{A, B, C, D, E, F\}$ is given with following set of functional dependencies:

$F = \{A \rightarrow B, AD \rightarrow C, B \rightarrow F, A \rightarrow E\}$. Which of the following is Candidate Key?

- A. A
- B. AC
- C. AD
- D. None of these

ugcnetcse-june2007-paper2

Answer key

36.0.223 UGC NET CSE | June 2007 | Part 2 | Question: 18

A subclass having more than one super class is called:

- A. Category
- B. Classification
- C. Combination
- D. Partial Participation

ugcnetcse-june2007-paper2

Answer key

36.0.224 UGC NET CSE | June 2007 | Part 2 | Question: 17

A recursive foreign key is a :

- A. references a relation
- B. references a table
- C. references its own relation
- D. references a foreign key

ugcnetcse-june2007-paper2

Answer key

36.0.225 UGC NET CSE | June 2007 | Part 2 | Question: 16

Which of the following statements is wrong?

- A. 2-phase Locking Protocols suffer from dead locks.
- B. Time-Stamp Protocols suffer from more aborts.

- C. Time-Stamp Protocols suffer from cascading roll back where as 2-phase locking Protocol donot.
D. None of these

ugcnetcse-june2007-paper2

Answer key 

36.0.226 UGC NET CSE | June 2007 | Part 2 | Question: 15



Runtime polymorphism can be achieved by:

- A. Accessing virtual function through the pointer of the base class
B. Accessing virtual function through the object
C. The derived class
D. None of these

ugcnetcse-june2007-paper2

Answer key 

36.0.227 UGC NET CSE | June 2007 | Part 2 | Question: 13



What keyword in class specification helps to hide data:

- A. Public B. Private C. Static D. Void

ugcnetcse-june2007-paper2

Answer key 

36.0.228 UGC NET CSE | June 2007 | Part 2 | Question: 12



In case of right shift bitwise operator in 'C' language, after shifting n bits, the leftmost n bits:

- A. are always filled with zeros B. are always filled with ones
C. are filled with zeros or ones and is machine dependent D. none of the above

ugcnetcse-june2007-paper2

Answer key 

36.0.229 UGC NET CSE | June 2007 | Part 2 | Question: 11



The following loop in 'C':

```
int i=0;
while (i++<0) i--;
```

- A. will terminate B. will go into an infinite loop
C. will give compilation error D. will never be executed

ugcnetcse-june2007-paper2

Answer key 

36.0.230 UGC NET CSE | June 2007 | Part 2 | Question: 10



What is the maximum counting speed of a 4-bit binary counter which is composed of Flip-Flop with a propagation delay of 25 ns ?

- A. 1MHz B. 10MHz C. 100MHz D. 4MHz

ugcnetcse-june2007-paper2

Answer key 

36.0.231 UGC NET CSE | June 2007 | Part 2 | Question: 9



If $A \oplus B = C$, then:

- A. $A \oplus C = B$
 C. $A \oplus B \oplus C = 1$
- B. $B \oplus C = A$
 D. $A \oplus B \oplus C = 0$

ugcnetcse-june2007-paper2

Answer key 

36.0.232 UGC NET CSE | June 2007 | Part 2 | Question: 8



How many 1's are present in the binary representation of $15 \times 256 + 5 \times 16 + 3$:

- A. 8 B. 9 C. 10 D. 11

ugcnetcse-june2007-paper2

Answer key 

36.0.233 UGC NET CSE | June 2007 | Part 2 | Question: 7



Which of the following expression remove hazard form : $xy + z\bar{x}$?

- A. $xy + z\bar{x}$ B. $xy + z\bar{x}$ C. $xy + z\bar{x} + yz$ D. $xy + z\bar{x} + wz$

ugcnetcse-june2007-paper2

36.0.234 UGC NET CSE | June 2007 | Part 2 | Question: 6



2's complement of -100 is:

- A. 00011100 B. 10011101 C. 10011100 D. 11100100

ugcnetcse-june2007-paper2

Answer key

36.0.235 UGC NET CSE | June 2007 | Part 2 | Question: 5



Which of the following language is regular:

- A. $L = \{a^n b^n | n \geq 1\}$
 B. $L = \{a^n b^m c^n d^m | n, m \geq 1\}$
 C. $L = \{a^n b^m | n, m \geq 1\}$
 D. $L = \{a^n b^m c^n | n, m \geq 1\}$

ugcnetcse-june2007-paper2

Answer key

36.0.236 UGC NET CSE | June 2007 | Part 2 | Question: 4



The regular expression given below describes:

$$r = (1 + 01)^*(0 + \lambda)$$

- A. Set of all string not containing '11'
 B. Set of all string not containing '00'
 C. Set of all string containing '01'
 D. Set of all string ending in '0'

ugcnetcse-june2007-paper2

Answer key 

36.0.237 UGC NET CSE | June 2007 | Part 2 | Question: 3

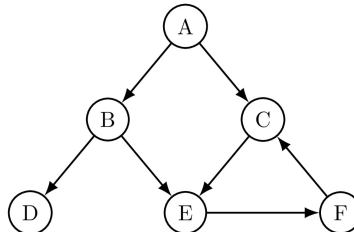
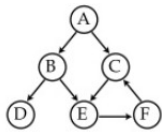


The maximum number of nodes in a binary tree of depth 10:

- A. 1024 B. $2^{10} - 1$ C. 1000 D. None of the above

36.0.238 UGC NET CSE | June 2007 | Part 2 | Question: 2

Depth ion travels of the following directed graph is:



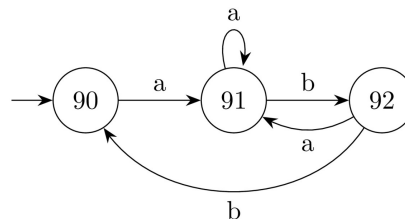
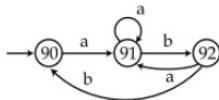
- A. ABCDEF
C. ACEBDF

- B. ABDEF C
D. None of the above

ugcnetcse-june2007-paper2

36.0.239 UGC NET CSE | June 2007 | Part 2 | Question: 1

The following determiniotic finite automata recognizes:



- A. Set of all strings containing 'ab'
C. Set of all strings ending in 'abab'

- B. Set of all strings containing 'aab'
D. None of the above

ugcnetcse-june2007-paper2

Answer key

36.0.240 UGC NET CSE | December 2007 | Part 2 | Question: 48

Consider a typical mobile communication system. If 840 frequencies are available, how many can be used in a given cell ?

- A. 140 B. 120 C. 84 D. 60

ugcnetcse-dec2007-paper2

36.0.241 UGC NET CSE | December 2007 | Part 2 | Question: 47

With reference to implementation of different association mining algorithms, identify the correct statement :

- A. The FP growth method was usually better than the best implementation of the apriori algorithm
B. Apriori algorithm is usually better than CHARM
C. Apriori algorithm is good when the support required is low
D. At very low support the number of frequent items becomes less

36.0.242 UGC NET CSE | December 2007 | Part 2 | Question: 46

Given a parallel algorithm A with computation time t , if parallel algorithm A performs m computational operation, then p processors can execute algorithm A in time given by :

- A. t/p B. mt/p C. $t + (m-t)/p$ D. $(m-t)/p$

ugcnetcse-dec2007-paper2

36.0.243 UGC NET CSE | December 2007 | Part 2 | Question: 43

An error message produced by an interactive system should have :

- A. always the error code B. the list of mistakes done by the user displayed
C. a non-judgmental approach D. the past records of the occurrence of the same mistake

ugcnetcse-dec2007-paper2

36.0.244 UGC NET CSE | December 2007 | Part 2 | Question: 39

An example of a directory found in most UNIX system is :

- A. `usr` B. `waitpid` C. `brk` D. `unmap`

ugcnetcse-dec2007-paper2

Answer key

36.0.245 UGC NET CSE | December 2007 | Part 2 | Question: 24

An example of a file extension is :

- A. `text` B. `pict` C. `mp3` D. `web`

ugcnetcse-dec2007-paper2

36.0.246 UGC NET CSE | December 2007 | Part 2 | Question: 18

If a relation is in 2NF and 3NF forms then :

- A. no non-prime attribute is functionally dependent on other non-prime attributes
B. no non-prime attribute is functionally dependent on prime attributes
C. all attributes are functionally independent
D. prime attribute is functionally independent of all non-prime attributes

ugcnetcse-dec2007-paper2

36.0.247 UGC NET CSE | December 2007 | Part 2 | Question: 10

Amongst the logic families *DTL*, *TTL*, *ECL* and *CMOS*, the family with the least power dissipation is :

- A. *CMOS* B. *DTL* C. *TTL* D. *ECL*

ugcnetcse-dec2007-paper2

Answer key

36.0.248 UGC NET CSE | December 2007 | Part 2 | Question: 9

The dual of the switching function $F = x + yz$ is given by :

- A. $x+yz$ B. $x(y+z)$ C. $(\sim x)+(\sim y)(\sim z)$ D. $(\sim x)((\sim y)+(\sim z))$

ugcnetcse-dec2007-paper2

Answer key

36.0.249 UGC NET CSE | December 2007 | Part 2 | Question: 6

A WFF that is equivalent to the $WFF x \Rightarrow y$ is :

- A. $y \Rightarrow x$ B. $\sim y \Rightarrow x$ C. $\sim y \Rightarrow \sim x$ D. $y \Rightarrow \sim x$

ugcnetcse-dec2007-paper2

Answer key

36.0.250 UGC NET CSE | December 2019 | Part 1 | Question: 50

Read the passage carefully and answer question that follow:

There is no doubt that the market as a reality and political economy as a theory played an important role in the liberal critique. But liberalism is neither the consequence nor the development of these; rather, the market played, in the liberal critique, the role of a “test, a locus of privileged experience where one can identify the effects of excessive governmentality and even weigh their significance; the analysis of the mechanisms of “dearth” or more generally, of the grain trade in the middle of the eighteenth century, was meant to show the point at which governing was always governing too much. Therefore, an analysis to make visible, in the form of evidence, the formation of the value and circulation of wealth – or, on the contrary, an analysis pre-supposing the intrinsic invisibility of the connection between individual profit-seeking and the growth of collective wealth-economics, in any case, shows a basic incompatibility between the optimal development of the economic process and maximization of government procedures. It is by this, more than the play of ideas, the French or English economists broke away from mercantilism and commercialism; they freed reflection on economic practice from the hegemony of the “reason of state” and from the saturation of governmental intervention. By using it as a measure of “governing too much”, they placed it at the limit of governmental action. Liberalism does not derive from juridical thought any more than it does from an economic analysis. It is not the idea of a political society, but the result of search for a liberal technology of a government.

The passage is indicative of the author’s preference to

- A. economic hegemony of individuals B. limit government control of economics
C. seek liberalism from juridical thought D. promote individual profits

ugcnetcse-dec2019-paper1

36.0.251 UGC NET CSE | December 2019 | Part 1 | Question: 49

Read the passage carefully and answer questions that follow:

There is no doubt that the market as a reality and political economy as a theory played an important role in the liberal critique. But liberalism is neither the consequence nor the development of these; rather, the market played, in the liberal critique, the role of a “test, a locus of privileged experience where one can identify the effects of excessive governmentality and even weigh their significance; the analysis of the mechanisms of “dearth” or more generally, of the grain trade in the middle of the eighteenth century, was meant to show the point at which governing was always governing too much. Therefore, an analysis to make visible, in the form of evidence, the formation of the value and circulation of wealth – or, on the contrary, an analysis pre-supposing the intrinsic invisibility of the connection between individual profit-seeking and the growth of collective wealth-economics, in any case, shows a basic incompatibility between the optimal development of the economic process and maximisation of government procedures. It is by this, more than the play of ideas, the French or English economists broke away from mercantilism and commercialism; they freed reflection on economic practice from the hegemony of the “reason of state” and from the saturation of governmental intervention. By using it as a measure of “governing too much”, they placed it at the limit of governmental action. Liberalism does not derive from juridical thought any more than it does from an economic analysis. It is not the idea of a political society, but the result of search for a liberal technology of a government.

What is incompatible with optimal economic development?

- A. Play of ideas
- C. Political society

- B. Absence of commercialism
- D. Excessive government procedures

ugcnetcse-dec2019-paper1

36.0.252 UGC NET CSE | December 2019 | Part 1 | Question: 48



Read the passage carefully and answer questions that follow:

There is no doubt that the market as a reality and political economy as a theory played an important role in the liberal critique. But liberalism is neither the consequence nor the development of these; rather, the market played, in the liberal critique, the role of a “test, a locus of privileged experience where one can identify the effects of excessive governmentality and even weigh their significance; the analysis of the mechanisms of “dearth” or more generally, of the grain trade in the middle of the eighteenth century, was meant to show the point at which governing was always governing too much. Therefore, an analysis to make visible, in the form of evidence, the formation of the value and circulation of wealth – or, on the contrary, an analysis pre-supposing the intrinsic invisibility of the connection between individual profit-seeking and the growth of collective wealth-economics, in any case, shows a basic incompatibility between the optimal development of the economic process and maximisation of government procedures. It is by this, more than the play of ideas, the French or English economists broke away from mercantilism and commercialism; they freed reflection on economic practice from the hegemony of the “reason of state” and from the saturation of governmental intervention. By using it as a measure of “governing too much”, they placed it at the limit of governmental action. Liberalism does not derive from juridical thought any more than it does from an economic analysis. It is not the idea of a political society, but the result of search for a liberal technology of a government.

What kind of evidence was needed to make the liberal critique visible?

- A. Circulation of wealth
- C. Dearth in supply of grain

- B. Pre-supposing individual profit
- D. Incompatibility of growth

ugcnetcse-dec2019-paper1

36.0.253 UGC NET CSE | December 2019 | Part 1 | Question: 47



Read the passage carefully and answer questions that follow:

There is no doubt that the market as a reality and political economy as a theory played an important role in the liberal critique. But liberalism is neither the consequence nor the development of these; rather, the market played, in the liberal critique, the role of a “test, a locus of privileged experience where one can identify the effects of excessive governmentality and even weigh their significance; the analysis of the mechanisms of “dearth” or more generally, of the grain trade in the middle of the eighteenth century, was meant to show the point at which governing was always governing too much. Therefore, an analysis to make visible, in the form of evidence, the formation of the value and circulation of wealth – or, on the contrary, an analysis pre-supposing the intrinsic invisibility of the connection between individual profit-seeking and the growth of collective wealth-economics, in any case, shows a basic incompatibility between the optimal development of the economic process and maximisation of government procedures. It is by this, more than the play of ideas, the French or English economists broke away from mercantilism and commercialism; they freed reflection on economic practice from the hegemony of the “reason of state” and from the saturation of governmental intervention. By using it as a measure of “governing too much”, they placed it at the limit of governmental action. Liberalism does not derive from juridical thought any more than it does from an economic analysis. It is not the idea of a political society, but the result of search for a liberal technology of a government.

The liberal critique examined the implications of

- A. market expansion
- C. growth of political economy

- B. too much governance
- D. politics of marketisation

ugcnetcse-dec2019-paper1

36.0.254 UGC NET CSE | December 2019 | Part 1 | Question: 46



Read the passage carefully and answer questions that follow:

There is no doubt that the market as a reality and political economy as a theory played an important role in the liberal critique. But liberalism is neither the consequence nor the development of these; rather, the market played, in the liberal critique, the role of a “test, a locus of privileged experience where one can identify the effects of excessive governmentality and even weigh their significance; the analysis of the mechanisms of “dearth” or more generally, of the grain trade in the middle of the eighteenth century, was meant to show the point at which governing was always governing too much. Therefore, an analysis to make visible, in the form of evidence, the formation of the value and

circulation of wealth – or, on the contrary, an analysis pre-supposing the intrinsic invisibility of the connection between individual profit-seeking and the growth of collective wealth-economics, in any case, shows a basic incompatibility between the optimal development of the economic process and maximisation of government procedures. It is by this, more than the play of ideas, the French or English economists broke away from mercantilism and commercialism; they freed reflection on economic practice from the hegemony of the “reason of state” and from the saturation of governmental intervention. By using it as a measure of “governing too much”, they placed it at the limit of governmental action. Liberalism does not derive from juridical thought any more than it does from an economic analysis. It is not the idea of a political society, but the result of search for a liberal technology of a government.

Which of the following played a role in the liberal critique?

- A. Liberalism as a consequence of market forces
- B. Liberalism as an offshoot of political economy
- C. Reality of market
- D. Political economy as a practice

ugcnetcse-dec2019-paper1

36.0.255 UGC NET CSE | December 2019 | Part 1 | Question: 45



Rashtriya Uchhatar Shiksha Abhiyan (RUSA) aims to achieve the following in Higher Education System:

- i. Equity
- ii. Access
- iii. 50% GER
- iv. Excellence

Choose the correct option from those given below:

- A. i, ii and iii
- B. i, iii and iv
- C. i, ii and iv
- D. ii, iii and iv

ugcnetcse-dec2019-paper1

Answer key

36.0.256 UGC NET CSE | December 2019 | Part 1 | Question: 44



Which of the following statements is true in the Indian context?

- A. Autonomous colleges can grant degrees independent of universities
- B. Autonomous colleges can grant only bachelor's degree independent of universities
- C. Except doctoral degrees, all other degrees and diplomas can be granted by autonomous colleges
- D. Whatever may be the degree or diploma, autonomous colleges can grant them under their own name but under the seal of an affiliated university

ugcnetcse-dec2019-paper1

Answer key

36.0.257 UGC NET CSE | December 2019 | Part 1 | Question: 43



According to GATS (General Agreement on Trade and Services), higher education should be a commodity in the

- A. domestic public sector
- B. domestic private sector
- C. non-trading sector
- D. global marketplace

ugcnetcse-dec2019-paper1

Answer key

36.0.258 UGC NET CSE | December 2019 | Part 1 | Question: 42



The major barriers for access to higher education in India are:

- i. more opportunities of employment for less educated
- ii. government policies
- iii. language of instruction
- iv. economic status

- v. competition from foreign universities
- vi. gender discrimination in society

Choose the correct answer from the options given below:

- A. i, ii and iii
- B. ii, iii and v
- C. iii, iv and vi
- D. iv, v and vi

ugcnetcse-dec2019-paper1

Answer key 

36.0.259 UGC NET CSE | December 2019 | Part 1 | Question: 41



Database WOS stands for

- A. Web of Science
- B. Web of Sources
- C. World of Science
- D. Web of Service

ugcnetcse-dec2019-paper1

Answer key 

36.0.260 UGC NET CSE | December 2019 | Part 1 | Question: 40



The most important pollutants that cause degradation of water quality in rivers and streams are

- i. Bacteria
- ii. Nutrients
- iii. Metals
- iv. Total dissolved solids
- v. Algae

Choose the most appropriate answer from the options given below:

- A. i, ii and iii
- B. i, ii and iv
- C. i, ii, iv and v
- D. i, ii, iii, iv and v

ugcnetcse-dec2019-paper1

Answer key 

36.0.261 UGC NET CSE | December 2019 | Part 1 | Question: 39



The most dominant source of Benzene emissions in ambient air is

- A. Cement industry
- B. Cigarettes
- C. Car exhausts
- D. Paints and varnish

ugcnetcse-dec2019-paper1

Answer key 

36.0.262 UGC NET CSE | December 2019 | Part 1 | Question: 38



The themes of some Sustainable Development Goals are

- a. Climate action
- b. Sustainable cities and communities
- c. Peace, justice and strong institutions
- d. Skill development and decent employment
- e. Green agriculture
- f. Responsible consumption and production

Choose the most appropriate from those given below:

- A. a, b, c, e and f
- B. b, c, e and f
- C. b, c, d, e and f
- D. a, b, c and f

ugcnetcse-dec2019-paper1

Answer key 

36.0.263 UGC NET CSE | December 2019 | Part 1 | Question: 37



Under Green India Mission, how many hectares of degraded forest land is to be brought under afforestation?

- A. 02 million hectares
- B. 06 million hectares
- C. 08 million hectares
- D. 20 million hectares

ugcnetcse-dec2019-paper1

Answer key

36.0.264 UGC NET CSE | December 2019 | Part 1 | Question: 36



Identify from the options given below, the co-benefit of Montreal Protocol

- A. Impetus to development of energy efficient systems
- B. Reduction in carbon dioxide (equivalent) emissions
- C. Convergence of efforts of international community in addressing air pollution
- D. Control of transboundary movement of hazardous waste

ugcnetcse-dec2019-paper1

Answer key

36.0.265 UGC NET CSE | December 2019 | Part 1 | Question: 35



The national agency for responding to computer security incidents as and when they occur, is

- A. CAT
- B. CDAC
- C. CCA
- D. ICERT

ugcnetcse-dec2019-paper1

Answer key

36.0.266 UGC NET CSE | December 2019 | Part 1 | Question: 34



PCI stands for

- A. Partial Component Interconnect
- B. Partial Component Interaction
- C. Peripheral Component Interconnect
- D. Peripheral Component Interaction

ugcnetcse-dec2019-paper1

Answer key

36.0.267 UGC NET CSE | December 2019 | Part 1 | Question: 33



_____ is a wireless technology built in electronic gadgets for transferring data over short distance

- A. WiFi
- B. Bluetooth
- C. Modem
- D. USB

ugcnetcse-dec2019-paper1

Answer key

36.0.268 UGC NET CSE | December 2019 | Part 1 | Question: 32



The term one gigabyte refers to:

- A. 1024 petabytes
- B. 1024 megabytes
- C. 1024 kilobytes
- D. 1024 bytes

ugcnetcse-dec2019-paper1

Answer key

36.0.269 UGC NET CSE | December 2019 | Part 1 | Question: 31



Which one of the following tasks is associated with changing appearance of a document in word processing?

A. Editing

B. Formatting

C. Writing

D. Printing

ugcnetcse-dec2019-paper1

Answer key **36.0.270 UGC NET CSE | December 2019 | Part 1 | Question: 29**

The following table gives the percentage of marks obtained by 6 students in 5 different subjects in an examination. The numbers in the parenthesis give the maximum marks in each subject. Answer the given questions based on the table:

Subject (Max. Marks)	Percentage (%) of marks obtained				
	Maths (150)	Chemistry (130)	Physics (120)	Biology (100)	English (100)
Ankit	90	50	80	60	70
Amar	100	60	70	40	80
Sanya	90	60	85	80	60
Rahul	80	75	65	90	90
Puneet	80	80	70	85	50
Pooja	70	75	50	80	40

How many students obtained 60% and above marks in all the subjects?

A. Three

B. Two

C. One

D. Zero

ugcnetcse-dec2019-paper1

Answer key **36.0.271 UGC NET CSE | December 2019 | Part 1 | Question: 28**

The following table gives the percentage of marks obtained by 6 students in 5 different subjects in an examination. The numbers in the parenthesis give the maximum marks in each subject. Answer the given questions based on the table:

Subject (Max. Marks)	Percentage (%) of marks obtained				
	Maths (150)	Chemistry (130)	Physics (120)	Biology (100)	English (100)
Ankit	90	50	80	60	70
Amar	100	60	70	40	80
Sanya	90	60	85	80	60
Rahul	80	75	65	90	90
Puneet	80	80	70	85	50
Pooja	70	75	50	80	40

What is the average marks obtained by all the six students in Chemistry?

A. 86.66

B. 76.66

C. 66.66

D. 60.00

ugcnetcse-dec2019-paper1

Answer key **36.0.272 UGC NET CSE | December 2019 | Part 1 | Question: 27**

The following table gives the percentage of marks obtained by 6 students in 5 different subjects in an examination. The numbers in the parenthesis give the maximum marks in each subject. Answer the given questions based on the table:

Subject (Max. Marks)	Percentage (%) of marks obtained				
Name of student	Maths (150)	Chemistry (130)	Physics (120)	Biology (100)	English (100)
Ankit	90	50	80	60	70
Amar	100	60	70	40	80
Sanya	90	60	85	80	60
Rahul	80	75	65	90	90
Puneet	80	80	70	85	50
Pooja	70	75	50	80	40

What was the aggregate of marks obtained by Sanya in all the five subjects?

- A. 375 B. 395 C. 455 D. 475

ugcnetcse-dec2019-paper1

Answer key 

36.0.273 UGC NET CSE | December 2019 | Part 1 | Question: 26



The following table gives the percentage of marks obtained by 6 students in 5 different subjects in an examination. The numbers in the parenthesis give the maximum marks in each subject. Answer the given questions based on the table:

Subject (Max. Marks)	Percentage (%) of marks obtained				
Name of student	Maths (150)	Chemistry (130)	Physics (120)	Biology (100)	English (100)
Ankit	90	50	80	60	70
Amar	100	60	70	40	80
Sanya	90	60	85	80	60
Rahul	80	75	65	90	90
Puneet	80	80	70	85	50
Pooja	70	75	50	80	40

What is the overall percentage of marks obtained by Amar?

- A. 70% B. 72% C. 60% D. 58.33%

ugcnetcse-dec2019-paper1

Answer key 

36.0.274 UGC NET CSE | December 2019 | Part 1 | Question: 25



According to classical Indian school of logic, what is the correct sequence of steps involved in Anum $\bar{a}$ na (influence)?

- A. Upanaya, Pratijñā, Hetu, Udāharana, Nigmana
 B. Pratijñā, Hetu, Upanaya, Udāharana, Nigmana
 C. Pratijñā, Upanaya, Hetu, Udāharana, Nigmana
 D. Pratijñā, Hetu, Udāharana, Upanaya, Nigmana

ugcnetcse-dec2019-paper1

36.0.275 UGC NET CSE | December 2019 | Part 1 | Question: 24



Which one of the following fallacious hetu (middle term) is not uniformly concomitant with the major term?

- A. Asatpratipaksa B. Auyatireki
 C. Anyonya-Asiddha D. Suyābhicāra

ugcnetcse-dec2019-paper1

Answer key 

Answer key 

Answer key 



Answer key 🖱️

36.0.282 UGC NET CSE | December 2019 | Part 1 | Question: 17

If 3% of $(a + b) = 7\%$ of (ab)
and 5% of $(a - b) = 4\%$ of (ab) ,
then what percentage of b is a ?

- A. $\frac{47}{23}\%$
B. $\frac{23}{47}\%$
C. $\frac{4700}{23}\%$
D. $\frac{2300}{47}\%$

ugcnetcse-dec2019-paper1

36.0.283 UGC NET CSE | December 2019 | Part 1 | Question: 16

In a certain coding language, 'AEIOU' is written as 'TNHDZ'. Using the same coding language, 'BFJPV' will be written as

- A. UOIEA B. AEIOU C. CGKQW D. WQKGC

ugcnetcse-dec2019-paper1

Answer key

36.0.284 UGC NET CSE | December 2019 | Part 1 | Question: 15

Given below are two Statements – One is labelled as Assertion (A) and other is labelled as Reason (R):

Assertion (A): Use of slang in formal teaching makes communication lively and interesting.

Reasons (R): Academic decency demands the avoidance of slang in the classroom environment.

In the light of the above statements, choose the correct option:

- A. Both (A) and (R) are true and (R) is the correct explanation of (A)
B. Both (A) and (R) are true and (R) is not the correct explanation of (A)
C. (A) is true, but (R) is false
D. (A) is false, but (R) is true

ugcnetcse-dec2019-paper1

36.0.285 UGC NET CSE | December 2019 | Part 1 | Question: 14

Which of the following modes of communication can be employed in a classroom for effective teaching?

- | | |
|------------------|-------------------|
| a. Top-down | b. Iconic |
| c. Associational | d. Dissociational |
| e. Symbolic | f. Abstract |

Choose the most appropriate option from the following:

- A. (a), (b) and (f) B. (c), (e) and (f)
C. (b), (c) and (e) D. (a), (c) and (d)

ugcnetcse-dec2019-paper1

Answer key

36.0.286 UGC NET CSE | December 2019 | Part 1 | Question: 13



A verbal communication technique used in teaching is

- A. Slow expression of words
- B. Varying the speed of voice and tone
- C. Presentation without pause
- D. Resorting to semantic jugglery

ugcnetcse-dec2019-paper1

Answer key

36.0.287 UGC NET CSE | December 2019 | Part 1 | Question: 12



Information overload in a classroom environment by a teacher will lead to

- A. High level participation
- B. Semantic precision
- C. Effective impression
- D. Delayed feedback

ugcnetcse-dec2019-paper1

Answer key

36.0.288 UGC NET CSE | December 2019 | Part 1 | Question: 11



When there is an animated discussion between a teacher and his or her students in the classroom, it can be classified as

- A. Horizontal communication
- B. Mechanical communication
- C. Linear communication
- D. Categorical communication

ugcnetcse-dec2019-paper1

Answer key

36.0.289 UGC NET CSE | December 2019 | Part 1 | Question: 10



A research scholar while submitting his/her thesis, did not acknowledge in the preface the help and support of the respondents of questionnaires. This will be called an instance of

- A. Technical lapse
- B. Wilful negligence
- C. Unethical act
- D. Lack of respect

ugcnetcse-dec2019-paper1

Answer key

36.0.290 UGC NET CSE | December 2019 | Part 1 | Question: 9



In which of the following formats 'Chapter Scheme' of the document has to be formally approved by a research degree committee in the university?

- A. Thesis/dissertation
- B. Seminar papers
- C. Research articles
- D. Research monographs

ugcnetcse-dec2019-paper1

Answer key

36.0.291 UGC NET CSE | December 2019 | Part 1 | Question: 8



Which of the following sampling procedures will be appropriate for conducting researches with empirico-inductive research paradigm?

- A. Simple random sampling procedure
- B. Systematic sampling procedure
- C. Stratified sampling procedure
- D. Any of the non-probability sampling procedures

ugcnetcse-dec2019-paper1

Answer key

36.0.292 UGC NET CSE | December 2019 | Part 1 | Question: 7



A researcher intends to make use of ICT in his/her research. What considerations should weigh most in such a decision?

- i. Appropriateness of the tool used
- ii. Cost involved in procuring it
- iii. Availability of tools in the department where research is to be undertaken
- iv. Willingness of his/her supervisor to offer help
- v. The company from which the ICT equipment has been procured

Choose your answer from the following options:

- A. (i), (ii) and (iii)
- B. (iii), (iv) and (v)
- C. (i), (iii) and (iv)
- D. (ii), (iii) and (v)

ugcnetcse-dec2019-paper1

Answer key 

36.0.293 UGC NET CSE | December 2019 | Part 1 | Question: 6



A university teacher plans to improve the study habits of students in his/her class. Which type of research paradigm will be helpful in this regard?

- A. Evaluative research paradigm
- B. Fundamental research paradigm
- C. Applied research paradigm
- D. Action research paradigm

ugcnetcse-dec2019-paper1

Answer key 

36.0.294 UGC NET CSE | December 2019 | Part 1 | Question: 5



Which of the following types of assessment is conducted periodically with an eye on standards?

- A. Formative assessment
- B. Summative assessment
- C. Portfolio assessment
- D. Performance assessment

ugcnetcse-dec2019-paper1

Answer key 

36.0.295 UGC NET CSE | December 2019 | Part 1 | Question: 4



In which of the following types of learning materials, the presentations are highly structured and individualised?

- A. Textbooks prescribed by the universities
- B. Journals and the articles recommended for readings
- C. Writings of great thinkers selected for reflective readings
- D. Programmed instructional and modular learning material

ugcnetcse-dec2019-paper1

Answer key 

36.0.296 UGC NET CSE | December 2019 | Part 1 | Question: 3



Which of the following are classroom related factors that influence effectiveness of teaching?

- i. Prior task related behavior of students
- ii. Adherence to linear pattern of communication by the teacher
- iii. Socio-economic status of the family to which learners belong
- iv. Inappropriate use of technological resources by the teacher
- v. Cultural background of students

Choose your answer from the following options:

- A. (i), (ii) and (iii)
- B. (ii), (iii) and (iv)
- C. (i), (ii) and (iv)
- D. (iii), (iv) and (v)

ugcnetcse-dec2019-paper1

Answer key 

36.0.297 UGC NET CSE | December 2019 | Part 1 | Question: 2

Which of the following statements differentiate teaching from learning?

- i. Teaching is a social act while learning is a personal act
- ii. Teaching implies learning
- iii. Teaching is like selling while learning is like buying
- iv. Teaching can occur without learning taking place
- v. In teaching, influence is directed towards learning and learner, while in learning it is usually towards oneself

Choose the correct answer from the following options:

- A. (i), (iii) and (v)
- B. (i), (ii) and (iii)
- C. (ii), (iii) and (iv)
- D. (iii), (iv) and (v)

ugcnetcse-dec2019-paper1

Answer key

36.0.298 UGC NET CSE | December 2019 | Part 1 | Question: 1

Which of the learning outcomes are intended in teaching organized at understanding level?

- A. Longer recall and retention of facts
- B. Seeking of relationships and patterns among facts
- C. Creative construction and critical interpretation of ideas
- D. Mastery of facts and information

ugcnetcse-dec2019-paper1

36.0.299 UGC NET CSE | December 2019 | Part 2 | Question: 100

An organization needs to maintain database having five attributes **A, B, C, D, E**. These attributes are functionally dependent on each other for which functionality dependency set F is given as: $F : \{A \rightarrow BC, D \rightarrow E, BC \rightarrow D, A \rightarrow D\}$. Consider a universal relation $R(A, B, C, D, E)$ with functional dependency set F . Also all attributes are simple and take atomic values only.

Assume that given table R is decomposed in two tables

$R_1(A, B, C)$ with functional dependency set $f_1 = \{A \rightarrow B, A \rightarrow C\}$ and

$R_2(A, D, E)$ with FD set $F_2 = \{A \rightarrow D, D \rightarrow E\}$

Which of the following option is true w.r.t. given decomposition?

- A. Dependency preservation property is followed
- B. R_1 and R_2 are both in $2NF$
- C. R_2 is in $2NF$ and R_3 is in $3NF$
- D. R_1 is in $3NF$ and R_2 is in $2NF$

ugcnetcse-dec2019-paper2

Answer key

36.0.300 UGC NET CSE | December 2019 | Part 2 | Question: 99

An organization needs to maintain database having five attributes **A, B, C, D, E**. These attributes are functionally dependent on each other for which functionality dependency set F is given as: $F : \{A \rightarrow BC, D \rightarrow E, BC \rightarrow D, A \rightarrow D\}$. Consider a universal relation $R(A, B, C, D, E)$ with functional dependency set F . Also all attributes are simple and take atomic values only.

Identify the redundant functional dependency in F

- A. $BC \rightarrow D$
- B. $D \rightarrow E$
- C. $A \rightarrow D$
- D. $A \rightarrow BC$

ugcnetcse-dec2019-paper2

Answer key

36.0.301 UGC NET CSE | December 2019 | Part 2 | Question: 98



An organization needs to maintain database having five attributes **A, B, C, D, E**. These attributes are functionally dependent on each other for which functionality dependency set F is given as: $F : \{A \rightarrow BC, D \rightarrow E, BC \rightarrow D, A \rightarrow D\}$. Consider a universal relation $R(A, B, C, D, E)$ with functional dependency set F . Also all attributes are simple and take atomic values only.

Identify the normal form in which relation R belong to

- A. 1 NF B. 2 NF C. 3 NF D. BCNF

ugcnetcse-dec2019-paper2

Answer key

36.0.302 UGC NET CSE | December 2019 | Part 2 | Question: 97



An organization needs to maintain database having five attributes **A, B, C, D, E**. These attributes are functionally dependent on each other for which functionality dependency set F is given as: $F : \{A \rightarrow BC, D \rightarrow E, BC \rightarrow D, A \rightarrow D\}$. Consider a universal relation $R(A, B, C, D, E)$ with functional dependency set F . Also all attributes are simple and take atomic values only.

Identify primary key of table R with functional dependency set F

- A. BC B. AD C. A D. AB

ugcnetcse-dec2019-paper2

Answer key

36.0.303 UGC NET CSE | December 2019 | Part 2 | Question: 96



An organization needs to maintain database having five attributes **A, B, C, D, E**. These attributes are functionally dependent on each other for which functionality dependency set F is given as: $F : \{A \rightarrow BC, D \rightarrow E, BC \rightarrow D, A \rightarrow D\}$. Consider a universal relation $R(A, B, C, D, E)$ with functional dependency set F . Also all attributes are simple and take atomic values only.

Minimal cover F' of functional dependency set F is

- A. $F' = \{A \rightarrow B, A \rightarrow C, BC \rightarrow D, D \rightarrow E\}$
 B. $F' = \{A \rightarrow BC, B \rightarrow D, D \rightarrow E\}$
 C. $F' = \{A \rightarrow B, A \rightarrow C, A \rightarrow D, D \rightarrow E\}$
 D. $F' = \{A \rightarrow B, A \rightarrow C, B \rightarrow D, C \rightarrow D, D \rightarrow E\}$

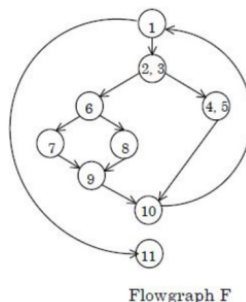
ugcnetcse-dec2019-paper2

Answer key

36.0.304 UGC NET CSE | December 2019 | Part 2 | Question: 95



A flow graph F with entry node (1) and exit node (11) is shown below:



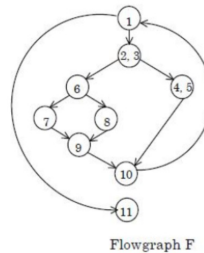
Flowgraph F

How many predicate nodes are there and what are their names?

- A. Three: (1,(2,3),6) B. Three: (1,4,6)
 C. Four: ((2,3),6,10,11) D. Four: ((2,3),6,9,10)

36.0.305 UGC NET CSE | December 2019 | Part 2 | Question: 94

A flow graph F with entry node (1) and exit node (11) is shown below:

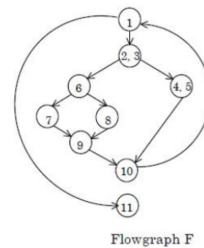


How many nodes are there in the longest independent path?

- A. 6 B. 7 C. 8 D. 9

36.0.306 UGC NET CSE | December 2019 | Part 2 | Question: 93

A flow graph F with entry node (1) and exit node (11) is shown below:

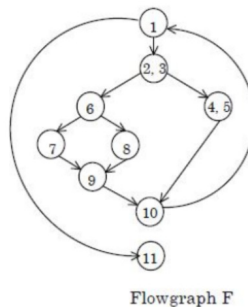


How many regions are there in flowgraph F ?

- A. 2 B. 3 C. 4 D. 5

36.0.307 UGC NET CSE | December 2019 | Part 2 | Question: 92

A flow graph F with entry node (1) and exit node (11) is shown below:



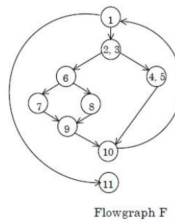
What is the cyclomatic complexity of flowgraph F ?

- A. 2 B. 3 C. 4 D. 5

36.0.308 UGC NET CSE | December 2019 | Part 2 | Question: 91



A flow graph F with entry node (1) and exit node (11) is shown below:



How many nodes are there in flowgraph F

- A. 9 B. 10 C. 11 D. 12

ugcnetcse-dec2019-paper2

Answer key

36.0.309 UGC NET CSE | December 2019 | Part 2 | Question: 90



The Data Encryption Standard (DES) has a function consists of four steps. Which of the following is correct order of these four steps?

- A. an expansion permutation, *S*-boxes, an XOR operation, a straight permutation
 B. an expansion permutation, an XOR operation, *S*-boxes, a straight permutation
 C. a straight permutation, *S*-boxes, an XOR operation, an expansion permutation
 D. a straight permutation, an XOR operation, *S*-boxes, an expansion permutation

ugcnetcse-dec2019-paper2

Answer key

36.0.310 UGC NET CSE | December 2019 | Part 2 | Question: 89

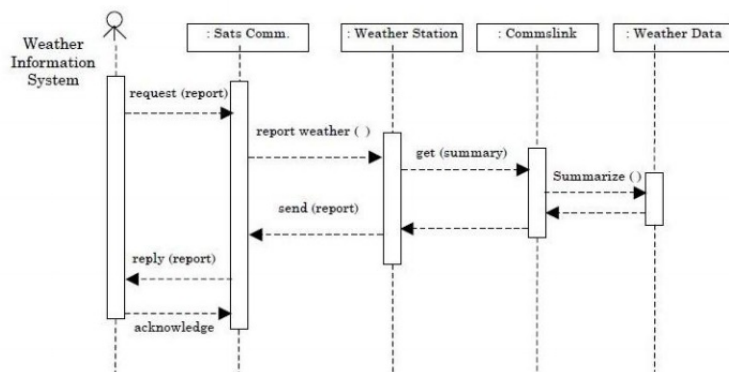
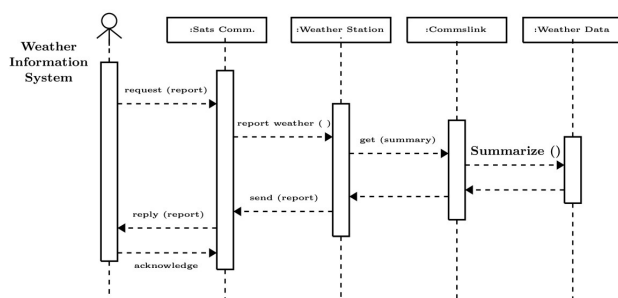


Figure 1



The sequence diagram given in Figure 1 for the Weather Information System takes place when an external system requests the summarized data from the weather station. The increasing order of lifeline for the objects in the system are:

- A. Sat comms → Weather station → Commslink → Weather data
- B. Sat comms → Comms link → Weather station → Weather data
- C. Weather data → Comms link → Weather station → Sat Comms
- D. Weather data → Weather station → Comms link → Sat Comms

ugcnetcse-dec2019-paper2

Answer key 

36.0.311 UGC NET CSE | December 2019 | Part 2 | Question: 88



Match List-I with List-II:

- | List I | List II |
|-----------------------------------|---|
| (a) Micro operation | (i) Specify micro operations |
| (b) Micro programmed control unit | (ii) Improve CPU utilization |
| (c) Interrupts | (iii) Control Memory |
| (d) Micro instruction | (iv) Elementary operation performed on data stored in registers |

Choose the correct option from those given below:

- A. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
- B. (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
- C. (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
- D. (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

ugcnetcse-dec2019-paper2

Answer key 

36.0.312 UGC NET CSE | December 2019 | Part 2 | Question: 87



Match List-I and List-II:

- | List I | List II |
|--------------------------------|--|
| (a) Isolated I/O | (i) same set of control signal for I/O and memory communication |
| (b) Memory mapped I/O | (ii) separate instructions for I/O and memory communication |
| (c) I/O interface | (iii) requires control signals to be transmitted between the communicating units |
| (d) Asynchronous data transfer | (iv) resolve the differences in central computer and peripherals |

Choose the correct option from those given below:

- A. (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
- B. (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
- C. (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
- D. (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)

ugcnetcse-dec2019-paper2

Answer key 

36.0.313 UGC NET CSE | December 2019 | Part 2 | Question: 86



An instruction is stored at location 500 with its address field at location 501. The address field has the value 400. A processor register R_1 contains the number 200. Match the addressing mode (List-I) given below with effective address (List-II) for the given instruction:

List I	List II
(a) Direct	(i) 200
(b) Register indirect	(ii) 902
(c) Index with R_1 as the index register	(iii) 400
(d) Relative	(iv) 600

Choose the correct option from those given below:

- A. (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
 B. (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
 C. (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)
 D. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

ugcnetcse-dec2019-paper2

Answer key 

36.0.314 UGC NET CSE | December 2019 | Part 2 | Question: 85



Match List – I with List-II:

List I	List II
(a) Frame attribute	(i) to create link in HTML
(b) <tr> tag	(ii) for vertical alignment of content in cell
(c) Valign attribute	(iii) to enclose each row in table
(d) <a> tag	(iv) to specify the side of the table frame that display border

Choose the correct option from those given below:

- A. (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
 B. (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)
 C. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
 D. (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

ugcnetcse-dec2019-paper2

Answer key 

36.0.315 UGC NET CSE | December 2019 | Part 2 | Question: 84



Match the Agile Process models with the task performed during the model:

List I	List II
(a) Scrum	(i) CRC cards
(b) Adaptive software development	(ii) Sprint backlog
(c) Extreme programming	(iii) <action> the <result> <by/for/of/to> a(n) <object>
(d) Feature-driven development	(iv) Time box release plan

Choose the correct option from those given below:

- A. (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
 B. (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)
 C. (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
 D. (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)

ugcnetcse-dec2019-paper2

Answer key 

36.0.316 UGC NET CSE | December 2019 | Part 2 | Question: 83



According to the ISO – 9126 Standard Quality Model, match the attributes given in List-I with their definitions in List-II:

List I	List II
(a) Functionality	(i) Relationship between level of performance and amount of resources
(b) Reliability	(ii) Characteristics related with achievement of purpose
(c) Efficiency	(iii) Effort needed to make for improvement
(d) Maintainability	(iv) Capability of software to maintain performance of software

Choose the correct options from the ones given below:

- A. (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
- B. (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
- C. (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
- D. (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)

ugcnetcse-dec2019-paper2

Answer key 

36.0.317 UGC NET CSE | December 2019 | Part 2 | Question: 82



Match List-I and List-II:

List I	List II
(a) Physical layer	(i) Provide token management service
(b) Transport layer	(ii) Concerned with transmitting raw bits over a communication channel
(c) Session Layer	(iii) Concerned with the syntax and semantics of the information transmitted
(d) Presentation layer	(iv) True end-to-end layer from source to destination

Choose the correct option from those given below:

- A. (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)
- B. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
- C. (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
- D. (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

ugcnetcse-dec2019-paper2

Answer key 

36.0.318 UGC NET CSE | December 2019 | Part 2 | Question: 81



Consider the following learning algorithms:

- | | |
|---|--|
| <ul style="list-style-type: none"> a. Logistic repression c. Linear repression <p>algorithms?</p> | <ul style="list-style-type: none"> b. Back propagation <p>Which of the following option represents classification</p> |
|---|--|
-
- | | |
|--|--|
| <ul style="list-style-type: none"> A. (a) and (b) only C. (b) and (c) only | <ul style="list-style-type: none"> B. (a) and (c) only D. (a), (b) and (c) |
|--|--|

ugcnetcse-dec2019-paper2

36.0.319 UGC NET CSE | December 2019 | Part 2 | Question: 80

Consider the following:

- a. Trapping at local maxima
 - c. Traversal along the ridge
- the hill climbing algorithm?

- b. Reaching a plateau

Which of the following option represents shortcomings of

- A. (a) and (b) only
- C. (b) and (c) only

- B. (a) and (c) only
- D. (a), (b) and (c)

ugcnetcse-dec2019-paper2

Answer key

36.0.320 UGC NET CSE | December 2019 | Part 2 | Question: 79

Consider the following statements with respect to network security:

- i. Message confidentiality means that the sender and the receiver expect privacy
- ii. Message integrity means that the data must arrive at the receiver exactly as they were sent.
- iii. Message authentication means the receiver is ensured that the message is coming from the intended sender

Which of the statements is (are) correct?

- A. Only (i) and (ii)
- C. Only (ii) and (iii)

- B. Only (i) and (iii)
- D. (i), (ii) and (iii)

ugcnetcse-dec2019-paper2

Answer key

36.0.321 UGC NET CSE | December 2019 | Part 2 | Question: 78

Consider the following statements:

- i. Windows Azure is a cloud-based operating system.
- ii. Google App Engine is an integrated set of online services for consumers to communicate and share with others.
- iii. Amazon Cloud Front is a web service for content delivery.

Which of the following statements is (are) correct?

- A. Only (i) and (ii)
- C. Only (ii) and (iii)

- B. Only (i) and (iii)
- D. (i), (ii) and (iii)

ugcnetcse-dec2019-paper2

Answer key

36.0.322 UGC NET CSE | December 2019 | Part 2 | Question: 77

Consider the following statements:

- i. Fiber optic cable is much lighter than copper cable
- ii. Fiber optic cable is not affected by power surges or electromagnetic interference
- iii. Optical transmission is inherently bidirectional

Which of the statements is (are) correct?

- A. Only (i) and (ii)
- C. Only (ii) and (iii)

- B. Only (i) and (iii)
- D. (i), (ii) and (iii)

ugcnetcse-dec2019-paper2

Answer key

36.0.323 UGC NET CSE | December 2019 | Part 2 | Question: 76

Consider the following statements:

- The running time of dynamic programming algorithm is always $\theta(\rho)$ where ρ is number of subproblems
- When a recurrence relation has a cyclic dependency, it is impossible to use that recurrence relation (unmodified) in a correct dynamic program.
- For a dynamic programming algorithm, computing all values in a bottom-up fashion is asymptotically faster than using recursion and memorization
- If a problem X can be reduced to a known NP-hard problem, then X must be NP-hard

Which of the statement(s) is (are) true?

- A. Only (ii) and (i) B. Only (ii) C. Only (ii) and (iii) D. Only (ii) and (iv)

ugcnetcse-dec2019-paper2

Answer key 

36.0.324 UGC NET CSE | December 2019 | Part 2 | Question: 75



The following multithreaded algorithm computes transpose of a matrix in parallel:

p Trans (X, Y, N)

if $N = 1$

then $Y[1, 1] \leftarrow X[1, 1]$

else partition X into four $(N/2) \times (N/2)$ submatrices $X_{11}, X_{12}, X_{21}, X_{22}$

partition Y into four $(N/2) \times (N/2)$ submatrices $Y_{11}, Y_{12}, Y_{21}, Y_{22}$

spawn p Trans $X_{11}, Y_{11}, N/2$

spawn p Trans $X_{12}, Y_{12}, N/2$

spawn p Trans $X_{21}, Y_{21}, N/2$

spawn p Trans $X_{22}, Y_{22}, N/2$

What is the asymptotic parallelism of the algorithm?

- A. T_1/T_∞ or $\theta(N^2/\lg N)$ B. T_1/T_∞ or $\theta(N/\lg N)$
 C. T_1/T_∞ or $\theta(\lg N/N^2)$ D. T_1/T_∞ or $\theta(\lg N/N)$

ugcnetcse-dec2019-paper2

Answer key 

36.0.325 UGC NET CSE | December 2019 | Part 2 | Question: 74



Identify the circumstances under which pre-emptive CPU scheduling is used:

- a. A process switches from Running state to Ready state b. A process switches from Waiting state to Ready state
 c. A process completes its execution d. A process switches from Ready to Waiting state

Choose the correct option:

- A. (a) and (b) only B. (a) and (d) only
 C. (c) and (d) only D. (a), (b), (c) only

ugcnetcse-dec2019-paper2

Answer key 

36.0.326 UGC NET CSE | December 2019 | Part 2 | Question: 73



Two concurrent executing transactions T_1 and T_2 are allowed to update same stock item say 'A' in an uncontrolled manner. In such scenario, following problems may occur:

- a. Dirty read problem b. Lost update problem
 c. Transaction failure d. Inconsistent database state

Which of the following option is correct if database system has no concurrency module and allows concurrent execution of above two transactions?

- A. (a), (b) and (c) only B. (c) and (d) only
 C. (a) and (b) only D. (a), (b) and (d) only

ugcnetcse-dec2019-paper2

Answer key 

36.0.327 UGC NET CSE | December 2019 | Part 2 | Question: 72



Which of the following are legal statements in *C* programming language?

- i. `int *P=&44;`
- ii. `int *P=&r;`
- iii. `int P=&a;`
- iv. `int P=a;`

Choose the correct option:

- A. (i) and (ii) B. (ii) and (iii) C. (ii) and (iv) D. (i) and (iv)

ugcnetcse-dec2019-paper2

Answer key 

36.0.328 UGC NET CSE | December 2019 | Part 2 | Question: 71



Which of the following statements are true regarding *C++*?

- i. Overloading gives the capability to an existing operator to operate on other data types
- ii. Inheritance in object oriented programming provides support to reusability
- iii. When object of a derived class is defined, first the constructor of derived class is executed then constructor of a base class is executed
- iv. Overloading is a type of polymorphism

Choose the correct option from those given below:

- A. (i) and (ii) only B. (i), (ii) and (iii) only
C. (i), (ii) and (iv) only D. (ii), (iii) and (iv) only

ugcnetcse-dec2019-paper2

Answer key 

36.0.329 UGC NET CSE | December 2019 | Part 2 | Question: 70



Consider the following statements with respect to approaches to fill area on raster systems:

P: To determine the overlap intervals for scan lines that cross the area

Q: To start from a given interior position and paint outward from this point until we encounter the specified boundary conditions.

Select the correct answer from the options given below:

- A. *P* only B. *Q* only C. Both *P* and *Q* D. Neither *P* nor *Q*

ugcnetcse-dec2019-paper2

Answer key 

36.0.330 UGC NET CSE | December 2019 | Part 2 | Question: 69



Which of the following binary codes for decimal digits are self complementing?

- i. 8421 code
- ii. 2421 code
- iii. excess-3 code
- iv. excess-3 gray code

Choose the correct option:

- A. (i) and (ii) B. (ii) and (iii) C. (iii) and (iv) D. (iv) and (i)

ugcnetcse-dec2019-paper2

Answer key 

36.0.331 UGC NET CSE | December 2019 | Part 2 | Question: 68



The Reduced Instruction Set Computer (RISC) characteristics are:

- i. Single cycle instruction execution
- ii. Variable length instruction formats
- iii. Instructions that manipulates operands in memory
- iv. Efficient instruction pipeline

Choose the correct characteristics from the options given below:

- A. (i) and (ii) B. (ii) and (iii) C. (i) and (iv) D. (iii) and (iv)

ugcnetcse-dec2019-paper2

Answer key 

36.0.332 UGC NET CSE | December 2019 | Part 2 | Question: 67



Consider the following statements with respect to duality in LPP:

- i. The final simplex table giving optimal solution of the primal also contains optimal solution of its dual in itself
- ii. If either the primal or the dual problem has a finite optimal solution, then the other problem also has a finite optimal solution
- iii. If either problem has an unbounded optimum solution, then the other problem has no feasible solution at all

Which of the statements is (are) correct?

- A. Only (i) and (ii) B. Only (i) and (iii)
C. only (ii) and (iii) D. (i), (ii) and (iii)

ugcnetcse-dec2019-paper2

Answer key 

36.0.333 UGC NET CSE | December 2019 | Part 2 | Question: 66



Consider the following statements:

S_1 : If a group $(G, *)$ is of order n , and $a \in G$ is such that $a^m = e$ for some integer $m \leq n$, then m must divide n .

S_2 : If a group $(G, *)$ is of even order, then there must be an element $a \in G$ such that $a \neq e$ and $a * a = e$.

Which of the statements is (are) correct?

- A. Only S_1 B. Only S_2 C. Both S_1 and S_2 D. Neither S_1 and S_2

ugcnetcse-dec2019-paper2

Answer key 

36.0.334 UGC NET CSE | December 2019 | Part 2 | Question: 65



Let the population of chromosomes in genetic algorithm is represented in terms of binary number. The strength of fitness of a chromosome in decimal form, x , is given by

$$Sf(x) = \frac{f(x)}{\sum f(x)} \text{ where } f(x) = x^2$$

The population is given by P where:

$$P = \{(01101, (11000), (01000), (10011)\}$$

The strength of fitness of chromosome (11000) is _____

A. 24

B. 576

C. 14.4

D. 49.2

ugcnetcse-dec2019-paper2

Answer key **36.0.335 UGC NET CSE | December 2019 | Part 2 | Question: 64**

The order of schema 10^*101^* and $??^*0^*?1$ are _____ and _____ respectively.

A. 5,3

B. 5,2

C. 7,5

D. 8,7

ugcnetcse-dec2019-paper2

Answer key **36.0.336 UGC NET CSE | December 2019 | Part 2 | Question: 63**

Let $\mathbf{W}_{ij}$ represents weight between node i at layer k and node j at layer $(k - 1)$ of a given multilayer perceptron. The weight updation using gradient descent method is given by

A. $\mathbf{W}_{ij}(t+1) = \mathbf{W}_{ij}(t) + \alpha \frac{\partial \mathbf{E}}{\partial \mathbf{W}_{ij}}, 0 \leq \alpha \leq 1$

B. $\mathbf{W}_{ij}(t+1) = \mathbf{W}_{ij}(t) - \alpha \frac{\partial \mathbf{E}}{\partial \mathbf{W}_{ij}}, 0 \leq \alpha \leq 1$

C. $\mathbf{W}_{ij}(t+1) = \alpha \frac{\partial \mathbf{E}}{\partial \mathbf{W}_{ij}}, 0 \leq \alpha \leq 1$

D. $\mathbf{W}_{ij}(t+1) = -\alpha \frac{\partial \mathbf{E}}{\partial \mathbf{W}_{ij}}, 0 \leq \alpha \leq 1$

Where α and E represents learning rate and Error in the output respectively.

ugcnetcse-dec2019-paper2

36.0.337 UGC NET CSE | December 2019 | Part 2 | Question: 62

A fuzzy conjunction operators, $t(x, y)$, and a fuzzy disjunction operator, $s(x, y)$, form a pair if they satisfy:

$$t(x, y) = 1 - s(1 - x, 1 - y).$$

If $t(x, y) = \frac{xy}{(x + y - xy)}$ then $s(x, y)$ is given by

A.

$$\frac{x + y}{1 - xy}$$

B.

$$\frac{x + y - 2xy}{1 - xy}$$

C.

$$\frac{x + y - xy}{1 - xy}$$

D. $\frac{x + y - xy}{1 + xy}$

ugcnetcse-dec2019-paper2

36.0.338 UGC NET CSE | December 2019 | Part 2 | Question: 61

Consider the following models:

M_1 : Mamdani model

M_2 : Takagi – Sugeno – Kang model

M_3 : Kosko's additive model (SAM)

Which of the following option contains examples of additive rule model?

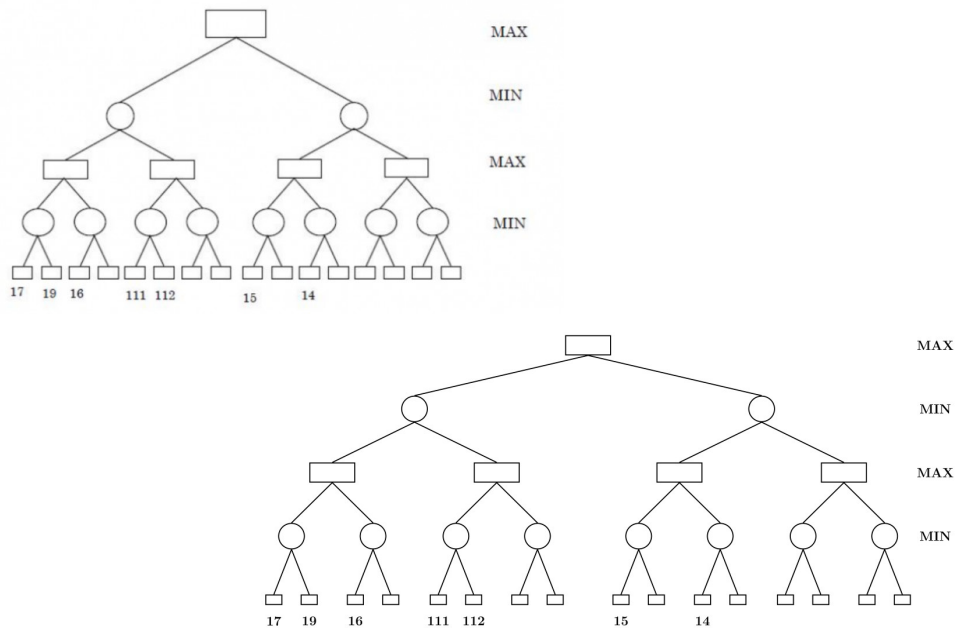
A. Only M_1 and M_2 B. Only M_2 and M_3 C. Only M_1 and M_3 D. M_1, M_2 and M_3

ugcnetcse-dec2019-paper2

Answer key 



Consider the game tree given below:



Here $\bigcirc$ and $\square$ represent MIN and MAX nodes respectively, The value of the root node of the game tree is:

- A. 14 B. 17 C. 111 D. 112

ugcnetcse-dec2019-paper2

Answer key



Consider the following statements:

$S_1 : \forall xP(x) \vee \forall xQ(x)$ and $\forall x(P(x) \vee Q(x))$ are not logically equivalent.

$S_2 : \exists xP(x) \wedge \exists xQ(x)$ and $\exists x(P(x) \wedge Q(x))$ are not logically equivalent

Which of the following statements is/are correct?

- A. Only S_1 B. Only S_2 C. Both S_1 and S_2 D. Neither S_1 nor S_2

ugcnetcse-dec2019-paper2



According to Dempster-Shafer theory for uncertainty management,

- A. $Bel(A) + Bel(\neg A) \leq 1$ B. $Bel(A) + Bel(\neg A) \geq 1$
C. $Bel(A) + Bel(\neg A) = 1$ D. $Bel(A) + Bel(\neg A) = 0$

Where $Bel(A)$ denotes Belief of event A .

ugcnetcse-dec2019-paper2

Answer key



The full form of ICANN is

- A. Internet Corporation for Assigned Names and Numbers
B. Internet Corporation for Assigned Numbers and Names
C. Institute of Cooperation for Assigned Names and Numbers
D. Internet Connection for Assigned Names and Numbers

Answer key **36.0.343 UGC NET CSE | December 2019 | Part 2 | Question: 56**

A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

- A. 1,000,000 bps B. 2,000,000 bps
C. 12,000,000 bps D. 1,200,00,000 bps

ugcnetcse-dec2019-paper2

Answer key **36.0.344 UGC NET CSE | December 2019 | Part 2 | Question: 55**

Piconet is a basic unit of a bluetooth system consisting of _____ master node and up to _____ active slave nodes.

- A. one, five B. one, seven C. two, eight D. one, eight

ugcnetcse-dec2019-paper2

Answer key **36.0.345 UGC NET CSE | December 2019 | Part 2 | Question: 54**

Consider a subnet with 720 routers. If a three-level hierarchy is chosen, with eight clusters, each containing 9 regions of 10 routers, then total number of entries in hierarchical table of each router is

- A. 25 B. 27 C. 53 D. 72

ugcnetcse-dec2019-paper2

Answer key **36.0.346 UGC NET CSE | December 2019 | Part 2 | Question: 53**

Which of the following class of IP address has the last address as 223.255.255.255?

- A. Class A B. Class B C. Class C D. Class D

ugcnetcse-dec2019-paper2

Answer key **36.0.347 UGC NET CSE | December 2019 | Part 2 | Question: 52**

Let $A = \{001, 0011, 11, 101\}$ and $B = \{01, 111, 111, 010\}$. Similarly, let $C = \{00, 001, 1000\}$ and $D = \{0, 11, 011\}$.

Which of the following pairs have a post-correspondence solution?

- A. Only pair (A, B) B. Only pair (C, D)
C. Both (A, B) and (C, D) D. Neither (A, B) nor (C, D)

ugcnetcse-dec2019-paper2

Answer key **36.0.348 UGC NET CSE | December 2019 | Part 2 | Question: 51**

Consider the following statements:

S_1 : There exists no algorithm for deciding if any two Turing machines M_1 and M_2 accept the same language

S_2 : Let M_1 and M_2 be arbitrary Turing machines. The problem to determine $L(M_1) \subseteq L(M_2)$ is undecidable

Which of the statements is (are) correct?

A. Only S_1

B. Only S_2

C. Both S_1 and S_2

D. Neither S_1 nor S_2

ugcnetcse-dec2019-paper2

Answer key 

36.0.349 UGC NET CSE | December 2019 | Part 2 | Question: 50



Consider the following language families:

$L_1 \equiv$ The context-free languages

$L_2 \equiv$ The context-sensitive languages

$L_3 \equiv$ The recursively enumerable languages

$L_4 \equiv$ The recursive languages

Which one of the following options is correct?

A. $L_1 \subseteq L_2 \subseteq L_3 \subseteq L_4$

B. $L_2 \subseteq L_1 \subseteq L_3 \subseteq L_4$

C. $L_1 \subseteq L_2 \subseteq L_4 \subseteq L_3$

D. $L_2 \subseteq L_1 \subseteq L_4 \subseteq L_3$

ugcnetcse-dec2019-paper2

Answer key 

36.0.350 UGC NET CSE | December 2019 | Part 2 | Question: 49



Let $G = (V, T, S, P)$ be any context-free grammar without any λ -productions or unit productions. Let K be the maximum number of symbols on the right of any production in P . The maximum number of production rules for any equivalent grammar in Chomsky normal form is given by:

A. $(K-1)|P| + |T| - 1$

B. $(K-1)|P| + |T|$

C. $K|P| + |T| - 1$

D. $K|P| + |T|$

Where $|\cdot|$ denotes the cardinality of the set.

ugcnetcse-dec2019-paper2

Answer key 

36.0.351 UGC NET CSE | December 2019 | Part 2 | Question: 47



Consider the following statements with respect to the language $L = \{a^n b^n \mid n \geq 0\}$

$S_1 : L^2$ is a context free language

$S_2 : L^k$ is context free language for any given $k \geq 1$

$S_3 : \bar{L}$ and L^* are context free languages

Which one of the following is correct?

A. only S_1 and S_2

B. only S_1 and S_3

C. only S_2 and S_3

D. S_1, S_2 and S_3

ugcnetcse-dec2019-paper2

Answer key 

36.0.352 UGC NET CSE | December 2019 | Part 2 | Question: 46



Consider $\Sigma = \{w, x\}$ and $T = \{x, y, z\}$. Define homomorphism h by:

$$h(x) = xzy$$

$$h(w) = zxyy$$

If L is the regular language denoted by $r = (w + x^*)(ww)^*$, then the regular language $h(L)$ is given by

- A. $(zxyy + xzy)(zxyy)$
 C. $(zxyy + xzy)(zxyy)^*$

- B. $(zxyy + (xzy)^*)(zxyy zxyy)^*$
 D. $(zxyy + (xzy)^*)(zxyy zxyy)$

ugcnetcse-dec2019-paper2

Answer key 

36.0.353 UGC NET CSE | December 2019 | Part 2 | Question: 45



Consider the following grammar:

- $S \rightarrow 0A \mid 0BB$
- $A \rightarrow 00A \mid \lambda$
- $B \rightarrow 1B \mid 11C$
- $C \rightarrow B$

Which language does this grammar generate?

- A. $L((00)^*0 + (11)^*1)$
 C. $L((00)^*0)$

- B. $L(0(11)^* + 1(00)^*)$
 D. $L(0(11)^*1)$

ugcnetcse-dec2019-paper2

Answer key 

36.0.354 UGC NET CSE | December 2019 | Part 2 | Question: 44



Consider the language $L = \{a^n b^{n-3} \mid n > 2\}$ on $\Sigma = \{a, b\}$. Which one of the following grammars generates the language L ?

- A. $S \rightarrow aA \mid a, A \rightarrow aAb \mid b$
 B. $S \rightarrow aaA \mid \lambda, A \rightarrow aAb \mid \lambda$
 C. $S \rightarrow aaaA \mid a, A \rightarrow aAb \mid \lambda$
 D. $S \rightarrow aaaA, A \rightarrow aAb \mid \lambda$

ugcnetcse-dec2019-paper2

Answer key 

36.0.355 UGC NET CSE | December 2019 | Part 2 | Question: 43



Consider the following grammars:

- $G_1 : S \rightarrow aSb \mid bSa \mid aa$
 $G_2 : S \rightarrow aSb \mid bSa \mid SS \mid \lambda$
 $G_3 : S \rightarrow aSb \mid bSa \mid SS \mid a$
 $G_4 : S \rightarrow aSb \mid bSa \mid SS \mid SSS \mid \lambda$

Which of the following is correct w.r.t. the above grammars?

- A. G_1 and G_3 are equivalent
 C. G_2 and G_4 are equivalent

- B. G_2 and G_3 are equivalent
 D. G_3 and G_4 are equivalent

ugcnetcse-dec2019-paper2

Answer key 

36.0.356 UGC NET CSE | December 2019 | Part 2 | Question: 42



The time complexity to multiply two polynomials of degree n using Fast Fourier transform method is:

- A. $\theta(n \lg n)$ B. $\theta(n^2)$ C. $\theta(n)$ D. $\theta(\lg n)$

ugcnetcse-dec2019-paper2

Answer key 

36.0.357 UGC NET CSE | December 2019 | Part 2 | Question: 41

When using Dijkstra's algorithm to find shortest path in a graph, which of the following statement is not true?

- A. It can find shortest path within the same graph data structure
- B. Every time a new node is visited, we choose the node with smallest known distance/ cost (weight) to visit first
- C. Shortest path always passes through least number of vertices
- D. The graph needs to have a non-negative weight on every edge

ugcnetcse-dec2019-paper2

Answer key

36.0.358 UGC NET CSE | December 2019 | Part 2 | Question: 40

Consider a weighted directed graph. The current shortest distance from source S to node x is represented by $d[x]$. Let $d[v] = 29$, $d[u] = 15$, $w[u, v] = 12$. What is the updated value of $d[v]$ based on current information?

- A. 29
- B. 27
- C. 25
- D. 17

ugcnetcse-dec2019-paper2

Answer key

36.0.359 UGC NET CSE | December 2019 | Part 2 | Question: 39

Give asymptotic upper and lower bound for $T(n)$ given below. Assume $T(n)$ is constant for $n \leq 2$.
 $T(n) = 4T(\sqrt{n}) + \lg^2 n$

- A. $T(n) = \theta(\lg(\lg^2 n) \lg n)$
- B. $T(n) = \theta(\lg^2 n \lg n)$
- C. $T(n) = \theta(\lg^2 n \lg \lg n)$
- D. $T(n) = \theta(\lg(\lg n) \lg n)$

ugcnetcse-dec2019-paper2

Answer key

36.0.360 UGC NET CSE | December 2019 | Part 2 | Question: 38

In a B-Tree, each node represents a disk block. Suppose one block holds 8192 bytes. Each key uses 32 bytes. In a B-tree of order M there are $M - 1$ keys. Since each branch is on another disk block, we assume a branch is of 4 bytes. The total memory requirement for a non-leaf node is

- A. $32M - 32$
- B. $36M - 32$
- C. $36M - 36$
- D. $32M - 36$

ugcnetcse-dec2019-paper2

Answer key

36.0.361 UGC NET CSE | December 2019 | Part 2 | Question: 37

What is the worst case running time of Insert and Extract-min, in an implementation of a priority queue using an unsorted array? Assume that all the insertions can be accommodated.

- A. $\theta(1), \theta(n)$
- B. $\theta(n), \theta(1)$
- C. $\theta(1), \theta(1)$
- D. $\theta(n), \theta(n)$

ugcnetcse-dec2019-paper2

Answer key

36.0.362 UGC NET CSE | December 2019 | Part 2 | Question: 36

A clique in an undirected graph $G = \langle V, E \rangle$ is a subset $V' \subseteq V$ of vertices, such that

- A. If $(u, v) \in E$ then $u \in V'$ and $v \in V'$
- B. If $(u, v) \in E$ then $u \in V'$ or $v \in V'$
- C. Each pair of vertices in V' is connected by an edge
- D. All pairs of vertices in V' are not connected by an edge

ugcnetcse-dec2019-paper2

Answer key 

36.0.363 UGC NET CSE | December 2019 | Part 2 | Question: 35



Let $a^{2^c} \bmod n = (a^c)^2 \bmod n$ and $a^{2^{c+1}} \bmod n = a \cdot (a^c)^2 \bmod n$. For $a = 7$, $b = 17$ and $n = 561$, What is the value of $a^b \bmod n$?

- A. 160
- B. 166
- C. 157
- D. 67

ugcnetcse-dec2019-paper2

Answer key 

36.0.364 UGC NET CSE | December 2019 | Part 2 | Question: 34



An _____ chart is a project schedule representation that presents project plan as a directed graph. The critical path is the _____ sequence of _____ tasks and it defines project _____.

- A. Activity, Shortest, Independent, Cost
- B. Activity, Longest, Dependent, Duration
- C. Activity, Longest, Independent, Duration
- D. Activity, Shortest, Dependent, Duration

ugcnetcse-dec2019-paper2

Answer key 

36.0.365 UGC NET CSE | December 2019 | Part 2 | Question: 33



In a system for a restaurant, the main scenario for placing order is given below:

- i. Customer reads menu
- ii. Customer places order
- iii. Order is sent to kitchen for preparation
- iv. Ordered items are served
- v. Customer requests for a bill for the order
- vi. Bill is prepared for this order
- vii. Customer is given the bill
- viii. Customer pays the bill

A sequence diagram for the scenario will have at least how many objects among whom the messages will be exchanged

- A. 3
- B. 4
- C. 5
- D. 6

ugcnetcse-dec2019-paper2

Answer key 

36.0.366 UGC NET CSE | December 2019 | Part 2 | Question: 32



Java Virtual Machine (JVM) is used to execute architectural neutral byte code. Which of the following is needed by the JVM for execution of Java Code?

- A. Class loader only
- B. Class loader and Java Interpreter
- C. Class loader, Java Interpreter and API
- D. Java Interpreter only

ugcnetcse-dec2019-paper2

Answer key 

36.0.367 UGC NET CSE | December 2019 | Part 2 | Question: 31

Which of the following methods are used to pass any number of parameters to the operating system through system calls?

- A. Registers
C. Stack
B. Block or table in main memory
D. Block in main memory and stack

ugcnetcse-dec2019-paper2

Answer key

36.0.368 UGC NET CSE | December 2019 | Part 2 | Question: 30

Given CPU time slice of 2ms and following list of processes.

Process	Burst time (ms)	Arrival time
p_1	3	0
p_2	4	2
p_3	5	5

Find average turnaround time and average waiting time using round robin CPU scheduling?

- A. 4,0
B. 5.66,1.66
C. 5.66,0
D. 7,2

ugcnetcse-dec2019-paper2

Answer key

36.0.369 UGC NET CSE | December 2019 | Part 2 | Question: 29

Which of the following interprocess communication model is used to exchange messages among co-operative processes?

- A. Shared memory model
C. Shared memory and message passing model
B. Message passing model
D. Queues

ugcnetcse-dec2019-paper2

Answer key

36.0.370 UGC NET CSE | December 2019 | Part 2 | Question: 28

Suppose a system has 12 magnetic tape drives and at time t_0 , three processes are allotted tape drives out of their need as given below:

	Maximum Needs	Current Needs
p_0	10	5
p_1	4	2
p_2	9	2

At time t_0 , the system is in safe state. Which of the following is safe sequence so that deadlock is avoided?

- A. $\langle p_0, p_1, p_2 \rangle$
B. $\langle p_1, p_0, p_2 \rangle$
C. $\langle p_2, p_1, p_0 \rangle$
D. $\langle p_0, p_2, p_1 \rangle$

ugcnetcse-dec2019-paper2

Answer key

36.0.371 UGC NET CSE | December 2019 | Part 2 | Question: 27

Consider a paging system where translation look aside buffer (TLB) a special type of associative memory is used with hit ratio of 80%.

Assume that memory reference takes 80 nanoseconds and reference time to TLB is 20 nanoseconds. What will be the effective memory access time given 80% hit ratio?

- A. 110 nanoseconds
- B. 116 nanoseconds
- C. 200 nanoseconds
- D. 100 nanoseconds

ugcnetcse-dec2019-paper2

Answer key

36.0.372 UGC NET CSE | December 2019 | Part 2 | Question: 26

Which of the following CPU scheduling algorithms is/are supported by LINUX operating system?

- A. Non-preemptive priority scheduling
- B. Preemptive priority scheduling and time sharing CPU scheduling
- C. Time sharing scheduling only
- D. Priority scheduling only

ugcnetcse-dec2019-paper2

Answer key

36.0.373 UGC NET CSE | December 2019 | Part 2 | Question: 25

Which of the following is not needed by an encryption algorithm used in Cryptography?

- A. KEY
- B. Message
- C. Ciphertext
- D. User details

ugcnetcse-dec2019-paper2

Answer key

36.0.374 UGC NET CSE | December 2019 | Part 2 | Question: 24

A counting semaphore is initialized to 8. 3 wait () operations and 4 signal () operations are applied. Find the current value of semaphore variable.

- A. 9
- B. 5
- C. 1
- D. 4

ugcnetcse-dec2019-paper2

Answer key

36.0.375 UGC NET CSE | December 2019 | Part 2 | Question: 21

Given two tables $R1(x, y)$ and $R2(y, z)$ with 50 and 30 number of tuples respectively. Find maximum number of tuples in the output of natural join between tables $R1$ and $R2$ i.e. $R1 * R2$? (*- Natural Join)

- A. 30
- B. 20
- C. 50
- D. 1500

ugcnetcse-dec2019-paper2

Answer key

36.0.376 UGC NET CSE | December 2019 | Part 2 | Question: 20

Which of the component module of DBMS does rearrangement and possible ordering of operations, eliminate redundancy in query and use efficient algorithms and indexes during the execution of a query?

- A. query compiler
- B. query optimizer
- C. Stored data manager
- D. Database processor

ugcnetcse-dec2019-paper2

Answer key

36.0.377 UGC NET CSE | December 2019 | Part 2 | Question: 19

If we want to resize a 1024×768 pixels image to one that is 640 pixels wide with the same aspect ratio, what would be the height of the resized image?

- A. 420 Pixels B. 460 Pixels C. 480 Pixels D. 540 Pixels

ugcnetcse-dec2019-paper2

Answer key

36.0.378 UGC NET CSE | December 2019 | Part 2 | Question: 18

Which of the following algorithms is not used for line clipping?

- A. Cohen-Sutherland algorithm B. Sutherland-Hodgeman algorithm
C. Liang-Barsky algorithm D. Nicholl-Lee-Nicholl algorithm

ugcnetcse-dec2019-paper2

36.0.379 UGC NET CSE | December 2019 | Part 2 | Question: 17

A rectangle is bound by the lines $x = 0; y = 0; x = 5$ and $y = 3$. The line segment joining $(-1, 0)$ and $(4, 5)$, if clipped against this window will connect the points _____.

- A. $(0, 1)$ and $(3, 3)$ B. $(0, 1)$ and $(2, 3)$ C. $(0, 1)$ and $(4, 5)$ D. $(0, 1)$ and $(3, 5)$

ugcnetcse-dec2019-paper2

Answer key

36.0.380 UGC NET CSE | December 2019 | Part 2 | Question: 16

Which tag is used to enclose any number of javascript statements in HTML document?

- A. `<code>` B. `<script>`
C. `<title>` D. `<body>`

ugcnetcse-dec2019-paper2

Answer key

36.0.381 UGC NET CSE | December 2019 | Part 2 | Question: 15

Let A be the base class in $C++$ and B be the derived class from A with protected inheritance. Which of the following statement is false for class B ?

- A. Member function of class B can access protected data of class A
B. Member function of class B can access public data of class A
C. Member function of class B cannot access private data of class A
D. Object of derived class B can access public base class data

ugcnetcse-dec2019-paper2

Answer key

36.0.382 UGC NET CSE | December 2019 | Part 2 | Question: 14

What is the output of the following C program?

```
#include <stdio.h>
main ()
{
    int i, j, x=0;
    for (i=0; i<5; ++i)
        for (j=0; j<i; ++j)
        {
            x+=(i+j-1);
            break;
        }
}
```

```
printf("%d", x);
}
```

- A. 6 B. 5 C. 4 D. 3

ugcnetcse-dec2019-paper2

Answer key 

36.0.383 UGC NET CSE | December 2019 | Part 2 | Question: 13



The following program is stored in the memory unit of the basic computer. Give the content of accumulator register in hexadecimal after the execution of the program.

Location	Instruction
010	CLA
011	ADD 016
012	BUN 014
013	HLT
014	AND 017
015	BUN 013
016	C1A5
017	93C6

- A. A1B4 B. 81B4 C. A184 D. 8184

ugcnetcse-dec2019-paper2

36.0.384 UGC NET CSE | December 2019 | Part 2 | Question: 12



Given following equation:

$$(142)_b + (112)_{b-2} = (75)_8, \text{ find base } b.$$

- A. 3 B. 6 C. 7 D. 5

ugcnetcse-dec2019-paper2

Answer key 

36.0.385 UGC NET CSE | December 2019 | Part 2 | Question: 11



A micro instruction format has microoperation field which is divided into 2 subfields F_1 and F_2 , each having 15 distinct microoperations, condition field CD for four status bits, branch field BR having four options used in conjunction with address field AD. The address space is of 128 memory words. The size of micro instruction is:

- A. 19 B. 18 C. 17 D. 20

ugcnetcse-dec2019-paper2

Answer key 

36.0.386 UGC NET CSE | December 2019 | Part 2 | Question: 10



A computer uses a memory unit of 512 K words of 32 bits each. A binary instruction code is stored in one word of the memory. The instruction has four parts: an addressing mode field to specify one of the two-addressing mode (direct and indirect), an operation code, a register code part to specify one of the 256 registers and an address part. How many bits are there in addressing mode part, opcode part, register code part and the address part?

- A. 1, 3, 9, 19 B. 1, 4, 9, 18 C. 1, 4, 8, 19 D. 1, 3, 8, 20

ugcnetcse-dec2019-paper2

Answer key 

36.0.387 UGC NET CSE | December 2019 | Part 2 | Question: 8

How many reflexive relations are there on a set with 4 elements?

- A. 2^4 B. 2^{12} C. 4^2 D. 2

ugcnetcse-dec2019-paper2

Answer key

36.0.388 UGC NET CSE | December 2019 | Part 2 | Question: 7

A tree has $2n$ vertices of degree 1, $3n$ vertices of degree 2, n vertices of degree 3. Determine the number of vertices and edges in tree.

- A. 12,11 B. 11,12 C. 10,11 D. 9,10

ugcnetcse-dec2019-paper2

Answer key

36.0.389 UGC NET CSE | December 2019 | Part 2 | Question: 5

Let P be the set of all people. Let R be a binary relation on P such that (a,b) is in R if a is a brother of b . Is R symmetric transitive, an equivalence relation, a partial order relation?

- A. NO, NO, NO, NO B. NO, NO, YES, NO
C. NO, YES, NO, NO D. NO, YES, YES, NO

ugcnetcse-dec2019-paper2

Answer key

36.0.390 UGC NET CSE | December 2019 | Part 2 | Question: 4

What are the greatest lower bound (GLB) and the least upper bound (LUB) of the sets $A = \{3, 9, 12\}$ and $B = \{1, 2, 4, 5, 10\}$ if they exist in poset $(z^+, /)$?

- A. $A(\text{GLB} - 3, \text{LUB} - 36)$; $B(\text{GLB} - 1, \text{LUB} - 20)$
B. $A(\text{GLB} - 3, \text{LUB} - 12)$; $B(\text{GLB} - 1, \text{LUB} - 10)$
C. $A(\text{GLB} - 1, \text{LUB} - 36)$; $B(\text{GLB} - 2, \text{LUB} - 20)$
D. $A(\text{GLB} - 1, \text{LUB} - 12)$; $B(\text{GLB} - 2, \text{LUB} - 10)$

ugcnetcse-dec2019-paper2

Answer key

36.0.391 UGC NET CSE | December 2019 | Part 2 | Question: 2

Consider the following Linear programming problem (LPP):

Maximize $z = x_1 + x_2$

Subject to the constraints:

$$x_1 + 2x_2 \leq 2000$$

$$x_1 + x_2 \leq 1500$$

$$x_2 \leq 600$$

$$\text{and } x_1, x_2 \geq 0$$

The solution of the above LPP is

- A. $x_1 = 750, x_2 = 750, z = 1500$ B. $x_1 = 500, x_2 = 1000, z = 1500$
C. $x_1 = 1000, x_2 = 500, z = 1500$ D. $x_1 = 900, x_2 = 600, z = 1500$

ugcnetcse-dec2019-paper2

Answer key

36.0.392 UGC NET CSE | December 2019 | Part 2 | Question: 1

A basic feasible solution of an $m \times n$ transportation problem is said to be non-degenerate, if basic feasible solution contains exactly _____ number of individual allocation in _____ positions.

- A. $m+n+1$, independent B. $m+n-1$, independent
C. $m+n-1$, appropriate D. $m-n+1$, independent

ugcnetcse-dec2019-paper2

Answer key

36.0.393 UGC NET CSE | June 2023 | Part 2: 100

IP datagram has arrived with following partial information in the header (in hexadecimal)

45000054000300002006.....

What is the header size?

- A. 10 bytes B. 20 bytes C. 30 bytes D. 40 bytes

ugcnetcse-june2023-paper2

36.0.394 UGC NET CSE | June 2023 | Part 2: 99

IP datagram has arrived with following partial information in the header (in hexadecimal)

45000054000300002006....

How many more routers can the packet travel to?

- A. 22 B. 26 C. 30 D. 32

ugcnetcse-june2023-paper2

36.0.395 UGC NET CSE | June 2023 | Part 2: 98

IP datagram has arrived with following partial information in the header (in hexadecimal)

45000054000300002006... ..

What is the protocol of the payload being carried by the packet?

- A. ICMP B. SCTP C. TCP protocol D. IGMP

ugcnetcse-june2023-paper2

36.0.396 UGC NET CSE | June 2023 | Part 2: 97

IP datagram has arrived with following partial information in the header (in hexadecimal)

45000054000300002006...

What is the efficiency of this datagram?

- A. 76.19% B. 80.50% C. 82.24% D. 85.45%

ugcnetcse-june2023-paper2

36.0.397 UGC NET CSE | June 2023 | Part 2: 96

IP datagram has arrived with following partial information in the header (in hexadecimal)

45000054000300002006....

What is the size of datagram?

- A. 64 bytes B. 74 bytes C. 84 bytes D. 104 bytes

36.0.398 UGC NET CSE | June 2023 | Part 2: 95

Consider the following program fragment that deals with a table T with 17 rows and 1024 columns, computing an average for each column and printing it to screen (i is row index and j is column index):

```
for j = [0 . . . 1023]{
temp = 0
for i = [0 . . . 16]
temp = temp + T[i][j]
print ( temp / 17.0 );}
```

$T[i][i]$ and $temp$ are 32 bit floating point values and memory is word addressable. The temporary variable $temp$ is kept in a processor register so access to $temp$ does not involve a memory reference. The main memory is page and holds 16 pages of size 1024 words, the page replacement policy is "least recently used", If T is stored in the virtual address space in row major format.

What is the main memory hit ratio?

- A. 0 B. 1 C. 2 D. 3

ugcnetcse-june2023-paper2

36.0.399 UGC NET CSE | June 2023 | Part 2: 94

Consider the following program fragment that deals with a table T with 17 rows and 1024 columns, computing an average for each column and printing it to screen (i is row index and j is column index):

```
for j = 0 . . 1023){
temp = 0;
for i = [0 . . . 16];
temp = temp + T[i][j]
print ( temp / 17.0); }
```

$T[i][i]$ and $temp$ are 32 bit floating point values and memory is word addressable. The temporary variable $temp$ is kept in a processor register so access to $temp$ does not involve a memory reference. The main memory is page and holds 16 pages of size 1024 words, the page replacement policy is "least recently used", If T is stored in the virtual address space in row major format.

Consider that T is stored in column major format, how many page faults will be encountered?

- A. 14 B. 15 C. 16 D. 17

ugcnetcse-june2023-paper2

36.0.400 UGC NET CSE | June 2023 | Part 2: 93

Consider the following program fragment that deals with a table T with 17 rows and 1024 columns, computing an average for each column and printing it to screen (i is row index and j is column index):

```
for j = [0 . . . 1023]{
temp = 0
for i = [0 . . . 16]
temp = temp + T[i][j]
print ( temp / 17.0); }
```

$T[i][j]$ and $temp$ are 32 bit floating point values and memory is word addressable. The temporary variable $temp$ is kept in a processor register so access to $temp$ does not involve a memory reference. The main memory is page and holds 16 pages of size 1024 words, the page replacement policy is "least recently used", If T is stored in the virtual address space in row major format.

How many page faults will be encountered?

A. 16,402

B. 17,408

C. 18,208

D. 18,608

ugcnetcse-june2023-paper2

36.0.401 UGC NET CSE | June 2023 | Part 2: 92

Consider the following program fragment that deals with a table T with 17 rows and 1024 columns, computing an average for each column and printing it to screen (i is row index and j is column index):

```
for j = [0...1023]{
temp = 0
for i = [0...16];
temp = temp + T[i][j]
print ( temp/ 17.0) ; }
```

$T[i][j]$ and $temp$ are 32 bit floating point values and memory is word addressable. The temporary variable $temp$ is kept in a processor register so access to $temp$ does not involve a memory reference. The main memory is page and holds 16 pages of size 1024 words, the page replacement policy is "least recently used", If T is stored in the virtual address space in row major format.

Consider again that T is stored in column-major format, what is the main memory hit ratio?

A. 80%

B. 95.6%

C. 97.8%

D. 99.9%

ugcnetcse-june2023-paper2

36.0.402 UGC NET CSE | June 2023 | Part 2: 91

Consider the following program fragment that deals with a table T with 17 rows and 1024 columns, computing an average for each column and printing it to screen (i is row index and j is column index):

```
for j = [0...1023]{
temp = 0
for i = [0...16]
temp = temp + T[i][j]
print ( temp/ 17.0);
```

$T[i][j]$ and $temp$ are 32 bit floating point values and memory is word addressable. The temporary variable $temp$ is kept in a processor register so access to $temp$ does not involve a memory reference. The main memory is page and holds 16 pages of size 1024 words, the page replacement policy is "least recently used", If T is stored in the virtual address space in row major format.

What is fault ratio of row major to column major arrangements ?

A. 1024 : 1

B. 1301 : 1

C. 1240 : 1

D. 9107 : 8

ugcnetcse-june2023-paper2

36.0.403 UGC NET CSE | June 2023 | Part 2: 90

The compiler for high level language that runs on one machine and produces code for other machine is called-

A. Cross compiler

B. Multipass compiler

C. Optimizing Compiler

D. One pass Compiler

ugcnetcse-june2023-paper2

Answer key



The maximum yield length of the following CNF CFG is

$S \rightarrow AB$

$A \rightarrow CD$

$B \rightarrow e$

$C \rightarrow a$

$D \rightarrow b$

A. 8

B. 7

C. 4

D. 5

ugcnetcse-june2023-paper2



Southerland Hodgeman method is used on Southerland Hodgeman method is used on

A. Smooth curves

B. Line segment

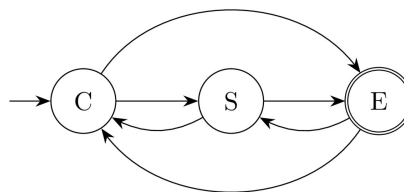
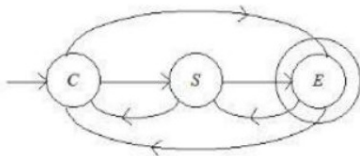
C. Convex polygons

D. Concave polygons

ugcnetcse-june2023-paper2



Consider the following finite automata F1 that accepts a language L



Let F2 be a finite automata which is obtained by reversal of F1. Then which of the following is correct?

A. $L(F1) \neq L(F2)$

B. $L(F1) = L(F2)$

C. $L(F1) \leq L(F2)$

D. $L(F1) \geq L(F2)$

ugcnetcse-june2023-paper2



Find the sum of all four digit numbers formed using the digits 1, 2, 4 and 6.

A. 86,658

B. 88,8858

C. 91,958

D. 93,358

ugcnetcse-june2023-paper2

36.0.408 UGC NET CSE | June 2023 | Part 2: 85

What is the output of following code?

```
main ()
{ static float a[]={13.24,1.5,4.5,5.4,3.5}
float * j, * k;
j= a};
k= a + 4
j=j * 2;
k = k / 2;
printf ("% f % f", * j,*k);
}
```

A. 13.25,4.5

B. 1.5,3.5

C. 13.24,1.5,4.5,5.4,3.5

D. Illegal use of pointer in main function

ugcnetcse-june2023-paper2

36.0.409 UGC NET CSE | June 2023 | Part 2: 84

Consider the following statements:

- i. Dynamic metrics are collected by measurements made of a program in execution
- ii. Static metrics are collected by measurements made of representations of the system
- iii. The assessment of software quality is an objective process
- iv. An important part of quality assurance in the selection of standards that should apply to the software development process.

Choose the correct answer from the options given below:

A. i, ii, iii only

B. ii, iii and iv only

C. i, iii and iv only

D. i, ii and iv only

ugcnetcse-june2023-paper2

36.0.410 UGC NET CSE | June 2023 | Part 2: 83

Let $R = \{x \mid x \in N, x \text{ is multiple of } 3 \text{ and } x \leq 100\}$ and $S = \{x, x \in N, x \text{ is a multiple of } 5 \text{ and } x < 100\}$. What is the number of elements in $(R \cap S) \times (S \cap R)$?

A. 36

B. 33

C. 20

D. 6

ugcnetcse-june2023-paper2

Answer key

36.0.411 UGC NET CSE | June 2023 | Part 2: 82

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: The AVL trees are more balanced as compared to Red Black trees, but they may cause more rotations during insertion and deletion

Reason R: A Red Black tree with n nodes has height that is greater than $2\log_2(n+1)$ and the AVL tree with n nodes has height less than $\log_\Phi(\sqrt{5}(n+2)) - 2$ (where Φ is golden ratio)

In the light of the above statements, choose the correct answer from the options given below.

- A. Both A and R are correct and R is the correct explanation of A
- B. Both A and R are correct and R is NOT the correct explanation of A
- C. A is true but R is false
- D. A is false but R is true

ugcnetcse-june2023-paper2

36.0.412 UGC NET CSE | June 2023 | Part 2: 81



Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Dendral is an expert system

Reason R: The rationality of an agent is not related to its reaction to the environment.

In the light of the above statements, choose the correct answer from the options given below.

- A. Both A and R are true and R is the correct explanation of A
- B. Both A and R are true but R is NOT the correct explanation of A
- C. A is true but R is false
- D. A is false but R is true

ugcnetcse-june2023-paper2

36.0.413 UGC NET CSE | June 2023 | Part 2: 80



An observational technique that can be used to understand operational process and help to derive requirement for software to support operational process is known as

- A. Requirement specification
- B. Structural specification
- C. Ethnography
- D. Natural language specification

ugcnetcse-june2023-paper2

36.0.414 UGC NET CSE | June 2023 | Part 2: 79



Consider following statements:

- A. A context free language is generated by LR(o) grammar if and only if it is accepted by a deterministic pushdown automata and has prefix property
- B. If M1 is the single tape TM simulating multilape TM M, then time taken by M1 to simulate n moves is (n^3)
- C. Push down automata behaves like a Turning machine when it has one auxiliary memory.
- D. $L = \{a^n b^n c^n : n \geq 1\}$ is not context free but context sensitive.

Choose the correct answer from the options given below:

- A. A, B and C only
- B. A, B only
- C. C, D only
- D. B, C only

ugcnetcse-june2023-paper2

36.0.415 UGC NET CSE | June 2023 | Part 2: 78



Match List I with List II

List I	List II
A. $A \Delta B$	I. $(A - B) \cup (A - C)$

B. $A - (B \cup C)$	II. $(A - B) \cap (A - C)$
C. $A - (B \cap C)$	III. $(A - B) \cup (B - A)$
D. $A \cap (B - C)$	IV. $(A \cap B) - (A \cap C)$

Choose the correct answer from the options given below: Choose the correct answer from the options given below:

- A. A-III B-II C-I D-IV
C. A-IV B-III C-I D-II

- B. A-II B-III C-IV D-I
D. A-IV B-I C-III D-II

ugcnetcse-june2023-paper2

Answer key 

36.0.416 UGC NET CSE | June 2023 | Part 2: 77



Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R .

Assertion A : I/O protection is ensured by a hardware trap

Reason R : I/O interrupt caused by the condition like I/O completion and device malfunction occurring within the I/O devices

In the light of the above statements, choose the correct answer from the options given below.

- A. Both A and R are true and R is the correct explanation of A
B. Both A and R are true but R is NOT the correct explanation of A
C. A is true but R is false
D. A is false but R is true

ugcnetcse-june2023-paper2

36.0.417 UGC NET CSE | June 2023 | Part 2: 76



Match List I with List II

List I	List II
A. Expert system	I. Decision tree
B. Fuzzy system	II. Scramble
C. Operator in genetic algorithm	III. Inference engine
D. Supervised technique	IV. Mycin

Choose the correct answer from the options given below: Choose the correct answer from the options given below:

- A. A-IV B-I C-III D-II
B. A-III B-IV C-II D-I
C. A-IV B-III C-II D-I
D. A-I B-II C-III D-IV

ugcnetcse-june2023-paper2

36.0.418 UGC NET CSE | June 2023 | Part 2: 75



What will be the output of the following code? What will be the output of the following code?

```
#include <stdio.h>
int main ( ) {
```

```
int a, b, c;
a=0x10; b=010;
c=a+b;
printf ( "%d", c);
return 0 ;
}
```

- A. 20 B. 24 C. Garbage D. error

ugcnetcse-june2023-paper2

Answer key 

36.0.419 UGC NET CSE | June 2023 | Part 2: 74



- i. The set of turning machine codes for TM's that accept all inputs that are palindromes (possible along with some other inputs) is decidable
- ii. The language of codes for TM's M that when started with blank tape, eventually write a 1 somewhere on the tape is undecidable
- iii. The language accepted by a TM M is $L(M)$ is always recursive
- iv. Post's correspondence problem is undecidable

Choose the correct answer from the options given below:

- A. i, ii and iii only B. ii, iii and iv only C. i and iii only D. ii and iv only

ugcnetcse-june2023-paper2

36.0.420 UGC NET CSE | June 2023 | Part 2: 73



Match **List I** with **List II**

List I	List II
A. Physical layer	I. Transforming the raw bits in the form of frame for transmission
B. Data Link Layer	II. Control and monitoring of subnet
C. Network layer	III. Transmission of raw bits over communication channel
D. Transport layer	IV. Datagrams transmission data through connection oriented or connectionless using datagram

Choose the correct answer from the options given below:

- A. A-III B-II C-I D-IV
 B. A-II B-III C-I D-IV
 C. A-III B-I C-II D-IV
 D. A-II B-IV C-I D-III

ugcnetcse-june2023-paper2

Answer key 

36.0.421 UGC NET CSE | June 2023 | Part 2: 72

Let N denote the set of all natural numbers and R be the relation on $N \times N$ defined by $(a, b)R(c, d)$, if $ad(b + c) = bc(a + d)$. Then R is

- A. Symmetric only
C. Transitive only
B. Reflexive only
D. An equivalence relation

ugcnetcse-june2023-paper2

36.0.422 UGC NET CSE | June 2023 | Part 2: 71

Given the basic ER diagram and relational model, which of the the following is incorrect?

- A. An attribute of an entity can have more than one value
B. An attribute of an entity can be composite
C. In a row of relational table, an attribute can have more than one value
D. In a row of a relational table, an attribute can have exactly one value or a NULL value

ugcnetcse-june2023-paper2

Answer key

36.0.423 UGC NET CSE | June 2023 | Part 2: 70

Which of the following is used to determine the cost performance index?

- A. Budgeted cost of work performed-budget at completion
B. Budgeted cost of work performed $\div$ budget at completion
C. Budgeted cost of work performed $\div$ Actual cost of work performed
D. Budgeted cost of work performed-Actual cost of work performed

ugcnetcse-june2023-paper2

36.0.424 UGC NET CSE | June 2023 | Part 2: 69

Let $(Z, +)$ denote the group of all integers under addition. Then the number of all automorphisms of $(Z, +)$ is

- A. 1
B. 2
C. 3
D. 4

ugcnetcse-june2023-paper2

36.0.425 UGC NET CSE | June 2023 | Part 2: 68

At a particular time of computation, the value of a counting semaphore is 7. Then 20p operations and 'x' V operations were completed on this semaphore. If the final value of semaphore is 5. x will be

- A. 15
B. 22
C. 18
D. 13

ugcnetcse-june2023-paper2

36.0.426 UGC NET CSE | June 2023 | Part 2: 67

Match **List I** with **List II**

List I	List II
A. Parallel FFT	I. $\theta(n^2)$
B. Iterative FFT	II. $\theta(n)$

C. Evaluation of polynomial at n points by Horner method	III. $\theta(\lg n)$
D. Product of two polynomials that are represented in point value form	IV. $\theta(n \lg n)$

Choose the correct answer from the options given below:

- A. A-III B-I C-II D-III
 B. A-II B-I C-III D-IV
 C. A-III B-IV C-I D-II
 D. A-II B-III C-IV D-I

ugcnetcse-june2023-paper2

36.0.427 UGC NET CSE | June 2023 | Part 2: 66



Which of the following transforms in 2 dimension is used to resize a 2-dimensional object?

- A. Translation B. Rotation C. Scaling D. Shearing

ugcnetcse-june2023-paper2

36.0.428 UGC NET CSE | June 2023 | Part 2: 65



let $R(A, B, C, D)$ be a relational schema with following function dependencies:

$A \rightarrow B, B \rightarrow C$

$C \rightarrow D$ and $D \rightarrow B$

The decomposition of R into

$(A, B) (B, C) (B, D)$

- A. gives a lossless join, and is dependency preserving
 B. gives lossless join, but is not dependency preserving
 C. does not give a lossless join, but is dependency preserving
 D. does not give a lossless join and is not dependency preserving

ugcnetcse-june2023-paper2

36.0.429 UGC NET CSE | June 2023 | Part 2: 64



Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Validity checks real need of system users

Reason R: Completeness checks system user defined requirements.

In the light of the above statements, choose the correct answer from the options given below.

- A. Both A and R are true and R is the correct explanation of A
 B. Both A and R are true but R is NOT the correct explanation of A
 C. A is true but R is false
 D. A is false but R is true

ugcnetcse-june2023-paper2



Match **List I** with **List II**

List I	List II
A. RAID level 2	I. Bit interleaved parity
B. RAID level 4	II. Block interleaved distributed parity
C. RAID level 5	III. Error correcting parity
D. RAID level 3	IV. Block interleaved parity

Choose the correct answer from the options given below: Choose the correct answer from the options given below:

- A. A-IV B-III C-I D-II
 B. A-III B-IV C-II D-I
 C. A-III B-I C-II D-IV
 D. A-I B-III C-IV D-I

ugcnetcse-june2023-paper2



If $A = \{4n + 2 \mid n \text{ is a natural number}\}$ and $B = \{3n \mid n \text{ is a natural number}\}$. Which of the following is correct for $A \cap B$?

- A. $\{12n^2 + 6n \mid n \text{ is a natural number}\}$
 B. $\{24n - 12 \mid n \text{ is a natural number}\}$
 C. $\{60n + 30 \mid n \text{ is a natural number}\}$
 D. $\{12n - 6 \mid n \text{ is a natural number}\}$

ugcnetcse-june2023-paper2

Answer key



The following table shows the time between failures for a software :

Error number	1	2	3	4	5
time since last failure (Hours)	6	4	8	5	6

The reliability of the system for one hour operation assuming an exponential model is-

- A. $e^{-9/29}$ B. $e^{-7/29}$ C. $e^{-5/29}$ D. $e^{-3/29}$

ugcnetcse-june2023-paper2



A B-tree used as an index for a large database table has four levels including the root node. If a new key is inserted in this index, then maximum number of nodes that could be newly created in the process is

- A. 5 B. 4 C. 3 D. 2

36.0.434 UGC NET CSE | June 2023 | Part 2: 59

Given below are two statements:

Which of the following concurrency control protocol ensures both conflict serializability and freedom from deadlock?

Statement I: Two phase locking

Statement II: Timestamp ordering

In the light of the above statements, choose the most appropriate answer from the options given below:

- A. Both Statement I and Statement II are correct
- B. Both Statement I and Statement II are incorrect
- C. Statement I is correct but Statement II is incorrect
- D. Statement I is incorrect but Statement II is correct

ugcnetcse-june2023-paper2

Answer key

36.0.435 UGC NET CSE | June 2023 | Part 2: 58

Suppose a circular queue of capacity $(n - 1)$ elements is implemented with an array of n elements. Assume that the insertion and deletion operations are carried out using REAR and FRONT as array index variable respectively. Initially,

REAR = FRONT = 0. The conditions to detect queue empty and queue full are

1. EMPTY: REAR == FRONT
FULL : $(\text{REAR} + 1) \bmod n == \text{FRONT}$
2. EMPTY: $(\text{FRONT} + 1) \bmod n == \text{REAR}$
FULL: $\text{REAR} + 1 \bmod n == \text{FRONT}$
3. EMPTY $\text{REAR} + 1 \bmod n == \text{FRONT}$
FULL: REAR == FRONT
4. EMPTY: REAR == FRONT
FULL: $(\text{FRONT} + 1) \bmod n == \text{REAR}$

ugcnetcse-june2023-paper2

36.0.436 UGC NET CSE | June 2023 | Part 2: 57

Match **List I** with **List II**

List I	List II
A. The running time of straight forward recursive method to compute nth Fibonacci number F_n	I. $O(n^2)$
B. The running time to compute F_n using memoization	II. $O(\lg n)$
C. The running time to compute Fibonacci number F_n using only integer addition and multiplication	III. $O(n)$
D. The running time to determine an optimal bitonic tour	IV. $\Theta(\phi^n)$

- A. A-I B-III C-IV D-II
 B. A-IV B-III C-II D-I
 C. A-I B-II C-IV D-III
 D. A-IV B-II C-III D-I

ugcnetcse-june2023-paper2

36.0.437 UGC NET CSE | June 2023 | Part 2: 56



Match **List I** with **List II**

List I	List II
A. CDMA	I. It provides mobile internet connection with faster data transfer rates
B. GSM	II. It allows user to connect to a network or to other devices over wireless channel
C. UMTS	III. Accessing mechanism for multiple transmitters over a single channel
D. WiFi	IV. It is a cellular technology that employs hybrid of FDMA and TDMA

- A. A-III B-IV C-II D-I
 B. A-III B-IV C-I D-II
 C. A-II B-III C-IV D-I
 D. A-II B-I C-IV D-III

ugcnetcse-june2023-paper2

Answer key

36.0.438 UGC NET CSE | June 2023 | Part 2: 55



Given below are two statements: one is labelled as Assertion **A** and the other is labelled as Reason **R**. A virtual memory system uses first-in first-out page replacement policy and allocates a fixed number of frames to a process

Assertion **A**: Increasing number of page frames allocated to a process sometimes increases the page fault rate.

Reason **R**: Some programs do not exhibit locality of reference.

In the light of the above statements, choose the correct answer from the options given below.

- A. Both **A** and **R** are true and **R** is the correct explanation of **A**
 B. Both **A** and **R** are true but **R** is NOT the correct explanation of **A**
 C. **A** is true but **R** is false
 D. **A** is false but **R** is true

ugcnetcse-june2023-paper2



Consider the following statements about the software product line system:

Statement I: At the interaction level, components provide an operator display interface and an interface with the communication system used.

Statement II: At the I/O management level, components handle operator authentication, report generator and query manager.

In the light of the above statements, choose the most appropriate answer from the options given below.

- A. Both Statement I and Statement II are correct
- B. Both Statement I and Statement II are incorrect
- C. Statement I is correct but Statement II is incorrect
- D. Statement I is incorrect but Statement II is correct

ugcnetcse-june2023-paper2



Match List I with List II

List I	List II
A. Critical region	I. Hoare's monitor
B. Wait/signal	II. Mutual exclusion
C. Working set	III. Principle of locality
D. Deadlock	IV. Circular wait

- A. A-IV B-I C-III D-II
- B. A-II B-I C-III D-IV
- C. A-I B-II C-III D-IV
- D. A-IV B-III C-I D-II

ugcnetcse-june2023-paper2



Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: A process involves a library function to create a thread.

Reason R: The threads make system calls to convey their resource and I/O requirement to the Kernel.

In the light of the above statements, choose the correct answer from the options given below.

- A. Both A and R are true and R is the correct explanation of A
- B. Both A and R are true but R is NOT the correct explanation of A
- C. A is true but R is false
- D. A is false but R is true

ugcnetcse-june2023-paper2

Answer key



Given below are two statements:

Statement I: subsystem models show logical grouping of objects into coherent subsystem

Statement II: State machine models show how objects change their states in response to events.

In the light of the above statements, choose the most appropriate answer from the options given below.

- A. Both Statement I and Statement II are correct
- B. Both Statement I and Statement II are incorrect
- C. Statement I is correct but Statement II is incorrect
- D. Statement I is incorrect but Statement II is correct

ugcnetcse-june2023-paper2

36.0.443 UGC NET CSE | June 2023 | Part 2: 50



The clipping process in computer graphics is used for

- A. Adding graphics
- B. Copying
- C. Zooming
- D. Removing objects and lines

ugcnetcse-june2023-paper2

36.0.444 UGC NET CSE | June 2023 | Part 2: 49



Consider the following language :

$$L = \{w \in \{a, b, c\}^* : n_a(w) + n_b(w) = n_c(w)\}$$

L is

- A. Context free but not linear
- B. Not context free
- C. Context free and linear
- D. Linear

ugcnetcse-june2023-paper2

36.0.445 UGC NET CSE | June 2023 | Part 2: 48



Consider the following statements:

- i. A database design is in BCNF if each member of the set of relation schemas that constitutes the design is in BCNF
- ii. A BCNF schema can have transitive dependency
- iii. It is always possible to obtain a 3NF design without sacrificing a lossless join

There are multivalued dependencies in 4NF

- A. i, ii and iii only
- B. ii, iii and iv only
- C. i, ii and iv only
- D. i, iii and iv only

ugcnetcse-june2023-paper2

36.0.446 UGC NET CSE | June 2023 | Part 2: 47



- i. If some NP-complete problem P is in $\mathbb{P}$ that $\mathbb{P} = \mathbb{N}$
- ii. TSP is in $\mathbb{NP}$
- iii. SAT is in $\mathbb{NP}$
- iv. Hamilton circuit problem is not NP-complete

Choose the correct answer from the options given below:

- A. i, ii and iii only
- B. ii, iii and iv only
- C. iii, iv and i only
- D. iv, i and ii only

ugcnetcse-june2023-paper2

36.0.447 UGC NET CSE | June 2023 | Part 2: 46

Consider a popular sports news site. At a given moment, 20,000 concurrent users submit a request (a transaction T) once every 2 minutes on average. Each transaction requires the webapp to download a new article that on average has 3k bytes in length. What is the throughput?

- A. 8 megabits per second
C. 6 megabits per second
- B. 4 megabits per second
D. 2 megabits per second

ugcnetcse-june2023-paper2

36.0.448 UGC NET CSE | June 2023 | Part 2: 45

What is the $(4 + 4)$ bit binary fixed point equivalent of $-(3.72)_{10}$?

- A. 0011.1100
B. 0011.1010
C. 1100.0100
D. 0011.1011

ugcnetcse-june2023-paper2

Answer key **36.0.449 UGC NET CSE | June 2023 | Part 2: 43**

Let $f(n)$ and $g(n)$ be asymptotically positive functions. The following conjectures are given

Statement I: $f(n) \geq \text{land } f(n) = O(g(n)) \Rightarrow g(n) = \Omega(f(n))$

Statement II: $f(n) = O(g(n)) \Rightarrow \lg(f(n)) = O(\lg(g(n)))$ where $\lg(g(n)) \geq 1$ for all sufficient large n .

In the light of the above statements, choose the most appropriate answer from the options given below.

- A. Both Statement I and Statement II are correct
B. Both Statement I and Statement II are incorrect
C. Statement I is correct but Statement II is incorrect
D. Statement I is incorrect but Statement II is correct

ugcnetcse-june2023-paper2

36.0.450 UGC NET CSE | June 2023 | Part 2: 42

Given below are two statements:

Which of the following statement/s is/are correct with respect to virtual memory

Statement I: Address translation is performed for every logical address used during the execution of a program

Statement II: A program can execute only when all of its components are loaded in the memory

In the light of the above statements, choose the most appropriate answer from the options given below.

- A. Both Statement I and Statement II are correct
B. Both Statement I and Statement II are incorrect
C. Statement I is correct but Statement II is incorrect
D. Statement I is incorrect but Statement II is correct

ugcnetcse-june2023-paper2

36.0.451 UGC NET CSE | June 2023 | Part 2: 41

How many integral solutions are there to $x + y + z + w = 29$, where $x \geq 1, y \geq 2, z \geq 3$ and $w \geq 0$?

- A. 2400
B. 2600
C. 2800
D. 3000

ugcnetcse-june2023-paper2

36.0.452 UGC NET CSE | June 2023 | Part 2: 40

Given below are two statements:

Statement I: If f and g are two functions and $f = O(g)$ but $g \neq o(f)$, we say that the growth rate of g is smaller than that of f

Statement II: The class of all decision problems decided by a TM in exponential time, that is $O(2^k)$, k being a constant.

In the light of the above statements, choose the most appropriate answer from the options given below.

- A. Both Statement I and Statement II are correct
- B. Both Statement I and Statement II are incorrect
- C. Statement I is correct but Statement II is incorrect
- D. Statement I is incorrect but Statement II is correct

ugcnetcse-june2023-paper2

36.0.453 UGC NET CSE | June 2023 | Part 2: 39



The total cost of retrieving records in sorted order using an unclustered B+ tree is (P-Average number of records per data page

N- Data pages

F- Ratio of the size of a data entry to the size of a data record)

- A. $(F * N) + P$
- B. $(F + P) * N$
- C. $F * N * P$
- D. $F + P/N$

ugcnetcse-june2023-paper2

36.0.454 UGC NET CSE | June 2023 | Part 2: 38



Which is not the component of the natural language understanding process?

- A. Morphological analysis
- B. Semantic analysis
- C. Pragmatic analysis
- D. Meaning analysis

ugcnetcse-june2023-paper2

36.0.455 UGC NET CSE | June 2023 | Part 2: 37



Match **List I** with **List II**

List I	List II
A. Scenario testing	I. To verify the I/O behavior of text object
B. Regression testing	II. user acceptance methodology
C. Component testing	III. No new bugs after changes in program
D. Beta testing	IV. The documentation of a use case

Choose the correct answer from the options given below:

- 1. A-IV B-III C-II D-I
- 2. A-II B-I C-III D-IV
- 3. A-IV B-III C-I D-II
- 4. A-III B-I C-IV D-II

ugcnetcse-june2023-paper2

Answer key

36.0.456 UGC NET CSE | June 2023 | Part 2: 36

Which of the following is not a property of a good system for representation of knowledge in a particular domain?

- A. Presentation adequacy
- B. inferential adequacy
- C. Inferential efficiency
- D. acquisitional efficiency

ugcnetcse-june2023-paper2

36.0.457 UGC NET CSE | June 2023 | Part 2: 35

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: It is possible to create doubly linked list using only one pointer with every node.

Reason R: By storing the XOR of the addresses of the previous and next nodes.

In the light of the above statements, choose the most appropriate answer from the options given below.

- A. Both A and R are true and R is the correct explanation of A
- B. Both A and R are true but R is NOT the correct explanation of A
- C. A is true but R is false
- D. A is false but R is true

ugcnetcse-june2023-paper2

36.0.458 UGC NET CSE | June 2023 | Part 2: 34

What is x in the following program?

```
# include < stdioh >
int main ( )
{typedef (*( * arrfptr[3]) ()) [10];
arrfptr x ;
return 0 ;
}
```

- A. x is a pointer
- B. x is a array of three pointer
- C. x is an array of three function pointer
- D. Error in x declaration

ugcnetcse-june2023-paper2

36.0.459 UGC NET CSE | June 2023 | Part 2: 33

Match List I with List II

List I	List II
A. PERT	I. used for nonrepetitive jobs
B. Optimistic time	II. Used for repetitive jobs
C. CPM	III. The shortest possible time to complete the activity if all goes well.
D. Pessimistic time	IV. The longest time that an activity takes

Choose the correct answer from the options given below:

- A. A-III B-IV C-II D-I
- B. A-IV B-I C-I D-III
- C. A-I B-III C-II D-III
- D. A-I B-III C-II D-IV

36.0.460 UGC NET CSE | June 2023 | Part 2: 32

Consider the rectangle with vertices $(0,0)$, $(0,2)$, $(3,0)$, $(3,2)$. There is scaling of 2 towards x-axis and 3 towards y-axis. The new coordinates of the rectangle are

- A. $(0,0)$, $(6,0)$, $(0,4)$, $(6,4)$ B. $(0,0)$, $(6,0)$, $(0,4)$, $(3,2)$
 C. $(0,0)$, $(6,0)$, $(0,6)$, $(6,6)$ D. $(0,0)$, $(4,0)$, $(0,6)$, $(4,6)$

ugcnetcse-june2023-paper2

36.0.461 UGC NET CSE | June 2023 | Part 2: 31

Let $G = \frac{R}{\{0\}}$ and $H = \{-1, 1\}$ be groups under the multiplication. Then, the map $\phi : G \rightarrow H$ defined by $\phi(x) = \frac{x}{|x|}$ is

- A. Not a homomorphism B. A one-one homomorphism, which is not onto
 C. An onto homomorphism, which is not one to one D. An homomorphism

ugcnetcse-june2023-paper2

36.0.462 UGC NET CSE | June 2023 | Part 2: 30

How will you free the memory allocated by the following program?

```
#include <stdio.h>
#include <stdlib.h>
#define MAXROW 3
#define MAXCOL 4
int main ( )
{
    int ** p, i, j;
    p= (int **) malloc (MAXROW * size of (int*));
    return 0 ;$
}
# include <stdio.h>
#include <stdlib.h>
#define MAXROW 3
#define MAXCOL 4
int main ( )
{
    int ** p $, \mathrm{i}, \mathrm{j}$;
    p=(int **) malloc (MAXROW * size of (int*));
    return 0;
}
```

- A. `memfree (int p);` B. `dealloc (p);`
 C. `malloc (p,0);` D. `free (p);`

ugcnetcse-june2023-paper2

36.0.463 UGC NET CSE | June 2023 | Part 2: 29

Which of the following is not a solution representation in a genetic algorithm?

- A. Binary valued B. Real valued C. Permutation D. Combinations

ugcnetcse-june2023-paper2

36.0.464 UGC NET CSE | June 2023 | Part 2: 28

Consider the following statements:

S1: LRU page replacement algorithm suffers from the belady's anomaly

S2: Shortest remaining time first scheduling may cause starvations

S3: Stack is shared by all threads in a process

- A. S1 S2 and S3 are true
- B. S1, S3 false and S2 is true
- C. S1, S2 are false and S3 is true
- D. S1, S2 and S3 are false

ugcnetcse-june2023-paper2

Answer key 

36.0.465 UGC NET CSE | June 2023 | Part 2: 26



A three dimensional array in C++ is declared as `int A[a][b][c]`. Consider that array elements are stored in row major order and indexing begin from 0. Here the address of an item at the location `A[r][s][t]` computed in terms of word length `w` of an integer is

- A. $\&A[0][0][0] + w(b * c * s + c * r + t)$
- B. $\&A[0][0][0] + w(b * c * r * s + c * s + t)$
- C. $\&A[0][0][0] + w(a * b * r * s + c * s + t)$
- D. $\&A[0][0][0] + w(a * b * s + c * r + t)$

ugcnetcse-june2023-paper2

Answer key 

36.0.466 UGC NET CSE | June 2023 | Part 2: 25



Match **List I** with **List II**

List I	List II
A. Bezier curves	I. 3 dim objects that have translation rotational symmetry
B. B-Splines	II. Fractal geometry
C. Sweep representations	III. Bias and tension
D. Natural objects	IV. painting drawing CAD

Choose the correct answer from the options given below:

- A. A-IV B-I C-III D-II
- B. A-II B-111 C-I V D-I
- C. A-IV B-III C-I D-II
- D. A-II B-IV C-I D-III

ugcnetcse-june2023-paper2

36.0.467 UGC NET CSE | June 2023 | Part 2: 24



Given below are two statements:

Statement I: Fuzzifier is a part of a fuzzy system

Statement II: Inference engine is a part of fuzzy system

In the light of the above statements, choose the most appropriate answer from the options given below.

- A. Both Statement I and Statement II are correct
- B. Both Statement I and Statement II are incorrect
- C. Statement I is correct but Statement II is incorrect
- D. Statement I is incorrect but Statement II is correct

36.0.468 UGC NET CSE | June 2023 | Part 2: 23

Which of the following is not a mutation operator in a genetic algorithm?

- | | |
|---------------------|---------------|
| A. Random resetting | B. Scramble |
| C. Inversion | D. Difference |

Choose the correct answer from the options given below:

- | | |
|-----------------|-----------------|
| A. A and B only | B. B and D only |
| C. C and D only | D. D only |

36.0.469 UGC NET CSE | June 2023 | Part 2: 20

256Mb DRAM is organized as a $32\text{M} \times 8$ memory externally and as a $16\text{K} \times 16\text{K}$ square array internally. Each row must be refreshed at least once every 50 mili second to forestall loss of data; refreshing one row takes 100 nanoseconds. What fraction of the total memory bandwidth is lost to refresh cycles?

- | | | | |
|---------|---------|---------|---------|
| A. 6.6% | B. 3.3% | C. 9.9% | D. 4.3% |
|---------|---------|---------|---------|

36.0.470 UGC NET CSE | June 2023 | Part 2: 18

Let $\oplus$ denote XOR operation. Let 1 and 0 denote the binary constants and F is the Boolean expression over two variables P and Q $F(P, Q) = ((1 \oplus P) \oplus (P \oplus Q)) \oplus ((P \oplus Q) \oplus (Q \oplus 0))$

Which of the following is equivalent expression to F ?

- | | | | |
|-----------------|------------|---------------------|----------------------------|
| A. $P \oplus Q$ | B. $P + Q$ | C. $\overline{P+Q}$ | D. $\overline{P \oplus Q}$ |
|-----------------|------------|---------------------|----------------------------|

Answer key

36.0.471 UGC NET CSE | June 2023 | Part 2: 17

A TCP server application is programmed to listen on port P on host S . A TCP client is connected to the TCP server over the network, Consider that while TCP connection is active the server is crashed and rebooted. Assume that the client does not use TCP keepalive timer. Which of the following behaviour /s is/are possible?

Statement I: The TCP application server on S can listen on P after reboot.

Statement II: If client sends a packet after the server reboot, it will receive the RST segment.

In the light of the above statements, choose the correct answer from the options given below.

- | |
|--|
| A. Both Statement I and Statement II are true |
| B. Both Statement I and Statement II are false |
| C. Statement I is true but Statement II is false |
| D. Statement I is false but Statement II is true |

36.0.472 UGC NET CSE | June 2023 | Part 2: 16

Consider two hosts P and Q that are connected through a router R . The maximum transfer unit (MTU) value of the link between P and R is 1500 bytes and between R and Q is 820 bytes. A TCP segment of size 1400 bytes is transferred from P to Q through R with IP identification value of 0×1234 . Assume that IP header size is 20 bytes. Further the packet is allowed to be fragmented that is Don't fragment (DF) flag in the IP Header is not set by P . Which of the following statement/s is/are true?

- | |
|--|
| A. Two fragments are created at R and IP datagram size carrying the second fragment is 620 bytes |
|--|

- B. If the second fragment is lost, then R resends the fragment with IP identification value of 0×1234
- C. If the second fragment lost, then P requires to resend the entire TCP segment.
- D. TCP destination port can be determined by analyzing the second fragment only.

Choose the correct answer from the options given below:

- A. A, B and C only
- B. A and C only
- C. C and D only
- D. B and D only

ugcnetcse-june2023-paper2

36.0.473 UGC NET CSE | June 2023 | Part 2: 15



In the standard Ethernet with transmission rate of 10Mbps, assume that the length of the medium is 2500 m and size of a frame is 512 bytes. The propagation speed of a signal in a cable is normally 2×10^8 m/s. The transmission delay and propagation delay are

- A. $25.25\mu s$ and $51.2\mu s$
- B. $51.2\mu s$ and $12.5\mu s$
- C. $10.24\mu s$ and $50.12\mu s$
- D. $12.5\mu s$ and $51.2\mu s$

ugcnetcse-june2023-paper2

36.0.474 UGC NET CSE | June 2023 | Part 2: 14



Consider a disk system with cylinders. The request to access the cylinders occurs in the following sequence:

4, 34, 10, 7, 19, 73, 2, 15, 6, 20

Assuming that the head is currently at cylinder 50, what is the time taken to satisfy all requests if it takes 1 ms to move from one cylinder to adjacent one and shortest seek time first policy is used?

- A. 119 ms
- B. 120 ms
- C. 142 ms
- D. 146 ms

ugcnetcse-june2023-paper2

Answer key

36.0.475 UGC NET CSE | June 2023 | Part 2: 13



Consider universe positive integer $X = \{1 \leq n \leq 8\}$, proposition $P = "n \text{ is an even integers}"$, $Q = "(3 \leq n \leq 7) \wedge (n \neq 6)"$. Then truth set of $P \leftrightarrow Q$ is

- A. $\{1, 4\}$
- B. $\{2, 6\}$
- C. $\{3, 4, 5\}$
- D. $\{1\}$

ugcnetcse-june2023-paper2

Answer key

36.0.476 UGC NET CSE | June 2023 | Part 2: 12



Which is not a basic approach to the problem of conflict resolution in a production system?

- A. Assigning a preference based on the rule that matched
- B. Assigning a preference based the object that matched
- C. Assigning a preference based on the action that the matched rule would perform
- D. Assigning a preference based on the action that the matched object would perform

36.0.477 UGC NET CSE | June 2023 | Part 2: 11

_____ is intended to show that a system both conforms to its specifications and meets the expectations of the system customer.

- | | |
|---------------------------|------------------------|
| A. Software specification | B. Software design |
| C. Software evaluation | D. Software validation |

ugcnetcse-june2023-paper2

Answer key

36.0.478 UGC NET CSE | June 2023 | Part 2: 10

A processor chip is used for application in which 30% of execution time is spent on floating point addition. For the new model of the processor, the design team has come up with redesign the floating point adder to make it twice as fast. What will be possible maximum speed up by this redesign?

- | | | | |
|--------|---------|---------|--------|
| A. 2.0 | B. 1.06 | C. 1.18 | D. 2.5 |
|--------|---------|---------|--------|

ugcnetcse-june2023-paper2

36.0.479 UGC NET CSE | June 2023 | Part 2: 8

In a multiuser operating system, 20 requests are made to use a particular resource per hour, on an average. The probability that no request is made in 45 minutes is

- | | | | |
|--------------|-------------|-----------------|------------------|
| A. e^{-15} | B. e^{-5} | C. $1 - e^{-5}$ | D. $1 - e^{-10}$ |
|--------------|-------------|-----------------|------------------|

ugcnetcse-june2023-paper2

Answer key

36.0.480 UGC NET CSE | June 2023 | Part 2: 7

Which phase of compiler checks the grammar of programming?

- | | |
|----------------------|----------------------|
| A. Code optimization | B. Semantic analysis |
| C. Code generators | D. Syntax analysis |

ugcnetcse-june2023-paper2

Answer key

36.0.481 UGC NET CSE | June 2023 | Part 2: 6

Consider the following statements

- A. C-Fuzzy means clustering is a supervised method of learning
- B. PCA is used for dimension reduction
- C. Apriori is not a supervised technique
- D. When a machine learning model becomes so specially tuned to its exact input data that it fails to generalize to other similar data it is called underfitting

Choose the correct answer from the options given below:

- A. A and B
- B. B and C
- C. C and D
- D. D and A

36.0.482 UGC NET CSE | June 2023 | Part 2: 5

Which of the following scenario may lead to an irrecoverable error in a database system?

- A. A transaction writes a data item after it is read by an uncommitted transaction
- B. A transaction reads a data item after it is read by an uncommitted transaction
- C. A transaction reads a data item after it is written by a committed transactions
- D. A transaction reads a data item after it is written by an uncommitted transaction.

ugcnetcse-june2023-paper2

Answer key

36.0.483 UGC NET CSE | June 2023 | Part 2: 4

Consider the following statements.

- A. The identity is unique in any monoid.
- B. A monoid is a group if there exists inverse of each element of monoid.
- C. Semi group has closure, associative and identity properties.
- D. Quasi group has closure property.

Choose the correct answer from the options given below :

- A. A, B and D only
- B. B, C and D only
- C. A, B and C only
- D. A, C and D only

ugcnetcse-june2023-paper2

Answer key

36.0.484 UGC NET CSE | June 2023 | Part 2: 3

There are M points on one straight line AB and n points on another straight line AC none of them being A . How many triangles can be formed with these points as vertices?

- A. $mn(m+n-2)$
- B. $\frac{1}{2}mn(m+n-2)$
- C. $\frac{1}{2}mn(m+n-1)$
- D. $mn(m+n-1)$

ugcnetcse-june2023-paper2

36.0.485 UGC NET CSE | June 2023 | Part 2: 2

Which of the following statement is correct?

- A. Ackermann's function is primitive recursive.
- B. $L = \{a^n b^k c^{n+k} : n \geq 0, k \geq 0\}$ is regular language.
- C. $L = \{a^n b^J : n = J^2\}$ is not context free language
- D. For every context sensitive language L not including λ , there exists some linear bounded automata M such that $L \neq L(M)$.

ugcnetcse-june2023-paper2

36.0.486 UGC NET CSE | June 2023 | Part 2: 1

What is the output of following code?

```
main ()
struct s1
{ char * z;
```

```

int i;
struct s1 * p;
}
static struct s1 a []={
    {"Nagpur", 1, a+1}
    {"Raipur", 2, a+2}
    {"Kanpur", 3, a}
};
struct s1* ptr =a;
printf ("%s %s %s\n", a[0].z. ptr $/\rightarrow$ z, a [2]. p $/\rightarrow$ z;)

```

- A. Nagpur Raipur Kanpur
C. Kanpur Kanpur Kanpur

- B. Nagpur Nagpur Nagpur
D. Error

ugcnetcse-june2023-paper2

Answer key 

36.0.487 UGC NET CSE | June 2008 | Part 2 | Question: 50



Identify the incorrect statement :

- A. The ATM adaptation layer is not service dependent.
B. Logical connections in ATM are referred to as virtual channel connections.
C. ATM is a streamlined protocol with minimal error and flow control capabilities.
D. ATM is also known as cell relay.

ugcnetcse-june2008-paper2

36.0.488 UGC NET CSE | June 2008 | Part 2 | Question: 49



Identify the incorrect statement :

- A. The overall strategy drives the e-commerce data warehousing strategy.
B. Data warehousing in an e-commerce environment should be done in a classical manner.
C. E-commerce opens up an entirely new world of web servers.
D. E-commerce security threats can be grouped into three major categories.

ugcnetcse-june2008-paper2

36.0.489 UGC NET CSE | June 2008 | Part 2 | Question: 48



In a mobile communication system, a geographic region is divided into cells. For each frequency set, there is a buffer _____ wide where that frequency is not used.

- A. one-cell B. two-cells C. three-cells D. four-cells

ugcnetcse-june2008-paper2

36.0.490 UGC NET CSE | June 2008 | Part 2 | Question: 47



With reference to cluster analysis in data mining, a distance measure that is NOT used is :

- A. Euclidean distance. B. Manhattan distance.
C. Chebychev's distance. D. Lee distance.

ugcnetcse-june2008-paper2

36.0.491 UGC NET CSE | June 2008 | Part 2 | Question: 45



In clean room software engineering :

- A. only eco-friendly hardware is used.
- B. only hired facilities are used for development.
- C. correctness of the code is verified before testing.
- D. implementation is done only after ensuring correctness.

ugcnetcse-june2008-paper2

36.0.492 UGC NET CSE | June 2008 | Part 2 | Question: 44



In software cost estimation, base estimation is related to :

- A. cost of similar projects already completed.
- B. cost of the base model of the present project.
- C. cost of the project with the base minimum profit.
- D. cost of the project under ideal situations.

ugcnetcse-june2008-paper2

36.0.493 UGC NET CSE | June 2008 | Part 2 | Question: 43



While designing the user interface, one should :

- A. use as many short cuts as possible.
- B. use as many defaults as possible.
- C. use as many visual layouts as possible.
- D. reduce the demand on short-term memory.

ugcnetcse-june2008-paper2

36.0.494 UGC NET CSE | June 2008 | Part 2 | Question: 42



In software development, value adjustment factors include the following among others :

- A. the criticality of the performance and reusability of the code.
- B. number of lines of code in the software.
- C. number of technical manpower and hardware costs.
- D. time period available and the level of user friendliness.

ugcnetcse-june2008-paper2

Answer key 

36.0.495 UGC NET CSE | June 2008 | Part 2 | Question: 41



Water fall model for software development is :

- | | |
|---------------------------|------------------------------|
| A. a top down approach. | B. a bottom up approach. |
| C. a sequential approach. | D. a consequential approach. |

ugcnetcse-june2008-paper2

36.0.496 UGC NET CSE | June 2008 | Part 2 | Question: 39



With 64 bit virtual addresses, a 4 KB page and 256MB of RAM, an inverted page table requires :

- A. 8192 entries.
- B. 16384 entries.
- C. 32768 entries.
- D. 65536 entries.

ugcnetcse-june2008-paper2

36.0.497 UGC NET CSE | June 2008 | Part 2 | Question: 38



An example of a memory management system call in UNIX is :

- A. fork.
- B. mmap.
- C. sigaction.
- D. execve.

ugcnetcse-june2008-paper2

36.0.498 UGC NET CSE | June 2008 | Part 2 | Question: 37



There are ' n ' processes in memory. A process spends a fraction ' p ' of its time waiting for I/O to complete. The CPU utilization is given by :

- A. p^n
- B. $1 - p^n$
- C. $(1 - p)^n$
- D. $1 - np$

ugcnetcse-june2008-paper2

Answer key

36.0.499 UGC NET CSE | June 2008 | Part 2 | Question: 36



An example of a non-preemptive CPU scheduling algorithm is :

- A. Shortest job first scheduling.
- B. Round robin scheduling.
- C. Priority scheduling.
- D. Fair share scheduling.

ugcnetcse-june2008-paper2

36.0.500 UGC NET CSE | June 2008 | Part 2 | Question: 35



A parse tree is an annotated parse tree if :

- A. it shows attribute values at each node.
- B. there are no inherited attributes.
- C. it has synthesized nodes as terminal nodes.
- D. every non-terminal nodes is an inherited attribute.

ugcnetcse-june2008-paper2

36.0.501 UGC NET CSE | June 2008 | Part 2 | Question: 34



Dead-code elimination in machine code optimization refers to :

- A. removal of all labels.
- B. removal of values that never get used.
- C. removal of function which are not involved.
- D. removal of a module after its use.

ugcnetcse-june2008-paper2

36.0.502 UGC NET CSE | June 2008 | Part 2 | Question: 32



In the indirect addressing scheme, the second part of an instruction contains :

- A. the operand in decimal form
- B. the address of the location where the value of the operand is stored
- C. the address of the location where the address of the operand is stored
- D. the operand in an encoded form

ugcnetcse-june2008-paper2

36.0.503 UGC NET CSE | June 2008 | Part 2 | Question: 31



Assembler program is :

- A. dependent on the operating system
- B. dependent on the compiler
- C. dependent on the hardware
- D. independent of the hardware

ugcnetcse-june2008-paper2

Answer key

36.0.504 UGC NET CSE | June 2008 | Part 2 | Question: 30



With reference to hierarchical routing, the optimum number of levels for an m router subnet is:

- A. m^2
- B. m
- C. $\ln m$
- D. $\sqrt{m}$

ugcnetcse-june2008-paper2

Answer key

36.0.505 UGC NET CSE | June 2008 | Part 2 | Question: 29



An example of a public key encryption algorithm is :

- A. Caesar cipher algorithm
- B. DES algorithm
- C. AES algorithm
- D. Knapsack algorithm

ugcnetcse-june2008-paper2

Answer key

36.0.506 UGC NET CSE | June 2008 | Part 2 | Question: 28



For a channel of 3KHz bandwidth and signal to noise ratio of 30 dB, the maximum data rate is:

- A. 3000bps
- B. 6000bps
- C. 15000bps
- D. 30000bps

ugcnetcse-june2008-paper2

Answer key

36.0.507 UGC NET CSE | June 2008 | Part 2 | Question: 27



For slotted ALOHA, the maximum channel utilization is :

- A. 100%
- B. 50%
- C. 36%
- D. 18%

36.0.508 UGC NET CSE | June 2008 | Part 2 | Question: 26

The ATM cells are _____ bytes long.

- A. 48 B. 53 C. 64 D. 69

36.0.509 UGC NET CSE | June 2008 | Part 2 | Question: 25

An example of a possible file attribute is:

- A. minimum size B. permanent flag
C. archive flag D. EBCDIC flag

36.0.510 UGC NET CSE | June 2008 | Part 2 | Question: 24

Files that are related to input/output and are used to model serial I/O devices such as terminals, printers and networks are called:

- A. regular files B. character special files
C. directories D. block special files

Answer key

36.0.511 UGC NET CSE | June 2008 | Part 2 | Question: 22

Which of the following can be the sequence of nodes examined in a binary search tree while searching for key 98?

- A. 100, 50, 75, 60, 98 B. 100, 120, 90, 95, 98
C. 200, 70, 100, 95, 98 D. 75, 150, 90, 80, 98

36.0.512 UGC NET CSE | June 2008 | Part 2 | Question: 21

Which of the following data structures is most efficient in terms of both space and time to reverse a string of characters?

- A. Linked list B. Stack C. Array D. Tree

Answer key

36.0.513 UGC NET CSE | June 2008 | Part 2 | Question: 20

Consider the query: SELECT student_name FROM student_data WHERE rollno (SELECT rollno FROM student_marks WHERE SEM1_MARK=SEM2_MARK); Which of the following is true?

- A. It gives the name of the student whose marks in semester 1 and semester 2 are same.
B. It gives all the names and roll nos of those students whose marks in semester 1 and semester 2 are same.
C. It gives the names of all the students whose marks in semester 1 and semester 2 are same.

D. It gives roll numbers of all students whose marks in semester 1 and semester 2 are same.

ugcnetcse-june2008-paper2

Answer key 

36.0.514 UGC NET CSE | June 2008 | Part 2 | Question: 19



Which of the following is true?

- A. A relation in 3NF is always in BCNF
- B. A relation in BCNF is always in 3NF
- C. BCNF and 3NF are totally different
- D. A relation in BCNF is in 2NF but not in 3NF

ugcnetcse-june2008-paper2

Answer key 

36.0.515 UGC NET CSE | June 2008 | Part 2 | Question: 18



If a relation is in 2NF then :

- A. every candidate key is a primary key
- B. every non-prime attribute is fully functionally dependent on each relation key
- C. every attribute is functionally independent
- D. every relational key is a primary key

ugcnetcse-june2008-paper2

Answer key 

36.0.516 UGC NET CSE | June 2008 | Part 2 | Question: 17



Which of the following set of keywords constitutes a mapping in SQL?

- | | |
|---------------------------|--------------------------|
| A. SELECT, FROM, TABLE | B. SELECT, FROM, WHERE |
| C. CONNECT, TABLE, CREATE | D. SELECT, TABLE, INSERT |

ugcnetcse-june2008-paper2

Answer key 

36.0.517 UGC NET CSE | June 2008 | Part 2 | Question: 16



A superkey for an entity consists of :

- | | |
|---------------------------|----------------------------|
| A. one attribute only | B. at least two attributes |
| C. at most two attributes | D. one or more attributes |

ugcnetcse-june2008-paper2

36.0.518 UGC NET CSE | June 2008 | Part 2 | Question: 15



Which of the following is true?

- A. A "static" member of a class cannot be inherited by its derived class.
- B. A "static" member of a class can be initialized only within the class it is a member of.
- C. A "static" member of a class can be initialized before an object of that class is created.
- D. Since "static" member of a class is actually a global element, it does not require a class/object qualifier to access it

independently of class/object.

ugcnetcse-june2008-paper2

Answer key 

36.0.519 UGC NET CSE | June 2008 | Part 2 | Question: 14



Function overloading is a concept in which :

- A. a function is used to implement lots of tasks at the same time.
- B. a function is called too many number of times by another function.
- C. a function provides common interface to the user to carry out possibly different functions in each call.
- D. a function is computationally too expensive for the system to handle.

ugcnetcse-june2008-paper2

Answer key 

36.0.520 UGC NET CSE | June 2008 | Part 2 | Question: 13



Consider the following C code :

```
{int a = 5, b=9;  
float r;  
r= b /a ;}
```

What is the value of r ?

- A. 1.8
- B. 1.0
- C. 2.0
- D. 0.0

ugcnetcse-june2008-paper2

Answer key 

36.0.521 UGC NET CSE | June 2008 | Part 2 | Question: 12



Consider the following declaration in C :

```
char a[];  
char * p;
```

Which of the following statement is not a valid statement?

- A. `p = a;`
- B. `p = a + 2;`
- C. `a = p;`
- D. `p = &a[2]`

ugcnetcse-june2008-paper2

Answer key 

36.0.522 UGC NET CSE | June 2008 | Part 2 | Question: 11



What is the effect of the following C code?

```
for(int i=1 ; i ≤ 5 ; i = i + 1/2)  
printf(" % d," , i);
```

- A. It prints 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5, and stops
- B. It prints 1, 2, 3, 4, 5, and stops
- C. It prints 1, 2, 3, 4, 5, and repeats forever
- D. It prints 1, 1, 1, 1, 1, and repeats forever

36.0.523 UGC NET CSE | June 2008 | Part 2 | Question: 10

The idempotent law in Boolean algebra says that:

- A. $\sim(\sim x) = x$ B. $x + x = x$ C. $x + xy = x$ D. $x(x + y) = x$

Answer key

36.0.524 UGC NET CSE | June 2008 | Part 2 | Question: 9

The characteristic equation of a T flip flop is given by :

- A. $Q_{N+1} = TQ_N$
 B. $Q_{N+1} = T + Q_N$
 C. $Q_{N+1} = T \oplus Q_N$
 D. $Q_{N+1} = \bar{T} + Q_N$

36.0.525 UGC NET CSE | June 2008 | Part 2 | Question: 8

The octal equivalent of the hexadecimal number FF is :

- A. 100 B. 150 C. 377 D. 737

Answer key

36.0.526 UGC NET CSE | June 2008 | Part 2 | Question: 7

Among the logic families RTL, TTL, ECL and CMOS, the fastest family is :

- A. ECL B. CMOS C. TTL D. RTL

Answer key

36.0.527 UGC NET CSE | June 2008 | Part 2 | Question: 6

An example of a tautology is :

- A. $x \vee y$ B. $x \vee (\sim y)$
 C. $x \vee (\sim x)$ D. $(x \Rightarrow y) \wedge (x \Leftarrow y)$

Answer key

36.0.528 UGC NET CSE | June 2008 | Part 2 | Question: 5

In a set of 8 positive integers, there always exists a pair of numbers having the same remainder when divided by :

- A. 7 B. 11 C. 13 D. 15

ugcnetcse-june2008-paper2

36.1 Abelian Group (1)**36.1.1 Abelian Group: UGC NET CSE | June 2008 | Part 2 | Question: 4**

The set of positive integers under the operation of ordinary multiplication is :

- A. not a monoid B. not a group
C. a group D. an Abelian group

ugcnetcse-june2008-paper2 abelian-group

Answer key

36.2 Acid Properties (1)**36.2.1 Acid Properties: UGC NET CSE | December 2009 | Part 2 | Question: 45**

The name of the transaction file shall be provided by the operator and the file that contains the edited transactions ready for execution shall be called

- (A) Batch. Exe
(B) Trans. Exe
(C) Opt. Exe
(D) Edit.Exe

ugcnetcse-dec2009-paper2 databases acid-properties

36.3 Array (1)**36.3.1 Array: UGC NET CSE | June 2008 | Part 2 | Question: 23**

Which of the following is true for a sorted list with ' n ' elements?

- A. Insertion in a sorted array takes constant time.
B. Insertion in a sorted linear linked list takes constant time.
C. Searching for a key in a sorted array can be done in $O(\log n)$ time.
D. Searching for a key in a sorted linear linked list can be done in $O(\log n)$ time.

ugcnetcse-june2008-paper2 sorting array time-complexity

Answer key

36.4 Artificial Intelligence (1)**36.4.1 Artificial Intelligence: UGC NET CSE | August 2016 | Part 3 | Question: 71**

Match each Artificial Intelligence term in List-I that best describes a given situation in List-II :

List-I

- I. Semantic Network
- II. Frame
- III. Declarative Knowledge
- IV. Primitive

List-II

- a. Knowledge about what to do as opposed to how to do it
- b. A premise of rule that is not concluded by any rule
- c. A method of knowledge representation that uses a graph
- d. A data structure representing stereotypical knowledge

Codes :

- A. I-d, II-a, III-b, IV-c
- C. I-d, II-c, III-b, IV-a

- B. I-d, II-c, III-a, IV-b
- D. I-c, II-d, III-a, IV-b

ugcnetcse-aug2016-paper3 artificial-intelligence

Answer key **36.5****Assembly (1)****36.5.1 Assembly: UGC NET CSE | June 2011 | Part 2 | Question: 34**

In which way(s) a macroprocessor for assembly language can be implemented?

- A. Independent two-pass processor
- B. Independent one-pass processor
- C. Expand macrocalls and substitute arguments
- D. All of the above

ugcnetcse-june2011-paper2 co-and-architecture microprocessors assembly

Answer key **36.6****Binary Heap (1)****36.6.1 Binary Heap: UGC NET CSE | June 2023 | Part 2: 21**

Consider the following statements about heap sort algorithm:

- A. The MAX-HEAPIFY procedure which runs in $O(\lg n)$ time, is the key to maintaining the max heap property
- B. The BUILD-MAX-HEAP procedure, which runs in $O(\lg n)$ time, produces max-heap from an unordered input array
- C. The MAX-HEAP-INSERT, which runs in $O(\lg n)$ time, implements the insertion operation
- D. The HEAP-INCREASE-KEY procedure runs in $O(\lg n)$ time, to set the key of new node of its correct value

Choose the correct answer from the options given below:

- A. A, B only
- B. A, C only
- C. B, D only
- D. A, B, C, D

ugcnetcse-june2023-paper2 binary-heap algorithms

Answer key **36.7****Binary Tree (1)****36.7.1 Binary Tree: UGC NET CSE | June 2011 | Part 2 | Question: 22**

Given a binary tree whose inorder and preorder traversal are given by

Inorder : EICFBGDJHK

Preorder : BCEIFDGHJK

The post order traversal of the above binary tree is

A. I E F C G J K H D B
C. I E F C G K J H D B

B. I E F C J G K H D B
D. I E F C G J K D B H

ugcnetcse-june2011-paper2 algorithms binary-tree

Answer key 

36.8

Boolean Algebra (1)

36.8.1 Boolean Algebra: UGC NET CSE | December 2019 | Part 2 | Question: 3



The Boolean expression $AB + \bar{A}\bar{B} + \bar{A}C + AC$ is unaffected by the value of the Boolean variable _____.

- A. A B. B C. C D. A, B and C

ugcnetcse-dec2019-paper2 digital-logic boolean-algebra

Answer key 

36.9

Combinatory (1)

36.9.1 Combinatory: UGC NET CSE | September 2013 | Part 2 | Question: 19



Find the number of ways to paint 12 offices so that 3 of them will be green, 2 of them pink, 2 of them yellow and the rest ones white.

- A. 55,440 B. 1,66,320 C. 4.790E+08 D. 39,91,680

ugcnetsep2013ii combinatory

Answer key 

36.10

Compilation Phases (1)

36.10.1 Compilation Phases: UGC NET CSE | December 2010 | Part 2 | Question: 33



Grammar of the programming is checked at _____ phase of compiler.

- A. Semantic analysis B. Code generation
C. Syntax analysis D. Code optimization

ugcnetcse-dec2010-paper2 compiler-design compilation-phases

Answer key 

36.11

Crypt Arithmetic (1)

36.11.1 Crypt Arithmetic: UGC NET CSE | September 2013 | Part 3 | Question: 3



Assume that each alphabet can have a value between 0 to 9 in a cryptoarithmic problem

$$\begin{array}{r} \text{CROSS} \\ + \text{ROADS} \\ \hline \text{DANGER} \end{array}$$

Which of the following statements is true?

- I. No two alphabets can have the same numeric value
II. Any two alphabets can have the same numeric value
III. $D=0$
IV. $D=1$

- A. I and III B. I and IV C. II and III D. II and IV

ugcnetcse-sep2013-paper3 cryptography crypt-arithmetic

Answer key 

36.12

Data Compression (1)

36.12.1 Data Compression: UGC NET CSE | Junet 2015 | Part 3 | Question: 65



The redundancy in images stems from

- A. pixel decolleration
- B. pixel colleration
- C. pixel quantization
- D. image size

ugcnetcse-june2015-paper3 data-compression digital-image-processing

Answer key

36.13 Data Flow Diagram (1)

36.13.1 Data Flow Diagram: UGC NET CSE | June 2011 | Part 2 | Question: 43



By means of a data flow diagram, the analyst can detect

- A. Task duplication
- B. Unnecessary delays
- C. Task overlapping
- D. All of the above

ugcnetcse-june2011-paper2 is&software-engineering data-flow-diagram

Answer key

36.14 Data Mart (1)

36.14.1 Data Mart: UGC NET CSE | December 2010 | Part 2 | Question: 46



A data mart may contain

- A. Summarised data
- B. De-normalized data
- C. Aggregate departmental data
- D. All of the above

ugcnetcse-dec2010-paper2 data-warehousing data-mart

Answer key

36.15 Data Mining (4)

36.15.1 Data Mining: UGC NET CSE | August 2016 | Part 2 | Question: 48



In Data mining, classification rules are extracted from _____.

- A. Data
- B. Information
- C. Decision Tree
- D. Database

ugcnetcse-aug2016-paper2 data-mining

Answer key

36.15.2 Data Mining: UGC NET CSE | August 2016 | Part 2 | Question: 50



In Data mining, _____ is a method of incremental conceptual clustering.

- A. STRING
- B. COBWEB
- C. CORBA
- D. OLAD

ugcnetcse-aug2016-paper2 data-mining

Answer key

36.15.3 Data Mining: UGC NET CSE | December 2010 | Part 2 | Question: 47



Data mining is

- A. The process of extracting valid, previously unknown, comprehensible and actionable information from large databases and using it to make crucial business decisions.
- B. A tool that allows end users direct access and manipulation of data from within datawarehousing environment without the intervention of customised programming activity.
- C. A tool that helps end users extract useful business information from large database.
- D. All of the above

Answer key **36.15.4 Data Mining: UGC NET CSE | December 2011 | Part 2 | Question: 47**

Web Mining is not used in which of the following areas ?

- | | |
|----------------------------------|-----------------------------------|
| A. Information filtering | B. Crime fighting on the internet |
| C. Online transaction processing | D. Click stream analysis. |

ugcnetcse-dec2011-paper2 data-mining

Answer key **36.16****Data Warehousing (2)****36.16.1 Data Warehousing: UGC NET CSE | August 2016 | Part 2 | Question: 47**

_____ is subject oriented, integrated, time variant, nonvolatile collection of data in support of management decisions.

- | | |
|-------------------|-------------------------------|
| A. Data mining | B. Web mining |
| C. Data warehouse | D. Database Management System |

ugcnetcse-aug2016-paper2 data-warehousing

Answer key **36.16.2 Data Warehousing: UGC NET CSE | June 2011 | Part 2 | Question: 46**

Data warehousing refers to

- | | |
|--|---------------------------------|
| A. Storing data offline at a separate site | B. Backing up data regularly |
| C. Is related to data mining | D. Uses tape as opposed to disk |

ugcnetcse-june2011-paper2 data-warehousing

Answer key **36.17****Deadlock Prevention Avoidance Detection (1)****36.17.1 Deadlock Prevention Avoidance Detection: UGC NET CSE | June 2008 | Part 2 | Question: 40**

A computer has 6 tape drives with ' n ' processes competing for them. Each process may need two drives. For which values of ' n ' is the system deadlock free?

- | | | | |
|------|------|------|------|
| A. 1 | B. 2 | C. 3 | D. 6 |
|------|------|------|------|

ugcnetcse-june2008-paper2 deadlock-prevention-avoidance-detection

36.18**Decomposition (1)****36.18.1 Decomposition: UGC NET CSE | June 2011 | Part 2 | Question: 20**

Decomposition help in eliminating some of the problems of bad design

- | | |
|---------------|---------------------|
| A. Redundancy | B. Inconsistencies |
| C. Anomalies | D. All of the above |

ugcnetcse-june2011-paper2 databases decomposition

Answer key **36.19****Digital Image Processing (3)**

36.19.1 Digital Image Processing: UGC NET CSE | August 2016 | Part 3 | Question: 60

If the histogram of an image is clustered towards origin on X-axis of a histogram plot then it indicates that the image is _____.

- A. Dark B. Good contrast C. Bright D. Very low contrast

ugcnetcse-aug2016-paper3 digital-image-processing

Answer key

36.19.2 Digital Image Processing: UGC NET CSE | December 2013 | Part 3 | Question: 31

Consider the formula in image processing $R_D = 1 - \frac{1}{C_R}$ where $C_R = \frac{n_1}{n_2}$. C_R is called as compression ratio n_1 and n_2 denotes the number of information carrying units in two datasets that represent the same information. In this situation R_D is called as relative ____ of the first data set.

- A. Data Compression B. Data Redundancy
C. Data Relation D. Data Representation

ugcnetcse-dec2013-paper3 digital-image-processing

Answer key

36.19.3 Digital Image Processing: UGC NET CSE | June 2012 | Part 3 | Question: 22

The transform which possess the highest 'energy compaction' property is

- A. Slant transform B. Cosine transform
C. Fourier transform D. Karhunen-Loeve transform

ugcnetcse-june2012-paper3 digital-image-processing

Answer key

36.20 Disk (1)**36.20.1 Disk: UGC NET CSE | June 2011 | Part 2 | Question: 36**

Block or Buffer caches are used to

- A. Improve disk performance B. Handle interrupts
C. Increase the capacity of main memory D. Speed up main memory Read operations

ugcnetcse-june2011-paper2 operating-system disk

Answer key

36.21 Disk Scheduling (1)**36.21.1 Disk Scheduling: UGC NET CSE | June 2011 | Part 2 | Question: 23**

The number of disk accesses performed by insertion operation in B-tree of height h is

- A. $O(1)$ B. $O(1gh)$ C. $O(h)$ D. None of these

ugcnetcse-june2011-paper2 operating-system disk-scheduling

Answer key

36.22 Duality (1)**36.22.1 Duality: UGC NET CSE | August 2016 | Part 3 | Question: 62**

Consider the following statements :

- (a) If primal (dual) problem has a finite optimal solution, then its dual (primal) problem has a finite optimal solution.
(b) If primal (dual) problem has an unbounded optimum solution, then its dual (primal) has no feasible solution at all.
(c) Both primal and dual problems may be infeasible. Which of the following is correct ?

- A. (a) and (b) only B. (a) and (c) only C. (b) and (c) only D. (a), (b) and (c)

ugcnetcse-aug2016-paper3 linear-programming duality

Answer key 

36.23 E Commerce (2)

36.23.1 E Commerce: UGC NET CSE | August 2016 | Part 2 | Question: 49



Discovery of cross sales opportunities is called as _____.

- A. Association B. Visualization C. Correlation D. Segmentation

ugcnetcse-aug2016-paper2 e-technologies e-commerce

Answer key 

36.23.2 E Commerce: UGC NET CSE | June 2011 | Part 2 | Question: 50



Which is the most valuable electronic commerce to the individual customer in long run ?

- A. Business to Customer B. Business to Business
C. Customer to Customer D. None of the above

ugcnetcse-june2011-paper2 e-technologies e-commerce

Answer key 

36.24 E Payment (1)

36.24.1 E Payment: UGC NET CSE | June 2011 | Part 2 | Question: 49



The principal electronic payment systems for electronic commerce is

- A. Credit Card B. Digital Wallet
C. Electronic Cheque D. All of the above

ugcnetcse-june2011-paper2 e-technologies e-payment

Answer key 

36.25 E Technologies (3)

36.25.1 E Technologies: UGC NET CSE | December 2010 | Part 2 | Question: 48



E-Choupal refers to

- A. Calling meeting of Surpanches electronically.
B. Conducting Panchayat meeting using latest electronic gadgets.
C. Providing window of information to villagers and farmers in NIC network.
D. None of the above

ugcnetcse-dec2010-paper2 e-technologies

Answer key 

36.25.2 E Technologies: UGC NET CSE | December 2011 | Part 2 | Question: 7



BCC in the internet refers to

- A. Black carbon copy B. Blind carbon copy
C. Blank carbon copy D. Beautiful carbon copy

ugcnetcse-dec2011-paper2 e-technologies

Answer key 

36.25.3 E Technologies: UGC NET CSE | June 2011 | Part 2 | Question: 48



"M-Commerce" refers to

- A. A myth which does not exist in reality
- B. The ability of business to reach potential customers wherever they are
- C. The ability to have large capacity of memory storage dealing trade and commerce
- D. None of the above

ugcnetcse-june2011-paper2 e-technologies

Answer key

36.26

Encryption Decryption (2)

36.26.1 Encryption Decryption: UGC NET CSE | September 2013 | Part 2 | Question: 3



Which of the following substitution technique have the relationship between a character in the plaintext and a character in the ciphertext as one-to-many?

- A. Monoalphabetic
- B. Polyalphabetic
- C. Transpositional
- D. None of the above

ugcnetsep2013iii cryptography encryption-decryption

Answer key

36.26.2 Encryption Decryption: UGC NET CSE | September 2013 | Part 3 | Question: 57



If user A wants to send an encrypted message to user B, the plain text of A is encrypted with the

- A. Public key of user A
- B. Public key of user B
- C. Private key of user A
- D. Private key of user B

ugcnetcse-sep2013-paper3 cryptography encryption-decryption

Answer key

36.27

Enterprise Resource Planning (2)

36.27.1 Enterprise Resource Planning: UGC NET CSE | December 2011 | Part 2 | Question: 44



Enterprise Resource Planning (ERP)

- A. Has existed for over a decade.
- B. Does not integrate well with the functional areas other than operations.
- C. Is inexpensive to implement.
- D. Automate and integrates the majority of business processes.

ugcnetcse-dec2011-paper2 enterprise-resource-planning

Answer key

36.27.2 Enterprise Resource Planning: UGC NET CSE | December 2011 | Part 2 | Question: 45



Which of the following is false concerning Enterprise Resource Planning (ERP) ?

- A. It attempts to automate and integrate the majority of business processes.
- B. It shares common data and practices across the enterprise.
- C. It is inexpensive to implement.
- D. It provides and access information in a real-time environment.

ugcnetcse-dec2011-paper2 enterprise-resource-planning

Answer key

36.28

Error Correction (1)

36.28.1 Error Correction: UGC NET CSE | June 2008 | Part 2 | Question: 3



If a code is t -error correcting, the minimum Hamming distance is equal to :

- A. $2t+1$ B. $2t$ C. $2t-1$ D. $t-1$

ugcnetcse-june2008-paper2 hamming-code error-correction

Answer key

36.29

File System (1)

36.29.1 File System: UGC NET CSE | June 2011 | Part 2 | Question: 37



A file organization component VSAM file is

- A. Relative records data set B. Keyed sequential data set
C. Entry sequential data set D. All of the above

ugcnetcse-june2011-paper2 operating-system file-system

Answer key

36.30

Fuzzy Logic (2)

36.30.1 Fuzzy Logic: UGC NET CSE | August 2016 | Part 3 | Question: 64



What are the following sequence of steps taken in designing a fuzzy logic machine ?

- A. Fuzzification $\rightarrow$ Rule evaluation $\rightarrow$ Defuzzification
B. Fuzzification $\rightarrow$ Defuzzification $\rightarrow$ Rule evaluation
C. Rule evaluation $\rightarrow$ Fuzzification $\rightarrow$ Defuzzification
D. Rule evaluation $\rightarrow$ Defuzzification $\rightarrow$ Fuzzification

ugcnetcse-aug2016-paper3 neural-network fuzzy-logic

Answer key

36.30.2 Fuzzy Logic: UGC NET CSE | September 2013 | Part 3 | Question: 26



Let A be a set of comfortable houses given as $A = \left\{ \frac{x_1}{0.8}, \frac{x_2}{0.9}, \frac{x_3}{0.1}, \frac{x_4}{0.7} \right\}$ and be the set of affordable houses $B = \left\{ \frac{x_1}{0.9}, \frac{x_2}{0.8}, \frac{x_3}{0.6}, \frac{x_4}{0.2} \right\}$ Then the set of comfortable and affordable houses is

- A. $\left\{ \frac{x_1}{0.8}, \frac{x_2}{0.8}, \frac{x_3}{0.1}, \frac{x_4}{0.2} \right\}$
B. $\left\{ \frac{x_1}{0.9}, \frac{x_2}{0.9}, \frac{x_3}{0.6}, \frac{x_4}{0.7} \right\}$
C. $\left\{ \frac{x_1}{0.8}, \frac{x_2}{0.8}, \frac{x_3}{0.6}, \frac{x_4}{0.7} \right\}$
D. $\left\{ \frac{x_1}{0.7}, \frac{x_2}{0.7}, \frac{x_3}{0.7}, \frac{x_4}{0.9} \right\}$

ugcnetcse-sep2013-paper3 neural-network fuzzy-logic

Answer key

36.31

Fuzzy Relations (1)

36.31.1 Fuzzy Relations: UGC NET CSE | August 2016 | Part 3 | Question: 66



Let R and S be two fuzzy relations defined as

$$R = \begin{matrix} & \begin{matrix} y_1 & y_2 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} 0.7 & 0.5 \\ 0.8 & 0.4 \end{bmatrix} \end{matrix}$$

and $S = \begin{matrix} & z_1 & z_2 & z_3 \\ y_1 & 0.9 & 0.6 & 0.2 \\ y_2 & 0.1 & 0.7 & 0.5 \end{matrix}$

Then, the resulting relation, T , which relates elements of universe of X to elements of universe of Z using max-product composition is given by

A. $T = \begin{matrix} & z_1 & z_2 & z_3 \\ x_1 & 0.68 & 0.89 & 0.39 \\ x_2 & 0.76 & 0.72 & 0.32 \end{matrix}$

B. $T = \begin{matrix} & z_1 & z_2 & z_3 \\ x_1 & 0.68 & 0.89 & 0.39 \\ x_2 & 0.72 & 0.76 & 0.32 \end{matrix}$

C. $T = \begin{matrix} & z_1 & z_2 & z_3 \\ x_1 & 0.63 & 0.42 & 0.25 \\ x_2 & 0.72 & 0.48 & 0.20 \end{matrix}$

D. $T = \begin{matrix} & z_1 & z_2 & z_3 \\ x_1 & 0.05 & 0.35 & 0.14 \\ x_2 & 0.04 & 0.28 & 0.16 \end{matrix}$

ugcnetcse-aug2016-paper3 neural-network fuzzy-relations

Answer key 

36.32

Fuzzy Set (2)

36.32.1 Fuzzy Set: UGC NET CSE | December 2013 | Part 3 | Question: 28



If A and B are two fuzzy sets with membership functions

$$\mu_A(x) = \{0.2, 0.5, 0.6, 0.1, 0.9\}$$

$$\mu_B(x) = \{0.1, 0.5, 0.2, 0.7, 0.8\}$$

Then the value of $\mu_{A \cap B}$ will be

A. $\{0.2, 0.5, 0.6, 0.7, 0.9\}$

B. $\{0.2, 0.5, 0.2, 0.1, 0.8\}$

C. $\{0.1, 0.5, 0.6, 0.1, 0.8\}$

D. $\{0.1, 0.5, 0.2, 0.1, 0.8\}$

ugcnetcse-dec2013-paper3 fuzzy-set

Answer key 

36.32.2 Fuzzy Set: UGC NET CSE | December 2013 | Part 3 | Question: 29



The height $h(A)$ of a fuzzy set A is defined as

$$h(A) = \sup_{x \in A} A(x)$$

$$x \in A$$

The fuzzy set A is called normal when

A. $h(A)=0$

B. $h(A)<0$

C. $h(A)=1$

D. $h(A)<0$

ugcnetcse-dec2013-paper3 fuzzy-set

Answer key 

36.33

General Awareness (33)

36.33.1 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 1



To organize discussion method in teaching effectively, which of the following conditions should be met?

- i. Topic be easy
- ii. Topic be declared in advance
- iii. Topic of common interest
- iv. Availability of more than one teacher
- v. Language facility of participants

Select appropriate answer from the options given below:

- A. ii, iii, and v
- B. i, ii, and iii
- C. i, ii and v
- D. iii, iv and v

ugcnetcse-june2019-paper1 general-awareness

Answer key 

36.33.2 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 10



Which of the following is a plagiarism checking website?

- A. <http://go.turnitin.com>
- B. <http://www.researchgate.com>
- C. <http://www.editorial.elsevier.com>
- D. <http://www.grammarly.com>

ugcnetcse-june2019-paper1 general-awareness

Answer key 

36.33.3 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 11



Michaelangelo is famous for having successfully interpreted the human body. His great achievement is that of the painting of David whose hands reach out as a sign of human capability and potential. It is assumed that the time he lived was ripe for exchange of knowledge, development in science and matured enough to advance the horizon of investigation in all fields. Renaissance humanism stressed on a serious rethink on the nature of art that focussed on accurate details. In painting and sculpture, artists focussed on not so casual but verifiable and minute details. Michaelangelo's paintings are no exception to it. In a study published in the journal of the Royal Society of medicine, a group of surgeons are of the opinion that the great master was "afflicted by an illness involving his joints". They have used his portraits as evidence to argue their view. During his life, he complained of what he felt to be 'gout'. Later he complained of his sore and stiff hands which the doctors would find to be natural for someone who was engaged in handmade art. The doctors found corroboration of those claims in portraits of the artist that show a hanging left hand with both degenerative and non-degenerative changes. They attribute the pain not just to arthritis, but to the stress of hammering and chiselling and note that though the master was seen hammering days before his death at an old age, he did not write or sign his own letters before his death. In recent times there have been attempts to diagnose famous artists with diseases that were not known during their time. This practice has raised many questions, especially on the issue of ethics in research. It is also inferred from authentic analysis that Michaelangelo persisted in his work until his last days. This theory would emphasize that his artistic subject defined his physical infirmities.

Michaelangelo lived during a time that lets us know that

- A. Human aspirants are limitless and open to new vistas of knowledge
- B. Cross cultural exchange in ideas is the only way for human progress
- C. It is progress of science and anatomy that contributes to civilizations exclusively
- D. Human beings possess language which is the only key to knowledge

ugcnetcse-june2019-paper1 general-awareness passage-reading

36.33.4 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 12



Michaelangelo is famous for having successfully interpreted the human body. His great achievement is that of the painting of David whose hands reach out as a sign of human capability and potential. It is assumed that the time he lived was ripe for exchange of knowledge, development in science and matured enough to advance the horizon of investigation in all fields. Renaissance humanism stressed on a serious rethink on the nature of art that focussed on accurate details. In painting and sculpture, artists focussed on not so casual but verifiable and minute details. Michaelangelo's paintings are no exception to it. In a study published in the journal of the Royal Society of medicine, a group of surgeons are of the opinion that the great master was "afflicted by an illness involving his joints". They have used his portraits as evidence to argue their view. During his life, he complained of what he felt to be 'gout'. Later he complained of his sore and stiff hands which the doctors would find to be natural for someone who was engaged in

handmade art. The doctors found corroboration of those claims in portraits of the artist that show a hanging left hand with both degenerative and non-degenerative changes. They attribute the pain not just to arthritis, but to the stress of hammering and chiselling and note that though the master was seen hammering days before his death at an old age, he did not write or sign his own letters before his death. In recent times there have been attempts to diagnose famous artists with diseases that were not known during their time. This practice has raised many questions, especially on the issue of ethics in research. It is also inferred from authentic analysis that Michaelangelo persisted in his work until his last days. This theory would emphasize that his artistic subject defined his physical infirmities.

Renaissance painting in Europe was sceptical of

- A. The obsessive medieval method of accuracy
- B. The classical simplicity and lack of control
- C. The case and decorative excess of earlier art
- D. Expressionist technique

ugcnetcse-june2019-paper1 general-awareness passage-reading

36.33.5 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 13



Michaelangelo is famous for having successfully interpreted the human body. His great achievement is that of the painting of David whose hands reach out as a sign of human capability and potential. It is assumed that the time he lived was ripe for exchange of knowledge, development in science and matured enough to advance the horizon of investigation in all fields. Renaissance humanism stressed on a serious rethink on the nature of art that focussed on accurate details. In painting and sculpture, artists focussed on not so casual but verifiable and minute details. Michaelangelo's paintings are no exception to it. In a study published in the journal of the Royal Society of medicine, a group of surgeons are of the opinion that the great master was "afflicted by an illness involving his joints". They have used his portraits as evidence to argue their view. During his life, he complained of what he felt to be 'gout'. Later he complained of his sore and stiff hands which the doctors would find to be natural for someone who was engaged in handmade art. The doctors found corroboration of those claims in portraits of the artist that show a hanging left hand with both degenerative and non-degenerative changes. They attribute the pain not just to arthritis, but to the stress of hammering and chiselling and note that though the master was seen hammering days before his death at an old age, he did not write or sign his own letters before his death. In recent times there have been attempts to diagnose famous artists with diseases that were not known during their time. This practice has raised many questions, especially on the issue of ethics in research. It is also inferred from authentic analysis that Michaelangelo persisted in his work until his last days. This theory would emphasize that his artistic subject defined his physical infirmities.

The controversy that the passage above refers to is whether

- A. Michaelangelo worked under duress
- B. Michaelangelo could contain his physical infirmity by artistic excellence
- C. Michaelangelo submitted to disease
- D. Michaelangelo survived different diseases before pursuing art

ugcnetcse-june2019-paper1 general-awareness passage-reading

36.33.6 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 14



Michaelangelo is famous for having successfully interpreted the human body. His great achievement is that of the painting of David whose hands reach out as a sign of human capability and potential. It is assumed that the time he lived was ripe for exchange of knowledge, development in science and matured enough to advance the horizon of investigation in all fields. Renaissance humanism stressed on a serious rethink on the nature of art that focussed on accurate details. In painting and sculpture, artists focussed on not so casual but verifiable and minute details. Michaelangelo's paintings are no exception to it. In a study published in the journal of the Royal Society of medicine, a group of surgeons are of the opinion that the great master was "afflicted by an illness involving his joints". They have used his portraits as evidence to argue their view. During his life, he complained of what he felt to be 'gout'. Later he complained of his sore and stiff hands which the doctors would find to be natural for someone who was engaged in handmade art. The doctors found corroboration of those claims in portraits of the artist that show a hanging left hand with both degenerative and non-degenerative changes. They attribute the pain not just to arthritis, but to the stress of hammering and chiselling and note that though the master was seen hammering days before his death at an old age, he did not write or sign his own letters before his death. In recent times there have been attempts to diagnose famous artists with diseases that were not known during their time. This practice has raised many questions, especially on the issue of ethics in research. It is also inferred from authentic analysis that Michaelangelo persisted in his work until his last days. This theory would emphasize that his artistic subject defined his physical infirmities.

What generalisations do people subscribe to?

- A. Establishing facts by *DNA* tests
- B. Inferring the essence of character from famous people's handwriting
- C. Carbon dating of the hair of celebrities to draw conclusion on their physical structure
- D. To retroactively diagnose famous artists and public figures of conditions that were not prevalent during their time

ugcnetcse-june2019-paper1 general-awareness passage-reading

36.33.7 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 15



Michaelangelo is famous for having successfully interpreted the human body. His great achievement is that of the painting of David whose hands reach out as a sign of human capability and potential. It is assumed that the time he lived was ripe for exchange of knowledge, development in science and matured enough to advance the horizon of investigation in all fields. Renaissance humanism stressed on a serious rethink on the nature of art that focussed on accurate details. In painting and sculpture, artists focussed on not so casual but verifiable and minute details. Michaelangelo's paintings are no exception to it. In a study published in the journal of the Royal Society of medicine, a group of surgeons are of the opinion that the great master was "afflicted by an illness involving his joints". They have used his portraits as evidence to argue their view. During his life, he complained of what he felt to be 'gout'. Later he complained of his sore and stiff hands which the doctors would find to be natural for someone who was engaged in handmade art. The doctors found corroboration of those claims in portraits of the artist that show a hanging left hand with both degenerative and non-degenerative changes. They attribute the pain not just to arthritis, but to the stress of hammering and chiselling and note that though the master was seen hammering days before his death at an old age, he did not write or sign his own letters before his death. In recent times there have been attempts to diagnose famous artists with diseases that were not known during their time. This practice has raised many questions, especially on the issue of ethics in research. It is also inferred from authentic analysis that Michaelangelo persisted in his work until his last days. This theory would emphasize that his artistic subject defined his physical infirmities. What actually may be concluded from the above passage?

- A. Physical infirmities do dissuade people with capabilities from excelling
- B. Excellence in any form triumphs over extraneous factors including physical ailments
- C. Michaelangelo's gout and other ailments lessened his efficiency
- D. The diseases Michaelangelo faced were due to constant hammering

ugcnetcse-june2019-paper1 general-awareness passage-reading

36.33.8 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 16



The dance of the honeybee conveying to other bees where nectar will be found is an example of

- A. Mass communication
- B. Group communication
- C. Interpersonal communication
- D. Intrapersonal communication

ugcnetcse-june2019-paper1 general-awareness

36.33.9 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 17



Choose the correct sequence of communication from the options given below:

- A. Information – exposure – persuasion – behavioural change
- B. Persuasion – information – behavioural change – exposure
- C. Exposure – information – persuasion – behavioural change
- D. Behavioural change – information – persuasion – exposure

ugcnetcse-june2019-paper1 general-awareness

36.33.10 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 18



Which of the following is a function of mass media?

- 1. To transmit culture
- 2. To formulate national policies
- 3. To help the judiciary take its decision

4. To stabilise the share market

ugcnetcse-june2019-paper1 general-awareness

36.33.11 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 19



In a classroom situation, a teacher organises group discussion to help arrive at a solution of a problem. In terms of a model communication used, it will be called

- A. A transaction model
- B. An interaction model
- C. A horizontal model
- D. A linear model

ugcnetcse-june2019-paper1 general-awareness

36.33.12 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 2



Who developed the theory of 'Multiple Intelligence'?

- A. Alfred Binet
- B. L. Thurstone
- C. Charles Spearman
- D. Howard Gardner

ugcnetcse-june2019-paper1 general-awareness

Answer key

36.33.13 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 3



From the list of the effective teaching behaviours, identify those which are called key behaviours.

- i. Direct, audible and oral delivery to all students
- ii. Encouraging students to elaborate on an answer
- iii. Warm and nurturing relationships with learners
- iv. Varying modes of presentation
- v. Preventing misbehaviour with a minimum of class disruption
- vi. Organising what is to come and summarising what has gone before

Select your answer from the options given below :

- A. i, iv and v
- B. i, ii and iii
- C. ii, iii and iv
- D. iv, v and vi

ugcnetcse-june2019-paper1 general-awareness

36.33.14 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 36



Which of the following is a type of malware intentionally inserted into a software system that will set off a malicious function when specified conditions are met?

- A. Worm
- B. Trojan
- C. Spyware
- D. Logic bomb

ugcnetcse-june2019-paper1 general-awareness

Answer key

36.33.15 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 37



Select the true statement about an Operating System (OS)?

- A. An OS controls peripherals, allocates memory and organises data into fields and records
- B. An OS provides protection against viruses and controls peripherals
- C. An OS controls peripheral, and allocates memory and processor time
- D. An OS controls the processor and peripherals and allows user to connect to the Internet

ugcnetcse-june2019-paper1 general-awareness

Answer key

36.33.16 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 38

Select the option that shows the storage devices in order of capacity from lowest to highest

- A. CD-ROM, DVD-ROM, Blu-ray
C. DVD-ROM, Blu-ray, CD-ROM
- B. Blu-ray, CD-ROM, DVD-ROM
D. DVD-ROM, CD-ROM, Blu-ray

ugcnetcse-june2019-paper1 general-awareness

Answer key

36.33.17 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 39

Which of the following statement(s) is/are True in respect of Wireless Technology?

P : Bluetooth is a wireless technology which can be used to connect a headset to a mobile phone.

Q : Bluetooth is a long range wireless technology and is a low cost means of data transfer.

- A. P only
B. Q only
C. Both P and Q
D. Neither P nor Q

ugcnetcse-june2019-paper1 general-awareness

Answer key

36.33.18 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 4

Which of the following statements explains the concepts of inclusive teaching?

- A. Teacher facilitates the learning of the gifted students
B. Teacher facilitates the learning of the weak students
C. Teacher takes support of parents of the students to make them learn
D. Teacher makes the students of different backgrounds to learn together in the same class

ugcnetcse-june2019-paper1 general-awareness

Answer key

36.33.19 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 40

Which one of the following pairs LEAST matches in respect of computers?

- A. 1 Giga Byte : $(1024) \times (1024) \times (1024) \times 8$ bits
B. CRT : Cathode Ray Tube
C. ROM : Rapid Online Memory
D. CPU : Central Processing Unit

ugcnetcse-june2019-paper1 general-awareness

Answer key

36.33.20 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 41

In the last few years, India has been affected by which of the following tropical cyclones?

- A. Gaja, Hudhud, Bhima
C. Gaja, Hudhud, Ockhi
- B. Hudhud, Bhima, Ockhi
D. Gaja, Bhima, Ockhi

ugcnetcse-june2019-paper1 general-awareness

36.33.21 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 42

Which of the following are priority areas in relation to the Sustainable Development Goals?

- i. No poverty
ii. Zero hunger
iii. Reducing urbanization
iv. Peace, justice and strong institutions

Choose the correct answer from the options given below :

- A. i, ii, iii B. i, iii, iv C. ii, iii, iv D. i, ii, iv

ugcnetcse-june2019-paper1 general-awareness

36.33.22 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 43



Assertion (A) : Methemoglobinemia is a condition in which blood is not able to carry and deliver enough oxygen to the body.

Reason (R) : Consuming drinking water with high nitrate levels may cause methemoglobinemia.

Choose the correct answer from the options given below :

- A. Both (A) and (R) are true and (R) is the correct explanation of (A)
B. Both (A) and (R) are true but (R) is not the correct explanation of (A)
C. (A) is true but (R) is false
D. (A) is false but (R) is true

ugcnetcse-june2019-paper1 general-awareness

36.33.23 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 44



Assertion (A) : High concentration of ozone in the lower troposphere is desirable.

Reason (R) : Ozone present in the atmosphere protects the living organisms on the surface of earth from the harmful ultra-violet radiation of the sun.

Choose the correct answer from the options given below :

- A. Both (A) and (R) are true and (R) is the correct explanation of (A)
B. Both (A) and (R) are true but (R) is not the correct explanation of (A)
C. (A) is true but (R) is false
D. (A) is false but (R) is true

ugcnetcse-june2019-paper1 general-awareness

36.33.24 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 45



Algal blooms in oligotrophic lakes are

- A. very frequent B. frequent C. very rare D. widespread

ugcnetcse-june2019-paper1 general-awareness

Answer key

36.33.25 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 46



In post independence India, which one of the following committee/commission's report deals with all levels of education in India?

- A. Sargeant Commission B. Hartog Committee
C. Kothari Commission D. Radhakrishnan Commission

ugcnetcse-june2019-paper1 general general-awareness

36.33.26 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 47



The Education Commission of India that first took serious note of the problem of Brain Drain was

- A. The Education Commission of India B. The University Education Commission
C. The Calcutta University Commission D. The Sargeant Commission

ugcnetcse-june2019-paper1 general-awareness

36.33.27 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 49



Which one of the following instructional designs is not a part of SWAYAM launched by Government of India?

- A. E-tutorial
- B. E-Content
- C. Physical interaction
- D. Discussion Forum

ugcnetcse-june2019-paper1 general-awareness

Answer key

36.33.28 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 5



Which among the following best describes the Emotional Intelligence of learners?

- i. Understand the emotion of other people and your own
- ii. Express oneself very strongly
- iii. Being rational in thinking
- iv. Adjusting one's emotion as per situation
- v. Being creative and open to criticism
- vi. Accepting other people as they are

Choose your answer from the options given below :

- A. i, iv, and vi
- B. iv, v and vi
- C. i, ii, and iii
- D. ii, iii and iv

ugcnetcse-june2019-paper1 general-awareness

Answer key

36.33.29 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 50



Which of the following NOT correctly matched?

- i. Gyan darshan – Satellite based educational T.V . Channel
- ii. Gyan vani – Educational FM Radio network
- iii. MOOCs – Massive Open Online Credits

Choose the correct answer from the options given below :

- A. Only i and ii
- B. Only ii and iii
- C. Only iii
- D. Only i and iii

ugcnetcse-june2019-paper1 general-awareness

Answer key

36.33.30 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 6



Bibliography given in a research report

- A. Helps those interested in further research
- B. Shows the vast knowledge of the researcher
- C. Makes the report authentic
- D. Is an optional part of the report

ugcnetcse-june2019-paper1 general-awareness

36.33.31 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 7



The research design is specifically related to which of the following features in research?

- i. Sample selection
- ii. Formulation of a plan
- iii. Deciding about the tool for data collection
- iv. Hypothesis making
- v. Choice of a field of inquiry

Select your answer from the options given below:

- A. ii, iii and iv
- B. i, ii and iii
- C. ii, iv and v
- D. iii, iv and v

36.33.32 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 8

Through which research method, the manipulation of an independent variable and its effect on dependent variable is examined with reference to a hypothesis under controlled conditions?

- A. Ex-post facto research
 B. Descriptive research
 C. Case study research
 D. Experimental research

ugcnetcse-june2019-paper1 general-awareness

36.33.33 General Awareness: UGC NET CSE | June 2019 | Part 1 | Question: 9

In which of the following research studies interpretation and meaning get more attention than formulation of generalisations?

- i. Historical studies
 ii. Survey studies
 iii. Philosophical studies
 iv. Ethnographic studies
 v. Hypothetico – deductive studies
 vi. Ex-post facto studies

Choose your answer from the options given below :

- A. i, ii and iii
 B. iv, v and vi
 C. ii, iv and v
 D. i, iii and iv

ugcnetcse-june2019-paper1 general-awareness

36.34 Hashing (2)**36.34.1 Hashing: UGC NET CSE | December 2010 | Part 2 | Question: 24**

Linear probing suffers from a problem known as

- A. Secondary clustering
 B. Primary clustering
 C. Both (A) and (B)
 D. None of these

ugcnetcse-dec2010-paper2 data-structures hashing

Answer key

36.34.2 Hashing: UGC NET CSE | June 2011 | Part 2 | Question: 24

Consider a hash table of size $m = 10000$ and the hash function $h(k) = \lfloor m(KA \bmod 1) \rfloor$ for $A = \frac{(\sqrt{5}-1)}{2}$. The location to the key $k = 123456$ is

- A. 46
 B. 47
 C. 41
 D. 43

ugcnetcse-june2011-paper2 data-structures hashing

Answer key

36.35 Image Compression (1)**36.35.1 Image Compression: UGC NET CSE | August 2016 | Part 3 | Question: 58**

Which of the following is not used in standard JPEG image compression ?

- A. Huffman coding
 B. Runlength encoding
 C. Zig-zag scan
 D. K-L Transform

ugcnetcse-aug2016-paper3 digital-image-processing image-compression

Answer key

36.36 Information System (1)

36.36.1 Information System: UGC NET CSE | December 2015 | Part 3 | Question: 48



As compared to rental and leasing methods to acquire computer systems for a Management Information System (MIS), purchase method has following advantage:

- A. It has high level of flexibility
- B. It does not require cash up-front
- C. It is a business investment
- D. Little risk of obsolescence

ugcnetcse-dec2015-paper3 information-system

Answer key

36.37 Interpreter (1)

36.37.1 Interpreter: UGC NET CSE | June 2011 | Part 2 | Question: 14



The scheme of which interpreter translates the source program is known as

- A. Paragraph by paragraph
- B. Instruction by instruction
- C. Line by line
- D. None of the above

ugcnetcse-june2011-paper2 compiler-design interpreter

Answer key

36.38 Is&software Engineering (1)

36.38.1 Is&software Engineering: UGC NET CSE | June 2011 | Part 2 | Question: 41



Which one of the items listed below is not one of the software engineering layers ?

- A. Process
- B. Manufacturing
- C. Method
- D. Tools

ugcnetcse-june2011-paper2 is&software-engineering

Answer key

36.39 Java (1)

36.39.1 Java: UGC NET CSE | June 2011 | Part 2 | Question: 15



Portable program means

- A. Program with wheels
- B. Independent from its authors
- C. Independent of platform
- D. None of the above

ugcnetcse-june2011-paper2 programming java

Answer key

36.40 Job (1)

36.40.1 Job: UGC NET CSE | June 2011 | Part 2 | Question: 32



A special software that is used to create a job queue is called

- A. Drive
- B. Spooler
- C. Interpreter
- D. Linkage editor

ugcnetcse-june2011-paper2 operating-system job

Answer key

36.41 Kruskals Algorithm (1)

36.41.1 Kruskals Algorithm: UGC NET CSE | June 2008 | Part 2 | Question: 2



The complexity of Kruskal's minimum spanning tree algorithm on a graph with ' n ' nodes and ' e ' edges is :

- A. $O(n)$
- B. $O(n \log n)$
- C. $O(e \log n)$
- D. $O(e)$

Answer key 

36.42

Linear Programming (3)

36.42.1 Linear Programming: UGC NET CSE | August 2016 | Part 3 | Question: 61



Consider the following linear programming problem :

$$\max. z = 0.50x_2 - 0.10x_1$$

Subject to the constraints

$$2x_1 + 5x_2 \leq 80$$

$$x_1 + x_2 \leq 20$$

$$\text{and } x_1, x_2 \geq 0$$

The total maximum profit (z) for the above problem is :

A. 6

B. 8

C. 10

D. 12

ugcnetcse-aug2016-paper3 linear-programming

Answer key 

36.42.2 Linear Programming: UGC NET CSE | January 2017 | Part 3 | Question: 70



Consider the following LPP:

$$\text{Min. } Z = x_1 + x_2 + x_3$$

$$\text{Subject to } 3x_1 + 4x_3 \leq 5$$

$$5x_1 + x_2 + 6x_3 = 7$$

$$8x_1 + 9x_3 \geq 2,$$

$$x_1, x_2, x_3 \geq 0$$

The standard form of this LPP shall be:

A.

$$\text{Min. } Z = x_1 + x_2 + x_3 + 0x_4 + 0x_5$$

$$\text{Subject to } 3x_1 + 4x_3 + x_4 = 5;$$

$$5x_1 + x_2 + 6x_3 = 7;$$

$$8x_1 + 9x_3 - x_5 = 2;$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$

$$\text{Min. } Z = x_1 + x_2 + x_3 + 0x_4 + 0x_5 - 1(x_6) - 1(x_7)$$

$$\text{Subject to } 3x_1 + 4x_3 + x_4 = 5;$$

$$\text{B. } 5x_1 + x_2 + 6x_3 + x_6 = 7;$$

$$8x_1 + 9x_3 - x_5 + x_7 = 2;$$

$$x_1 \text{ to } x_7 \geq 0$$

C.

$$\text{Min. } Z = x_1 + x_2 + x_3 + 0x_4 + 0x_5 + 0x_6$$

$$\text{Subject to } 3x_1 + 4x_3 + x_4 = 5;$$

$$5x_1 + x_2 + 6x_3 = 7;$$

$$8x_1 + 9x_3 - x_5 + x_6 = 2;$$

$$x_1 \text{ to } x_6 \geq 0$$

$$\begin{aligned} \text{Min. } Z &= x_1 + x_2 + x_3 + 0x_4 + 0x_5 + 0x_6 + 0x_7 \\ \text{Subject to } &3x_1 + 4x_3 + x_4 = 5; \\ \text{D. } &5x_1 + x_2 + 6x_3 + x_6 = 7 \\ &8x_1 + 9x_3 - x_5 + x_7 = 2; \\ &x_1 \text{ to } x_7 \geq 0 \end{aligned}$$

ugcnetcse-jan2017-paper3 linear-programming

Answer key 

36.42.3 Linear Programming: UGC NET CSE | June 2013 | Part 3 | Question: 23



At any iteration of simplex method if $\Delta_j(Z_j - C_j)$ corresponding to any non-basic variable X_j is obtained as zero, the solution under the test is

- A. Degenerate solution
- B. Unbounded solution
- C. Alternative solution
- D. Optimal solution

ugcnetcse-june2013-paper3 linear-programming

Answer key 

36.43

Linker (2)

36.43.1 Linker: UGC NET CSE | June 2011 | Part 2 | Question: 31



Object modules generated by assembler that contains unresolved external references are resolved for two or more object module by a/an

- A. Operating system
- B. Loader
- C. Linker
- D. Compiler

ugcnetcse-june2011-paper2 operating-system linker

Answer key 

36.43.2 Linker: UGC NET CSE | June 2011 | Part 2 | Question: 35



Which of the following can be accessed by transfer vector approach of linking ?

- A. External data segments
- B. External subroutine
- C. Data located in other procedure
- D. All of the above

ugcnetcse-june2011-paper2 operating-system linker

Answer key 

36.44

Linux (1)

36.44.1 Linux: UGC NET CSE | June 2011 | Part 2 | Question: 40



Which command allows you to view your file 24 lines at a time ?

- A. More
- B. Cat
- C. Pg
- D. None of the above

ugcnetcse-june2011-paper2 unix linux

36.45

Management Information System (2)

36.45.1 Management Information System: UGC NET CSE | June 2011 | Part 2 | Question: 45



Many causes of the software crisis can be traced to mythology based on

- A. Management Myths
- B. Customer Myths
- C. Practitioner Myths
- D. All of the above

ugcnetcse-june2011-paper2 management-information-system

Answer key 



Match the following :

List-I

- (a) Intelligence
- (b) Knowledge
- (c) Information
- (d) Data

List-II

- (i) Contextual, tacit, transfer needs learning
- (ii) Scattered facts, easily transferrable
- (iii) Judgemental
- (iv) Codifiable, endorsed with relevance and purpose

Codes :

- A. (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)
- C. (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

- B. (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
- D. (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

ugcnetcse-june2015-paper3 management-information-system

Answer key

36.46

Microprocessors (1)

36.46.1 Microprocessors: UGC NET CSE | June 2014 | Part 3 | Question: 45



One of the main features that distinguish microprocessor from micro-computers is

- A. Words are usually larger in microprocessors.
- B. Words are shorter in microprocessors.
- C. Microprocessor does not contain *I/O* devices.
- D. None of the above.

ugcnetjune2014iii microprocessors

Answer key

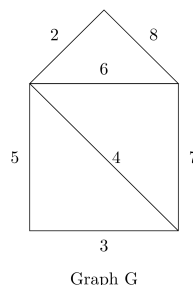
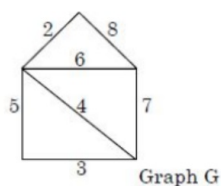
36.47

Minimum Spanning Tree (1)

36.47.1 Minimum Spanning Tree: UGC NET CSE | December 2019 | Part 2 | Question: 6



The weight of minimum spanning tree in graph G , calculated using Kruskal's algorithm is:



A. 14

B. 15

C. 17

D. 18

ugcnetcse-dec2019-paper2 algorithms minimum-spanning-tree

Answer key

36.48

Mis (1)

36.48.1 Mis: UGC NET CSE | December 2010 | Part 2 | Question: 49



The basic concepts of ethics in information society is/are

- A. Responsibility
C. Liability
- B. Accountability
D. All of the above

ugcnetcse-dec2010-paper2 mis

Answer key

36.49

Mobile Communication (2)

36.49.1 Mobile Communication: UGC NET CSE | January 2017 | Part 2 | Question: 48



Which of the following is/are true w.r.t. applications of mobile computing?

- a. Travelling of salesman
A. a true; b false.
- b. Location awareness services
B. Both a and b are true.
- C. Both a and b are false.
D. a false; b true.

ugcnetjan2017ii mobile-communication

Answer key

36.49.2 Mobile Communication: UGC NET CSE | January 2017 | Part 2 | Question: 49



In 3G network, W-CDMA is also known as UMTS. The minimum spectrum allocation required for W-CDMA is

- A. 2 MHz
B. 20 KHz
C. 5 KHz
D. 5 MHz

ugcnetjan2017ii mobile-communication csma-cd

Answer key

36.50

Multiple Document Interface (1)

36.50.1 Multiple Document Interface: UGC NET CSE | August 2016 | Part 3 | Question: 68



Which of the following statements is incorrect for a Windows Multiple Document Interface (MDI) ?

- A. Each document in an MDI application is displayed in a separate child window within the client area of the application's main window.
- B. An MDI application has three kinds of windows namely a frame window, an MDI client window and number of child windows.
- C. An MDI application can support more than one kind of document.
- D. An MDI application displays output in the client area of the frame window.

ugcnetcse-aug2016-paper3 windows-programming multiple-document-interface

36.51

Natural Join (1)

36.51.1 Natural Join: UGC NET CSE | August 2016 | Part 3 | Question: 9



Suppose database table $T_1(P, R)$ currently has tuples (10, 5), (15, 8), (25, 6) and table $T_2(A, C)$ currently has (10, 6), (25, 3), (10, 5). Consider the following three relational algebra queries RA_1 , RA_2 and RA_3 :

$RA_1 : T_1 \bowtie_{T_1.P=T_2.A} T_2$ where $\bowtie$ is natural join symbol

$RA_2 : T_1 \bowtie_{T_1.P=T_2.A} T_2$ where $\bowtie$ is left outer join symbol

$RA_3 : T_1 \bowtie_{T_1.P=T_2.A} T_2$ and $T_1.R=T_2.C$

The number of tuples in the resulting table of RA_1 , RA_2 and RA_3 are given by :

- A. 2, 4, 2 respectively
B. 2, 3, 2 respectively

C. 3,3,1 respectively

D. 3,4,1 respectively

ugcnetcse-aug2016-paper3 databases natural-join

Answer key 

36.52

Natural Language Processing (1)

36.52.1 Natural Language Processing: UGC NET CSE | June 2014 | Part 3 | Question: 29



STRIPS addresses the problem of efficiently representing and implementation of a planner. It is not related to which one of the following?

A. SHAKEY

B. SRI

C. NLP

D. None of these

ugcnetjune2014iii natural-language-processing

Answer key 

36.53

Network Switching (1)

36.53.1 Network Switching: UGC NET CSE | June 2011 | Part 2 | Question: 30



Which of the following switching techniques is most suitable for interactive traffic ?

A. Circuit switching

B. Message switching

C. Packet switching

D. All of the above

ugcnetcse-june2011-paper2 computer-networks network-switching

Answer key 

36.54

Neural Network (6)

36.54.1 Neural Network: UGC NET CSE | June 2013 | Part 3 | Question: 73



A fuzzy set A on R is _____ iff $A(\lambda x_1 + (1 - \lambda)x_2) \geq \min[A(x_1), A(x_2)]$ for all $x_1, x_2 \in R$ and all $\lambda \in [0, 1]$ where minimum denotes the minimum operator.

A. Support

B. α - cut

C. Convex

D. Concave

ugcnetcse-june2013-paper3 neural-network fuzzy-set

Answer key 

36.54.2 Neural Network: UGC NET CSE | June 2013 | Part 3 | Question: 74



If A and B are two fuzzy sets with membership functions

$$\mu_A(x) = \{0.6, 0.5, 0.1, 0.7, 0.8\}$$

$$\mu_B(x) = \{0.9, 0.2, 0.6, 0.8, 0.5\}$$

Then the value of $\mu_{\overline{A \cup B}}(x)$ will be

A. $\{0.9, 0.5, 0.6, 0.8, 0.8\}$

B. $\{0.6, 0.2, 0.1, 0.7, 0.5\}$

C. $\{0.1, 0.5, 0.4, 0.2, 0.2\}$

D. $\{0.1, 0.5, 0.4, 0.2, 0.3\}$

ugcnetcse-june2013-paper3 neural-network fuzzy-set

Answer key 

36.54.3 Neural Network: UGC NET CSE | Junet 2015 | Part 3 | Question: 70



Consider the two class classification task that consists of the following points:

Class C_1 : $[1 \ 1.5] \quad [1 \ -1.5]$

Class C_2 : $[-2 \ 2.5]$ $[-2 \ -2.5]$

The decision boundary between the two classes using single perceptron is given by:

- A. $x_1 + x_2 + 1.5 = 0$ B. $x_1 + x_2 - 1.5 = 0$ C. $x_1 + 1.5 = 0$ D. $x_1 - 1.5 = 0$

ugcnetcse-june2015-paper3 fuzzy-set neural-network

Answer key 

36.54.4 Neural Network: UGC NET CSE | Junet 2015 | Part 3 | Question: 71



Let A and B be two fuzzy integers defined as:

$A = \{(1, 0.3), (2, 0.6), (3, 1), (4, 0.7), (5, 0.2)\}$

$B = \{(10, 0.5), (11, 1), (12, 0.5)\}$

Using fuzzy arithmetic operation given by

$$\mu_{A+B(z)} = \bigoplus_{x+y=z} (\mu_A(x) \otimes \mu_B(y))$$

$f(A + B)$ is _____. [Note: $\oplus \equiv \max$; $\otimes \equiv \min$]

- A. $\{(11, 0.8), (13, 1), (15, 1)\}$
 B. $\{(11, 0.3), (12, 0.5), (13, 1), (14, 1), (15, 1), (16, 0.5), (17, 0.2)\}$
 C. $\{(11, 0.3), (12, 0.5), (13, 0.6), (14, 1), (15, 1), (16, 0.5), (17, 0.2)\}$
 D. $\{(11, 0.3), (12, 0.5), (13, 0.6), (14, 1), (15, 0.7), (16, 0.5), (17, 0.2)\}$

ugcnetcse-june2015-paper3 fuzzy-set neural-network

Answer key 

36.54.5 Neural Network: UGC NET CSE | Junet 2015 | Part 3 | Question: 72



Suppose the function y and a fuzzy integer number around - 4 for x are given as $y = (x - 3)^2 + 2$. Around - 4 = $\{(2, 0.3), (3, 0.6), (4, 1), (5, 0.6), (6, 0.3)\}$ respectively. Then $f(\text{Around-4})$ is given by

- A. $\{(2, 0.6), (3, 0.3), (6, 1), (11, 0.3)\}$ B. $\{(2, 0.6), (3, 1), (6, 1), (11, 0.3)\}$
 C. $\{(2, 0.6), (3, 1), (6, 0.6), (11, 0.3)\}$ D. $\{(2, 0.6), (3, 0.3), (6, 0.6), (11, 0.3)\}$

ugcnetcse-june2015-paper3 fuzzy-set neural-network

Answer key 

36.54.6 Neural Network: UGC NET CSE | September 2013 | Part 3 | Question: 27



Support of a fuzzy set $A = \left\{ \frac{x_1}{0.2}, \frac{x_2}{0.15}, \frac{x_3}{0.9}, \frac{x_4}{0.95}, \frac{x_5}{0.15} \right\}$ within a universal set X is given as

- A. $\left\{ \frac{x_1}{0.15}, \frac{x_2}{0.15}, \frac{x_3}{0.15}, \frac{x_4}{0.15}, \frac{x_5}{0.15} \right\}$
 B. $\left\{ \frac{x_1}{0.95}, \frac{x_2}{0.95}, \frac{x_3}{0.95}, \frac{x_4}{0.95}, \frac{x_5}{0.95} \right\}$
 C. $\{x_3, x_4\}$
 D. $\{x_1, x_2, x_3, x_4, x_5\}$

ugcnetcse-sep2013-paper3 neural-network fuzzy-set

Answer key 

36.55

Non Gatecse (9)

36.55.1 Non Gatecse: UGC NET CSE | January 2017 | Part 2 | Question: 1



Consider a sequence F_{00} defined as :

$$F_{00}(0) = 1, F_{00}(1) = 1$$

$$F_{00}(n) = \frac{10 \cdot F_{00}(n-1) + 100}{F_{00}(n-2)} \text{ for } n \geq 2$$

Then what shall be the set of values of the sequence F_{00} ?

- A. (1, 110, 1200) B. (1, 110, 600, 1200)
C. (1, 2, 55, 110, 600, 1200) D. (1, 55, 110, 600, 1200)

ugcnetjan2017ii non-gatecse

Answer key 

36.55.2 Non Gatecse: UGC NET CSE | January 2017 | Part 3 | Question: 13



Which of the following statement(s) is/are correct?

- A. Persistence is the term used to describe the duration of phosphorescence.
B. The control electrode is used to turn the electron beam on and off.
C. The electron gun creates a source of electrons which are focused into a narrow beam directed at the face of CRT.
D. All of the above

ugcnetcse-jan2017-paper3 non-gatecse

Answer key 

36.55.3 Non Gatecse: UGC NET CSE | January 2017 | Part 3 | Question: 14



A segment is any object described by GKS commands and data that start with CREATE SEGMENT and Terminates with CLOSE SEGMENT command. What functions can be performed on these segments?

- A. Translation and Rotation B. Panning and Zooming
C. Scaling and Shearing D. Translation, Rotation, Panning and Zooming

ugcnetcse-jan2017-paper3 non-gatecse

Answer key 

36.55.4 Non Gatecse: UGC NET CSE | January 2017 | Part 3 | Question: 15



Match the following :

List-I

- a. Glass
b. Conductive coating
c. Liquid crystal
d. Polarized film

List-II

- i. Contains liquid crystal and serves as a bonding surface for a conductive coating
ii. Acts as a conductor so that a voltage can be applied across the liquid crystal
iii. A substance which will polarize light when a voltage is applied to it
iv. A transparent sheet that polarizes light

Codes :

- A. a-i, b-ii, c-iii, d-iv B. a-i, b-iii, c-ii, d-iv
C. a-iv, b-iii, c-ii, d-i D. a-iv, b-ii, c-i, d-iii

ugcnetcse-jan2017-paper3 non-gatecse

Answer key 

36.55.5 Non Gatecse: UGC NET CSE | January 2017 | Part 3 | Question: 57



In Artificial Intelligence(AI), what is present in the planning graph?

- A. Sequence of levels B. Literals
C. Variables D. Heuristic estimates

ugcnetcse-jan2017-paper3 non-gatecse

Answer key 

36.55.6 Non Gatecse: UGC NET CSE | January 2017 | Part 3 | Question: 58

What is the best method to go for the game playing problem?

- A. Optimal Search
- B. Random Search
- C. Heuristic Search
- D. Stratified Search

ugcnetcse-jan2017-paper3 non-gatecse

Answer key

36.55.7 Non Gatecse: UGC NET CSE | January 2017 | Part 3 | Question: 73

Which of the following neural networks uses supervised learning?

- A. Multilayer perception
 - B. Self organizing feature map
 - C. Hopfield network
-
- A. (A) only
 - B. (B) only
 - C. (A) and (B) only
 - D. (A) and (C) only

ugcnetcse-jan2017-paper3 non-gatecse

Answer key

36.55.8 Non Gatecse: UGC NET CSE | January 2017 | Part 3 | Question: 74

Unix command to change the case of first three lines of file “shortlist” from lower to upper

- A. `$ tr '[a-z]' '[A-Z]' shortlist | head - 3`
- B. `$ head - 3 shortlist | tr '[a-z]' '[A-Z]'`
- C. `$ tr head - 3 shortlist '[A-Z]' '[a-z]'`
- D. `$tr shortlist head - 3 '[a-z]' '[A-Z]'`

ugcnetcse-jan2017-paper3 non-gatecse

Answer key

36.55.9 Non Gatecse: UGC NET CSE | January 2017 | Part 3 | Question: 75

Match the following vi commands in Unix :

List-I**List-II**

- | | |
|--------|--|
| a. :w | i. saves the file and quits editing mode |
| b. :x | ii. escapes unix shell |
| c. :q | iii. saves file and remains in editing mode |
| d. :sh | iv. quits editing mode and no change are saved to the file |

Codes :

- A. a-ii, b-iii, c-i, d-iv
- B. a-iv, b-iii, c-ii, d-i
- C. a-iii, b-iv, c-i, d-ii
- D. a-iii, b-i, c-iv, d-ii

ugcnetcse-jan2017-paper3 non-gatecse

Answer key

36.56

Number System (1)

36.56.1 Number System: UGC NET CSE | December 2005 | Part 2 | Question: 12



The bitwise OR of 35 with 7 in C will be :

- A. 35 B. 7 C. 42 D. 39

ugcnetcse-dec2005-paper2 digital-logic number-system

Answer key

36.57

Oops Concept (1)

36.57.1 Oops Concept: UGC NET CSE | August 2016 | Part 2 | Question: 14



In C++, polymorphism requires :

- A. Inheritance only B. Virtual functions only
C. References only D. Inheritance, Virtual functions and references

ugcnetcse-aug2016-paper2 c++ oops-concept

Answer key

36.58

Operator Overloading (1)

36.58.1 Operator Overloading: UGC NET CSE | June 2011 | Part 2 | Question: 13



The goal of operator overloading is

- A. To help the user of a class B. To help the developer of a class
C. To help define friend function D. None of the above

ugcnetcse-june2011-paper2 object-oriented-programming operator-overloading

Answer key

36.59

Parsing (1)

36.59.1 Parsing: UGC NET CSE | June 2008 | Part 2 | Question: 33



At the end of parsing,

- A. tokens are identified. B. set of instructions are identified.
C. the syntactic groups are identified. D. machine instructions are identified.

ugcnetcse-june2008-paper2 parsing

Answer key

36.60

Pipelining (1)

36.60.1 Pipelining: UGC NET CSE | December 2019 | Part 2 | Question: 9



A non-pipelined system takes 30ns to process a task. The same task can be processed in a four-segment pipeline with a clock cycle of 10ns. Determine the speed up of the pipeline for 100 tasks.

- A. 3 B. 4 C. 3.91 D. 2.91

ugcnetcse-dec2019-paper2 co-and-architecture pipelining

Answer key

36.61

Process Scheduling (2)

36.61.1 Process Scheduling: UGC NET CSE | June 2011 | Part 2 | Question: 38



A relationship between processes such that each has some part (critical section) which must not be executed while the critical section of another is being executed, is known as

- A. Semaphore
- B. Mutual exclusion
- C. Multiprogramming
- D. Message passing

ugcnetcse-june2011-paper2 operating-system process-scheduling

Answer key

36.61.2 Process Scheduling: UGC NET CSE | June 2011 | Part 2 | Question: 39



How many states can a process be in ?

- A. 3
- B. 4
- C. 2
- D. 5

ugcnetcse-june2011-paper2 operating-system process-scheduling

Answer key

36.62 Prolog (1)

36.62.1 Prolog: UGC NET CSE | August 2016 | Part 3 | Question: 75



Which formal system provides the semantic foundation for Prolog ?

- A. Predicate calculus
- B. Lambda calculus
- C. Hoare logic
- D. Propositional logic

ugcnetcse-aug2016-paper3 artificial-intelligence prolog

Answer key

36.63 Quadratic Equations (1)

36.63.1 Quadratic Equations: UGC NET CSE | June 2013 | Part 3 | Question: 11



The golden ratio φ and its conjugate $\bar{\varphi}$ both satisfy the equation

- A. $x^3 - x - 1 = 0$
- B. $x^3 + x - 1 = 0$
- C. $x^2 - x - 1 = 0$
- D. $x^2 + x - 1 = 0$

ugcnetcse-june2013-paper3 quadratic-equations

Answer key

36.64 Queue (1)

36.64.1 Queue: UGC NET CSE | June 2011 | Part 2 | Question: 25



When the priority queue is represented by max heap, the insertion and deletion of an element can be performed in (queue containing n elements)

- A. $\theta(n)$ and $\theta(1)$ respectively
- B. $\theta(n)$ and $\theta(n)$ respectively
- C. $\theta(1)$ and $\theta(1)$ respectively
- D. None of the above

ugcnetcse-june2011-paper2 data-structures queue binary-heap

Answer key

36.65 Rdbms (2)

36.65.1 Rdbms: UGC NET CSE | August 2016 | Part 2 | Question: 16



DBMS provides the facility of accessing data from a database through

- A. DDL
- B. DML
- C. DBA
- D. Schema

ugcnetcse-aug2016-paper2 databases rdbms

(c) The difference between the project operator (π) in relational algebra and the SELECT keyword in SQL is that if the resulting table/set has more than one occurrences of the same tuple, then π will return only one of them, while SQL SELECT will return all. One can determine that :

- D. (a), (b) and (c) are true.



Then, the resulting relation, T , which relates elements of universe x to the elements of universe z using max-min composition is given by:

- $$T = \begin{matrix} & \begin{matrix} z_1 & z_2 & z_3 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} 0.6 & 0.5 & 0.5 \\ 0.7 & 0.7 & 0.7 \end{bmatrix} \end{matrix}$$

36.67.1 Semantic Nets: UGC NET CSE | August 2016 | Part 3 | Question: 72

In Artificial Intelligence , a semantic network

- A. Is a graph-based method of knowledge representation where nodes represent concepts and arcs represent relations between concepts.
- B. Is a graph-based method of knowledge representation where nodes represent relations between concepts and arcs represent concepts.
- C. Represents an entity as a set of slots and associated rules.
- D. Is a subset of first-order logic.

ugcnetcse-aug2016-paper3 artificial-intelligence semantic-nets

Answer key

36.68**Shell (1)****36.68.1 Shell: UGC NET CSE | September 2013 | Part 3 | Question: 4**

Which of the following is not a part of an expert system shell?

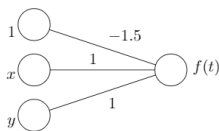
- A. Knowledge base
- B. Interference engine
- C. Explanation facility
- D. None of the above

ugcnetcse-sep2013-paper3 unix linux shell

Answer key

36.69**Single Layer Perceptron (1)****36.69.1 Single Layer Perceptron: UGC NET CSE | June 2013 | Part 3 | Question: 75**

Consider a single perception with weights as given in the following figure:



and $f(t)$ is defined as

$$f(t) = \begin{cases} 1, & t > 0 \\ 0, & t \leq 0 \end{cases}$$

The above perception can solve

- A. OR problem
- B. AND problem
- C. XOR problem
- D. All of the above

ugcnetcse-june2013-paper3 neural-network single-layer-perceptron

Answer key

36.70**Software Development Life Cycle Models (1)****36.70.1 Software Development Life Cycle Models: UGC NET CSE | June 2011 | Part 2 | Question: 42**

What is the first stage in program development ?

- A. Specification and design
- B. System Analysis
- C. Testing
- D. None of the above

ugcnetcse-june2011-paper2 is&software-engineering software-development-life-cycle-models

Answer key

36.71**Software Engineering Framework (1)**

36.71.1 Software Engineering Framework: UGC NET CSE | June 2011 | Part 2 | Question: 44



Which of these are the 5 generic software engineering framework activities ?

- A. Communication, planning, modelling, construction, deployment
- B. Communication, risk management, measurement, production, reviewing
- C. Analysis, designing, programming, Debugging, maintenance
- D. Analysis, planning, designing, programming, testing

ugcnetcse-june2011-paper2 is&software-engineering software-engineering-framework

Answer key

36.72 Software Testing (1)

36.72.1 Software Testing: UGC NET CSE | Junet 2015 | Part 3 | Question: 43



Verification:

- A. refers to the set of activities that ensure that software correctly implements a specific function
- B. Gives answer to the question - Are we building the right product
- C. requires execution of software
- D. both A and B

software-testing non-gatecse ugcnetcse-june2015-paper3

Answer key

36.73 System Software Compilers (1)

36.73.1 System Software Compilers: UGC NET CSE | August 2016 | Part 3 | Question: 45



Match each software term in List-I to its description in List-II :

List-I

List-II

- | | |
|-------------------------|---|
| I. Wizards | a. forms that provides structure for a document. |
| II. Templates | b. A series of commands grouped into a single command. |
| III. Macros | c. A single program that incorporates most commonly used tools. |
| IV. Integrated Software | d. step by step guide in application software. |
| V. Software Suite | e. Bundled group of software program |

Codes :

- | | |
|--------------------------------|--------------------------------|
| A. a-ii, b-iii, c-iv, d-i, e-v | B. a-ii, b-i, c-iv, d-iii, e-v |
| C. a-iv, b-iii, c-v, d-i, e-ii | D. a-iv, b-iii, c-ii, d-v, e-i |

ugcnetcse-aug2016-paper3 system-software-compilers macros

Answer key

36.74 Transaction and Concurrency (3)

36.74.1 Transaction and Concurrency: UGC NET CSE | June 2011 | Part 2 | Question: 16



Which of the following is the recovery management technique in DDBMS ?

- | | |
|---------------------------|---------------------|
| A. 2PC (Two Phase Commit) | B. Backup |
| C. Immediate update | D. All of the above |

ugcnetcse-june2011-paper2 databases transaction-and-concurrency

Answer key

36.74.2 Transaction and Concurrency: UGC NET CSE | June 2011 | Part 2 | Question: 18

The basic variants of time-stampbased method of concurrency control are

- A. Total time stamp-ordering
 B. Partial time stamp ordering
 C. Multiversion Time stamp ordering
 D. All of the above

ugcnetcse-june2011-paper2 databases transaction-and-concurrency

Answer key

36.74.3 Transaction and Concurrency: UGC NET CSE | June 2011 | Part 2 | Question: 19

A transaction can include following basic database access operations :

- A. Read_item(X) B. Write_item(X) C. Both (A) and (B) D. None of these

ugcnetcse-june2011-paper2 databases transaction-and-concurrency

Answer key

36.75 Transportation Problem (3)**36.75.1 Transportation Problem: UGC NET CSE | August 2016 | Part 3 | Question: 63**

Consider the following statements :

- (a) Assignment problem can be used to minimize the cost.
 (b) Assignment problem is a special case of transportation problem.
 (c) Assignment problem requires that only one activity be assigned to each resource.

Which of the following options is correct ?

- A. (a) and (b) only B. (a) and (c) only C. (b) and (c) only D. (a), (b) and (c)

ugcnetcse-aug2016-paper3 linear-programming transportation-problem

Answer key

36.75.2 Transportation Problem: UGC NET CSE | June 2013 | Part 3 | Question: 24

A basic feasible solution to a m-origin, n-destination transportation problem is said to be _____ if the number of positive allocations are less than $m+n-1$.

- A. degenerate B. non- degenerate C. unbounded D. unbalanced

ugcnetcse-june2013-paper3 linear-programming transportation-problem

Answer key

36.75.3 Transportation Problem: UGC NET CSE | June 2013 | Part 3 | Question: 25

The total transportation cost in an initial basic feasible solution to the following transportation problem using Vogel's Approximation method is

	W1	W2	W3	W4	W5	Supply
F1	4	2	3	2	6	8
F2	5	4	5	2	1	12
F3	6	5	4	7	3	14
Demand	4	4	6	8	8	

- A. 76 B. 80 C. 90 D. 96

ugcnetcse-june2013-paper3 linear-programming transportation-problem

Answer key

36.76 Tree (2)

36.76.1 Tree: UGC NET CSE | June 2008 | Part 2 | Question: 1

Which of the following does not define a tree?

- A. A tree is a connected acyclic graph.
- B. A tree is a connected graph with $n - 1$ edges where ' n ' is the number of vertices in the graph.
- C. A tree is an acyclic graph with $n - 1$ edges where ' n ' is the number of vertices in the graph.
- D. A tree is a graph with no cycles.

ugcnetcse-june2008-paper2 directed-acyclic-graph tree

Answer key

36.76.2 Tree: UGC NET CSE | June 2011 | Part 2 | Question: 21

The number of different trees with 8 nodes is

- A. 256
- B. 255
- C. 248
- D. None of these

ugcnetcse-june2011-paper2 algorithms tree

Answer key

36.77**Unix (4)****36.77.1 Unix: UGC NET CSE | August 2016 | Part 3 | Question: 67**

Consider the following operations to be performed in Unix : "The pipe sorts all files in the current directory modified in the month of "June" by order of size and prints them to the terminal screen. The sort option skips ten fields then sorts the lines in numeric order." Which of the following Unix command will perform above set of operations ?

- A. `ls -l | grep "June" | sort + 10n`
- B. `ls -l | grep "June" | sort + 10r`
- C. `ls -l | grep -v "June" | sort + 10n`
- D. `ls -l | grep -n "June" | sort + 10x`

ugcnetcse-aug2016-paper3 unix

Answer key

36.77.2 Unix: UGC NET CSE | August 2016 | Part 3 | Question: 69

Which of the following statement(s) is/are True regarding 'nice' command of UNIX ?

- I. It is used to set or change the priority of a process.
- II. A process's nice value can be set at the time of creation.
- III. 'nice' takes a command line as an argument.

- A. *I,II* only
- B. *II,III* only
- C. *I,II,III*
- D. *I,III* only

ugcnetcse-aug2016-paper3 unix

Answer key

36.77.3 Unix: UGC NET CSE | December 2010 | Part 2 | Question: 37

Which of the following command the file names in multiple columns ?

- A. `IS -X`
- B. `IS`
- C. `IS -1`
- D. `IS -f -X`

ugcnetcse-dec2010-paper2 unix

36.77.4 Unix: UGC NET CSE | September 2013 | Part 2 | Question: 48



Which of the following set of Unix commands will always display “WELCOME”?

- A. export title=WELCOME; Echo \$title
- B. title=WELCOME; export \$title ; sh -c “echo \$title”
- C. title=WELCOME; export title ; sh -c “echo \$title”
- D. title=WELCOME; echo \$title

ugcnetsep2013ii unix

Answer key

36.78

Virtual Memory (1)

36.78.1 Virtual Memory: UGC NET CSE | June 2011 | Part 2 | Question: 47



Virtual memory is

- A. Related to virtual reality
- B. A form of ROM
- C. A form of RAM
- D. None of the above

ugcnetcse-june2011-paper2 co-and-architecture virtual-memory

Answer key

36.79

Windows (5)

36.79.1 Windows: UGC NET CSE | December 2008 | Part 2 | Question: 50



(I) Each object in the active directory of windows 2000 has an access control list.

(II) The scheme is a blueprint of all objects in the domain of windows 2000.

Which of the following is true?

- A. only (I)
- B. only (II)
- C. both (I) and (II)
- D. None of the above

ugcnetcse-dec2008-paper2 windows

Answer key

36.79.2 Windows: UGC NET CSE | June 2007 | Part 2 | Question: 37



Windows is a _____ operating system:

- A. Non-preemptive
- B. Preemptive
- C. Multi-user
- D. Real time

ugcnetcse-june2007-paper2 operating-system windows

Answer key

36.79.3 Windows: UGC NET CSE | June 2014 | Part 3 | Question: 38



A part of Windows 2000 operating system that is not portable is

- A. Device Management
- B. Virtual Memory Management
- C. Processor Management
- D. User Interface

ugcnetjune2014iii windows

Answer key

36.79.4 Windows: UGC NET CSE | June 2014 | Part 3 | Question: 47

All the classes necessary for windows programming are available in the module :

- A. win.txt B. win.main C. win.std D. MFC

ugcnetjune2014iii windows

Answer key

36.79.5 Windows: UGC NET CSE | June 2014 | Part 3 | Question: 48

Windows 32 API supports

- A. 16-bit Windows B. 32-bit Windows
C. 64-bit Windows D. All of the above

ugcnetjune2014iii windows

Answer key

36.80**Windows Programming (2)****36.80.1 Windows Programming: UGC NET CSE | June 2009 | Part 2 | Question: 50**

We cannot delete the _____ icon but we can made it invisible.

- A. Recycle
B. My Computer
C. Internet explorer
D. None of the above

ugcnetcse-june2009-paper2 windows-programming

36.80.2 Windows Programming: UGC NET CSE | Junet 2015 | Part 3 | Question: 74

WOW32 is a

- A. Win 32 API library for creating processes and threads
B. Special kind of file system to the NT name space
C. Kernel-mode objects accessible through Win 32 API
D. Special execution environment used to run 16 bit Windows applications on 32-bit machines

ugcnetcse-june2015-paper3 windows-programming

Answer key

Answer Keys

36.0.1	D	36.0.2	D	36.0.3	B	36.0.4	A	36.0.5	C
36.0.6	A	36.0.7	D	36.0.8	D	36.0.9	D	36.0.10	D
36.0.11	C	36.0.12	B	36.0.13	B	36.0.14	B	36.0.15	A
36.0.16	B	36.0.17	D	36.0.18	A	36.0.19	D	36.0.20	B
36.0.21	B	36.0.22	D	36.0.23	D	36.0.24	C	36.0.25	B
36.0.26	D	36.0.27	B	36.0.28	A	36.0.29	C	36.0.30	C
36.0.31	C	36.0.32	Q-Q	36.0.33	C	36.0.34	A	36.0.35	C
36.0.36	B	36.0.37	D	36.0.38	Q-Q	36.0.39	Q-Q	36.0.40	Q-Q
36.0.41	Q-Q	36.0.42	Q-Q	36.0.43	Q-Q	36.0.44	Q-Q	36.0.45	Q-Q

36.0.46	D	36.0.47	Q-Q	36.0.48	Q-Q	36.0.49	Q-Q	36.0.50	Q-Q
36.0.51	Q-Q	36.0.52	Q-Q	36.0.53	C	36.0.54	Q-Q	36.0.55	Q-Q
36.0.56	B	36.0.57	C	36.0.58	C	36.0.59	A	36.0.60	C
36.0.61	A	36.0.62	D	36.0.63	A	36.0.64	Q-Q	36.0.65	Q-Q
36.0.66	Q-Q	36.0.67	Q-Q	36.0.68	Q-Q	36.0.69	Q-Q	36.0.70	Q-Q
36.0.71	Q-Q	36.0.72	Q-Q	36.0.73	Q-Q	36.0.74	Q-Q	36.0.75	Q-Q
36.0.76	Q-Q	36.0.77	Q-Q	36.0.78	Q-Q	36.0.79	Q-Q	36.0.80	Q-Q
36.0.81	Q-Q	36.0.82	Q-Q	36.0.83	Q-Q	36.0.84	Q-Q	36.0.85	Q-Q
36.0.86	Q-Q	36.0.87	Q-Q	36.0.88	Q-Q	36.0.89	Q-Q	36.0.90	Q-Q
36.0.91	Q-Q	36.0.92	Q-Q	36.0.93	C	36.0.94	Q-Q	36.0.95	Q-Q
36.0.96	Q-Q	36.0.97	Q-Q	36.0.98	Q-Q	36.0.99	Q-Q	36.0.100	Q-Q
36.0.101	Q-Q	36.0.102	D	36.0.103	Q-Q	36.0.104	Q-Q	36.0.105	Q-Q
36.0.106	Q-Q	36.0.107	Q-Q	36.0.108	Q-Q	36.0.109	Q-Q	36.0.110	Q-Q
36.0.111	Q-Q	36.0.112	Q-Q	36.0.113	Q-Q	36.0.114	Q-Q	36.0.115	Q-Q
36.0.116	Q-Q	36.0.117	Q-Q	36.0.118	Q-Q	36.0.119	Q-Q	36.0.120	Q-Q
36.0.121	Q-Q	36.0.122	Q-Q	36.0.123	Q-Q	36.0.124	Q-Q	36.0.125	Q-Q
36.0.126	Q-Q	36.0.127	Q-Q	36.0.128	Q-Q	36.0.129	Q-Q	36.0.130	Q-Q
36.0.131	Q-Q	36.0.132	Q-Q	36.0.133	Q-Q	36.0.134	Q-Q	36.0.135	Q-Q
36.0.136	Q-Q	36.0.137	Q-Q	36.0.138	Q-Q	36.0.139	Q-Q	36.0.140	Q-Q
36.0.141	Q-Q	36.0.142	Q-Q	36.0.143	Q-Q	36.0.144	Q-Q	36.0.145	Q-Q
36.0.146	Q-Q	36.0.147	Q-Q	36.0.148	Q-Q	36.0.149	Q-Q	36.0.150	Q-Q
36.0.151	Q-Q	36.0.152	Q-Q	36.0.153	Q-Q	36.0.154	Q-Q	36.0.155	Q-Q
36.0.156	Q-Q	36.0.157	Q-Q	36.0.158	Q-Q	36.0.159	Q-Q	36.0.160	Q-Q
36.0.161	Q-Q	36.0.162	Q-Q	36.0.163	Q-Q	36.0.164	Q-Q	36.0.165	Q-Q
36.0.166	Q-Q	36.0.167	Q-Q	36.0.168	Q-Q	36.0.169	Q-Q	36.0.170	Q-Q
36.0.171	Q-Q	36.0.172	Q-Q	36.0.173	Q-Q	36.0.174	Q-Q	36.0.175	Q-Q
36.0.176	Q-Q	36.0.177	Q-Q	36.0.178	Q-Q	36.0.179	Q-Q	36.0.180	Q-Q
36.0.181	Q-Q	36.0.182	Q-Q	36.0.183	Q-Q	36.0.184	Q-Q	36.0.185	Q-Q
36.0.186	Q-Q	36.0.187	Q-Q	36.0.188	Q-Q	36.0.189	Q-Q	36.0.190	Q-Q
36.0.191	Q-Q	36.0.192	Q-Q	36.0.193	Q-Q	36.0.194	Q-Q	36.0.195	Q-Q
36.0.196	Q-Q	36.0.197	Q-Q	36.0.198	Q-Q	36.0.199	Q-Q	36.0.200	Q-Q
36.0.201	Q-Q	36.0.202	Q-Q	36.0.203	Q-Q	36.0.204	Q-Q	36.0.205	Q-Q
36.0.206	Q-Q	36.0.207	Q-Q	36.0.208	Q-Q	36.0.209	Q-Q	36.0.210	C
36.0.211	Q-Q	36.0.212	Q-Q	36.0.213	Q-Q	36.0.214	Q-Q	36.0.215	Q-Q
36.0.216	Q-Q	36.0.217	Q-Q	36.0.218	Q-Q	36.0.219	Q-Q	36.0.220	Q-Q
36.0.221	Q-Q	36.0.222	Q-Q	36.0.223	Q-Q	36.0.224	Q-Q	36.0.225	Q-Q
36.0.226	Q-Q	36.0.227	Q-Q	36.0.228	Q-Q	36.0.229	Q-Q	36.0.230	Q-Q
36.0.231	Q-Q	36.0.232	Q-Q	36.0.233	Q-Q	36.0.234	Q-Q	36.0.235	C
36.0.236	Q-Q	36.0.237	Q-Q	36.0.238	Q-Q	36.0.239	Q-Q	36.0.240	Q-Q

36.0.241	Q-Q	36.0.242	Q-Q	36.0.243	Q-Q	36.0.244	Q-Q	36.0.245	Q-Q
36.0.246	Q-Q	36.0.247	Q-Q	36.0.248	Q-Q	36.0.249	Q-Q	36.0.250	Q-Q
36.0.251	Q-Q	36.0.252	Q-Q	36.0.253	Q-Q	36.0.254	Q-Q	36.0.255	Q-Q
36.0.256	Q-Q	36.0.257	Q-Q	36.0.258	Q-Q	36.0.259	Q-Q	36.0.260	Q-Q
36.0.261	Q-Q	36.0.262	Q-Q	36.0.263	Q-Q	36.0.264	Q-Q	36.0.265	Q-Q
36.0.266	Q-Q	36.0.267	Q-Q	36.0.268	B	36.0.269	Q-Q	36.0.270	Q-Q
36.0.271	Q-Q	36.0.272	Q-Q	36.0.273	Q-Q	36.0.274	Q-Q	36.0.275	Q-Q
36.0.276	Q-Q	36.0.277	Q-Q	36.0.278	Q-Q	36.0.279	Q-Q	36.0.280	Q-Q
36.0.281	Q-Q	36.0.282	Q-Q	36.0.283	Q-Q	36.0.284	Q-Q	36.0.285	Q-Q
36.0.286	Q-Q	36.0.287	Q-Q	36.0.288	Q-Q	36.0.289	Q-Q	36.0.290	Q-Q
36.0.291	Q-Q	36.0.292	Q-Q	36.0.293	Q-Q	36.0.294	Q-Q	36.0.295	Q-Q
36.0.296	Q-Q	36.0.297	Q-Q	36.0.298	Q-Q	36.0.299	Q-Q	36.0.300	Q-Q
36.0.301	Q-Q	36.0.302	Q-Q	36.0.303	Q-Q	36.0.304	Q-Q	36.0.305	Q-Q
36.0.306	Q-Q	36.0.307	Q-Q	36.0.308	Q-Q	36.0.309	Q-Q	36.0.310	Q-Q
36.0.311	Q-Q	36.0.312	Q-Q	36.0.313	Q-Q	36.0.314	Q-Q	36.0.315	Q-Q
36.0.316	Q-Q	36.0.317	Q-Q	36.0.318	Q-Q	36.0.319	Q-Q	36.0.320	Q-Q
36.0.321	Q-Q	36.0.322	Q-Q	36.0.323	Q-Q	36.0.324	Q-Q	36.0.325	Q-Q
36.0.326	Q-Q	36.0.327	Q-Q	36.0.328	Q-Q	36.0.329	Q-Q	36.0.330	Q-Q
36.0.331	Q-Q	36.0.332	Q-Q	36.0.333	Q-Q	36.0.334	Q-Q	36.0.335	Q-Q
36.0.336	Q-Q	36.0.337	Q-Q	36.0.338	Q-Q	36.0.339	Q-Q	36.0.340	Q-Q
36.0.341	Q-Q	36.0.342	Q-Q	36.0.343	Q-Q	36.0.344	Q-Q	36.0.345	Q-Q
36.0.346	Q-Q	36.0.347	Q-Q	36.0.348	Q-Q	36.0.349	Q-Q	36.0.350	Q-Q
36.0.351	Q-Q	36.0.352	Q-Q	36.0.353	Q-Q	36.0.354	Q-Q	36.0.355	Q-Q
36.0.356	Q-Q	36.0.357	Q-Q	36.0.358	Q-Q	36.0.359	Q-Q	36.0.360	Q-Q
36.0.361	Q-Q	36.0.362	Q-Q	36.0.363	Q-Q	36.0.364	Q-Q	36.0.365	Q-Q
36.0.366	Q-Q	36.0.367	Q-Q	36.0.368	Q-Q	36.0.369	Q-Q	36.0.370	Q-Q
36.0.371	Q-Q	36.0.372	Q-Q	36.0.373	Q-Q	36.0.374	Q-Q	36.0.375	Q-Q
36.0.376	Q-Q	36.0.377	Q-Q	36.0.378	Q-Q	36.0.379	Q-Q	36.0.380	Q-Q
36.0.381	Q-Q	36.0.382	Q-Q	36.0.383	Q-Q	36.0.384	Q-Q	36.0.385	Q-Q
36.0.386	Q-Q	36.0.387	Q-Q	36.0.388	Q-Q	36.0.389	Q-Q	36.0.390	Q-Q
36.0.391	Q-Q	36.0.392	Q-Q	36.0.393	Q-Q	36.0.394	Q-Q	36.0.395	Q-Q
36.0.396	Q-Q	36.0.397	Q-Q	36.0.398	Q-Q	36.0.399	Q-Q	36.0.400	Q-Q
36.0.401	Q-Q	36.0.402	Q-Q	36.0.403	Q-Q	36.0.404	Q-Q	36.0.405	Q-Q
36.0.406	Q-Q	36.0.407	Q-Q	36.0.408	Q-Q	36.0.409	Q-Q	36.0.410	Q-Q
36.0.411	Q-Q	36.0.412	Q-Q	36.0.413	Q-Q	36.0.414	Q-Q	36.0.415	Q-Q
36.0.416	Q-Q	36.0.417	Q-Q	36.0.418	Q-Q	36.0.419	Q-Q	36.0.420	Q-Q
36.0.421	Q-Q	36.0.422	Q-Q	36.0.423	Q-Q	36.0.424	Q-Q	36.0.425	Q-Q
36.0.426	Q-Q	36.0.427	Q-Q	36.0.428	Q-Q	36.0.429	Q-Q	36.0.430	Q-Q
36.0.431	Q-Q	36.0.432	Q-Q	36.0.433	Q-Q	36.0.434	Q-Q	36.0.435	Q-Q

36.0.436	Q-Q	36.0.437	Q-Q	36.0.438	Q-Q	36.0.439	Q-Q	36.0.440	Q-Q
36.0.441	Q-Q	36.0.442	Q-Q	36.0.443	Q-Q	36.0.444	Q-Q	36.0.445	Q-Q
36.0.446	Q-Q	36.0.447	Q-Q	36.0.448	Q-Q	36.0.449	Q-Q	36.0.450	Q-Q
36.0.451	Q-Q	36.0.452	Q-Q	36.0.453	Q-Q	36.0.454	Q-Q	36.0.455	Q-Q
36.0.456	Q-Q	36.0.457	Q-Q	36.0.458	Q-Q	36.0.459	Q-Q	36.0.460	Q-Q
36.0.461	Q-Q	36.0.462	Q-Q	36.0.463	Q-Q	36.0.464	Q-Q	36.0.465	Q-Q
36.0.466	Q-Q	36.0.467	Q-Q	36.0.468	Q-Q	36.0.469	Q-Q	36.0.470	Q-Q
36.0.471	Q-Q	36.0.472	Q-Q	36.0.473	Q-Q	36.0.474	Q-Q	36.0.475	Q-Q
36.0.476	Q-Q	36.0.477	Q-Q	36.0.478	Q-Q	36.0.479	Q-Q	36.0.480	Q-Q
36.0.481	Q-Q	36.0.482	Q-Q	36.0.483	Q-Q	36.0.484	Q-Q	36.0.485	Q-Q
36.0.486	Q-Q	36.0.487	Q-Q	36.0.488	Q-Q	36.0.489	Q-Q	36.0.490	Q-Q
36.0.491	Q-Q	36.0.492	Q-Q	36.0.493	Q-Q	36.0.494	Q-Q	36.0.495	Q-Q
36.0.496	Q-Q	36.0.497	Q-Q	36.0.498	Q-Q	36.0.499	Q-Q	36.0.500	Q-Q
36.0.501	Q-Q	36.0.502	Q-Q	36.0.503	Q-Q	36.0.504	Q-Q	36.0.505	Q-Q
36.0.506	Q-Q	36.0.507	Q-Q	36.0.508	Q-Q	36.0.509	Q-Q	36.0.510	Q-Q
36.0.511	Q-Q	36.0.512	Q-Q	36.0.513	Q-Q	36.0.514	Q-Q	36.0.515	Q-Q
36.0.516	Q-Q	36.0.517	Q-Q	36.0.518	Q-Q	36.0.519	Q-Q	36.0.520	Q-Q
36.0.521	Q-Q	36.0.522	Q-Q	36.0.523	Q-Q	36.0.524	Q-Q	36.0.525	Q-Q
36.0.526	Q-Q	36.0.527	Q-Q	36.0.528	Q-Q	36.1.1	Q-Q	36.2.1	Q-Q
36.3.1	Q-Q	36.4.1	Q-Q	36.5.1	Q-Q	36.6.1	Q-Q	36.7.1	Q-Q
36.8.1	Q-Q	36.9.1	B	36.10.1	C	36.11.1	B	36.12.1	B
36.13.1	Q-Q	36.14.1	Q-Q	36.15.1	Q-Q	36.15.2	Q-Q	36.15.3	Q-Q
36.15.4	Q-Q	36.16.1	Q-Q	36.16.2	Q-Q	36.17.1	Q-Q	36.18.1	Q-Q
36.19.1	Q-Q	36.19.2	B	36.19.3	D	36.20.1	Q-Q	36.21.1	Q-Q
36.22.1	Q-Q	36.23.1	Q-Q	36.23.2	Q-Q	36.24.1	Q-Q	36.25.1	Q-Q
36.25.2	Q-Q	36.25.3	Q-Q	36.26.1	B	36.26.2	B	36.27.1	Q-Q
36.27.2	Q-Q	36.28.1	Q-Q	36.29.1	Q-Q	36.30.1	Q-Q	36.30.2	A
36.31.1	Q-Q	36.32.1	D	36.32.2	C	36.33.1	A	36.33.2	A;D
36.33.3	A	36.33.4	C	36.33.5	B	36.33.6	D	36.33.7	B
36.33.8	B	36.33.9	C	36.33.10	A	36.33.11	A	36.33.12	D
36.33.13	A	36.33.14	D	36.33.15	C	36.33.16	A	36.33.17	A
36.33.18	D	36.33.19	C	36.33.20	C	36.33.21	D	36.33.22	A
36.33.23	D	36.33.24	C	36.33.25	C	36.33.26	A	36.33.27	C
36.33.28	A	36.33.29	C	36.33.30	A	36.33.31	B	36.33.32	D
36.33.33	D	36.34.1	Q-Q	36.34.2	Q-Q	36.35.1	Q-Q	36.36.1	D
36.37.1	Q-Q	36.38.1	Q-Q	36.39.1	Q-Q	36.40.1	Q-Q	36.41.1	Q-Q
36.42.1	Q-Q	36.42.2	A	36.42.3	C	36.43.1	Q-Q	36.43.2	Q-Q
36.44.1	Q-Q	36.45.1	Q-Q	36.45.2	2	36.46.1	C	36.47.1	Q-Q
36.48.1	Q-Q	36.49.1	B	36.49.2	D	36.50.1	Q-Q	36.51.1	Q-Q

36.52.1	C
36.54.4	D
36.55.3	D
36.55.8	B
36.59.1	Q-Q
36.63.1	C
36.67.1	Q-Q
36.72.1	D
36.75.1	Q-Q
36.77.1	Q-Q
36.79.1	Q-Q
36.80.1	Q-Q

36.53.1	C
36.54.5	C
36.55.4	A
36.55.9	D
36.60.1	Q-Q
36.64.1	D
36.68.1	A
36.73.1	Q-Q
36.75.2	A
36.77.2	Q-Q
36.79.2	Q-Q
36.80.2	D

36.54.1	C
36.54.6	D
36.55.5	A
36.56.1	Q-Q
36.61.1	Q-Q
36.65.1	Q-Q
36.69.1	B
36.74.1	Q-Q
36.75.3	B
36.77.3	Q-Q
36.79.3	B

36.54.2	C
36.55.1	A
36.55.6	C
36.57.1	Q-Q
36.61.2	Q-Q
36.65.2	Q-Q
36.70.1	Q-Q
36.74.2	Q-Q
36.76.1	Q-Q
36.77.4	C
36.79.4	A

36.54.3	C
36.55.2	D
36.55.7	A
36.58.1	Q-Q
36.62.1	Q-Q
36.66.1	C
36.71.1	Q-Q
36.74.3	Q-Q
36.76.2	Q-Q
36.78.1	Q-Q
36.79.5	D